



US012401912B2

(12) **United States Patent**
Rivard et al.

(10) **Patent No.:** **US 12,401,912 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE**

(71) Applicant: **Duelight LLC**, Sunnyvale, CA (US)

(72) Inventors: **William Rivard**, Menlo Park, CA (US);
Adam Feder, Mountain View, CA (US); **Brian Kindle**, Sunnyvale, CA (US)

(73) Assignee: **DUELIGHT LLC**, Sunnyvale, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **19/015,114**

(22) Filed: **Jan. 9, 2025**

(65) **Prior Publication Data**

US 2025/0150719 A1 May 8, 2025

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/388,158, filed on Nov. 8, 2023, which is a continuation-in-part (Continued)

(51) **Int. Cl.**
H04N 23/743 (2023.01)
H04N 23/56 (2023.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04N 23/743** (2023.01); **H04N 23/56** (2023.01); **H04N 23/62** (2023.01); **H04N 23/631** (2023.01);
(Continued)

(58) **Field of Classification Search**

CPC H04N 5/35554; H04N 5/2256; H04N 5/2355; H04N 5/2621; H04N 5/2624;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,057,809 A 11/1977 Nakamoto et al.
4,091,374 A 5/1978 Muller et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 101290388 A 10/2008
CN 101408709 A 4/2009
(Continued)

OTHER PUBLICATIONS

Notice of Allowance from U.S. Appl. No. 16/278,543, dated Apr. 3, 2020.

(Continued)

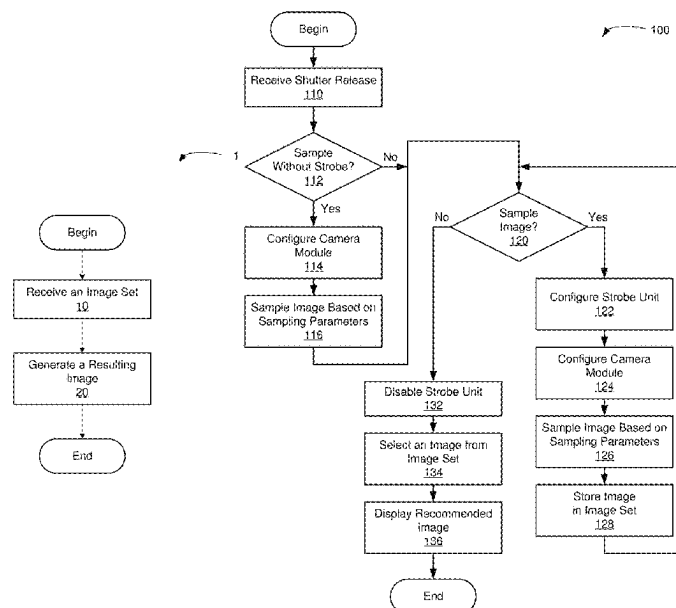
Primary Examiner — Mekonnen D Dagnew

(74) *Attorney, Agent, or Firm* — Zilka-Kotab, P.C.

(57) **ABSTRACT**

A system, method, and computer program product for generating a resulting image from a set of images is disclosed. The method comprises receiving an image set that includes a first image of a photographic scene based on a first set of sampling parameters and a second image of the photographic scene based on a second set of sampling parameters, and generating a resulting image based on the first image and the second image according to depth values in a selection depth map. Each distinct depth value in the selection depth map corresponds to a different image in the image set.

69 Claims, 19 Drawing Sheets



Related U.S. Application Data

of application No. 17/868,536, filed on Jul. 19, 2022, now abandoned, which is a continuation of application No. 17/518,473, filed on Nov. 3, 2021, now abandoned, which is a continuation of application No. 16/677,385, filed on Nov. 7, 2019, now Pat. No. 11,394,895, which is a continuation of application No. 16/211,931, filed on Dec. 6, 2018, now Pat. No. 10,491,834, which is a continuation of application No. 15/863,785, filed on Jan. 5, 2018, now Pat. No. 10,178,323, which is a continuation of application No. 15/289,039, filed on Oct. 7, 2016, now Pat. No. 9,894,289, which is a continuation of application No. 14/543,782, filed on Nov. 17, 2014, now Pat. No. 9,509,919.

(51) **Int. Cl.***H04N 23/62* (2023.01)*H04N 23/63* (2023.01)*H04N 23/72* (2023.01)*H04N 23/74* (2023.01)*H04N 23/741* (2023.01)(52) **U.S. Cl.**CPC *H04N 23/72* (2023.01); *H04N 23/74* (2023.01); *H04N 23/741* (2023.01)(58) **Field of Classification Search**

CPC .. H04N 5/265; H04N 5/3745; H04N 5/37452; H04N 5/37455; H04N 5/378

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,371,928	A	2/1983	Barlow et al.	5,726,670	A	3/1998	Tabata et al.
4,425,031	A	1/1984	Tamura et al.	5,734,760	A	3/1998	Yoshida
4,470,676	A	9/1984	Kinoshita et al.	5,737,547	A	4/1998	Zuravleff et al.
4,638,365	A	1/1987	Kato	5,754,705	A	5/1998	Okino
4,712,136	A	12/1987	Tsunekawa et al.	5,757,369	A	5/1998	Ohsawa et al.
4,720,723	A	1/1988	Harunari et al.	5,764,246	A	6/1998	Wataya et al.
4,734,762	A	3/1988	Aoki et al.	5,784,569	A	7/1998	Miller et al.
4,811,086	A	3/1989	Hieda	5,790,234	A	8/1998	Matsuyama
4,821,099	A	4/1989	Sakamoto	5,790,692	A	8/1998	Price et al.
4,832,518	A	5/1989	Moriyama	5,812,799	A	9/1998	Zuravleff et al.
4,873,561	A	10/1989	Wen	5,818,977	A	10/1998	Tansley
4,884,972	A	12/1989	Gasper	5,835,639	A	11/1998	Honsinger et al.
4,980,773	A	12/1990	Suda et al.	5,859,921	A	1/1999	Suzuki
5,101,253	A	3/1992	Mizutani et al.	5,867,215	A	2/1999	Kaplan
5,109,236	A	4/1992	Watanabe et al.	5,867,735	A	2/1999	Zuravleff et al.
5,115,124	A	5/1992	Muto et al.	5,872,867	A	2/1999	Bergen
5,126,777	A	6/1992	Akashi et al.	5,889,956	A	3/1999	Hauser et al.
5,132,783	A	7/1992	Hieda	5,900,909	A	5/1999	Parulski et al.
5,146,316	A	9/1992	Suzuki	5,909,594	A	6/1999	Ross et al.
5,151,796	A	9/1992	Ito et al.	5,917,494	A	6/1999	Arai et al.
5,175,615	A	12/1992	Ohara	5,983,261	A	11/1999	Riddle
5,185,668	A	2/1993	Ohta	5,986,668	A	11/1999	Szeliski et al.
5,200,828	A	4/1993	Jang et al.	5,987,164	A	11/1999	Szeliski et al.
5,262,870	A	11/1993	Nakamura et al.	5,987,186	A	11/1999	Oida et al.
5,282,024	A	1/1994	Takei	6,018,599	A	1/2000	Kawai
5,291,151	A	3/1994	Fujiwara et al.	6,026,188	A	2/2000	Dionysian
5,317,406	A	5/1994	Kobayashi et al.	6,038,074	A	3/2000	Kitaguchi et al.
5,321,528	A	6/1994	Nakamura	6,038,333	A	3/2000	Wang
5,363,209	A	11/1994	Eschbach et al.	6,041,351	A	3/2000	Kho
5,384,904	A	1/1995	Sprague et al.	6,055,326	A	4/2000	Chang et al.
5,553,864	A	9/1996	Sitrick	6,061,696	A	5/2000	Lee et al.
5,559,770	A	9/1996	Hiroki et al.	6,085,241	A	7/2000	Otis
5,572,633	A	11/1996	Lo et al.	6,092,137	A	7/2000	Huang et al.
5,579,530	A	11/1996	Solomon et al.	6,115,025	A	9/2000	Buxton et al.
5,633,677	A	5/1997	Okino et al.	6,115,065	A	9/2000	Yadid-Pecht et al.
5,689,437	A	11/1997	Nakagawa	6,115,717	A	9/2000	Mehrotra et al.
5,698,844	A	12/1997	Shinohara et al.	6,137,468	A	10/2000	Martinez et al.
5,699,108	A	12/1997	Katayama et al.	6,148,092	A	11/2000	Qian
				6,184,516	B1	2/2001	Sawada et al.
				6,184,940	B1	2/2001	Sano
				6,208,349	B1	3/2001	Davidson et al.
				6,241,609	B1	6/2001	Rutgers
				6,243,430	B1	6/2001	Mathe
				6,246,226	B1	6/2001	Kawase et al.
				6,262,769	B1	7/2001	Anderson et al.
				6,293,284	B1	9/2001	Rigg
				6,301,440	B1	10/2001	Bolle et al.
				6,326,978	B1	12/2001	Robbins
				6,332,033	B1	12/2001	Qian
				6,348,697	B1	2/2002	Kitajima
				6,365,950	B1	4/2002	Sohn
				6,385,169	B1	5/2002	Wang
				6,385,580	B1	5/2002	Lyberg et al.
				6,442,294	B1	8/2002	Nishizawa et al.
				6,453,068	B1	9/2002	Li
				6,473,159	B1	10/2002	Wakui et al.
				6,498,926	B1	12/2002	Ciccarelli et al.
				6,530,639	B1	3/2003	Matsuda
				6,532,011	B1	3/2003	Francini et al.
				6,539,129	B1	3/2003	Yushiya
				6,546,150	B2	4/2003	Inui
				6,577,613	B1	6/2003	Ramanathan
				6,594,279	B1	7/2003	Nguyen et al.
				6,597,399	B2	7/2003	Horii
				6,600,160	B2	7/2003	Kobayashi et al.
				6,608,622	B1	8/2003	Katayama et al.
				6,627,896	B1	9/2003	Hashimoto et al.
				6,642,962	B1	11/2003	Lin et al.
				6,650,774	B1	11/2003	Szeliski
				6,658,457	B2	12/2003	Nishikawa et al.
				6,662,233	B1	12/2003	Skarpness et al.
				6,704,007	B1	3/2004	Clapper
				6,704,844	B2	3/2004	Arimilli et al.
				6,734,905	B2	5/2004	Fossum et al.
				6,735,566	B1	5/2004	Brand
				6,744,471	B1	6/2004	Kakinuma et al.
				6,748,443	B1	6/2004	Parry et al.
				6,757,795	B2	6/2004	Barri et al.
				6,760,748	B1	7/2004	Hakim

(56)

References Cited

U.S. PATENT DOCUMENTS

6,778,948 B1	8/2004	Kanetaka et al.	7,730,422 B2	6/2010	Russo
6,787,778 B2	9/2004	Kobayashi et al.	7,750,913 B1	7/2010	Parenteau et al.
6,788,338 B1	9/2004	Dinev et al.	7,760,246 B2	7/2010	Dalton et al.
6,789,155 B2	9/2004	Jeddeloh	7,764,880 B2	7/2010	Hamada
6,810,031 B1	10/2004	Hegde et al.	7,768,920 B2	8/2010	Goshen et al.
6,842,265 B1	1/2005	Votipka et al.	7,783,126 B2	8/2010	Yamashita et al.
6,862,374 B1	3/2005	Nagai et al.	7,800,618 B1	9/2010	Westen
6,864,886 B1	3/2005	Cavallaro et al.	7,802,866 B2	9/2010	Kanno et al.
6,885,761 B2	4/2005	Kage	7,825,974 B2	11/2010	Itano et al.
6,906,332 B2	6/2005	Tashiro et al.	7,827,490 B2	11/2010	Kapur et al.
6,920,619 B1	7/2005	Milekic	7,835,586 B2	11/2010	Porikli
6,944,319 B1	9/2005	Huang et al.	7,842,927 B2	11/2010	Tashiro et al.
6,950,132 B1	9/2005	Kozuka	7,844,076 B2	11/2010	Corcoran et al.
6,952,015 B2	10/2005	Kameshima	7,847,259 B2	12/2010	Tashiro et al.
6,959,157 B2	10/2005	Nakayama	7,852,379 B2	12/2010	Aoki et al.
6,961,088 B2	11/2005	Kameshima et al.	7,903,115 B2	3/2011	Platzter et al.
6,975,750 B2	12/2005	Yan et al.	7,907,791 B2	3/2011	Kinrot et al.
6,989,863 B1	1/2006	Takahashi	7,908,410 B2	3/2011	Gildfind et al.
6,993,769 B2	1/2006	Simonson et al.	7,911,506 B2	3/2011	Suzuki
6,996,186 B2	2/2006	Ngai et al.	7,923,695 B2	4/2011	Ishii et al.
7,027,054 B1	4/2006	Cheiky et al.	7,923,696 B2	4/2011	Kameshima
7,030,868 B2	4/2006	Clapper	7,949,201 B2	5/2011	Suzuki
7,030,912 B1	4/2006	Honma	7,952,077 B2	5/2011	Tashiro et al.
7,030,922 B2	4/2006	Sakuragi	7,952,621 B2	5/2011	Yamauchi
7,076,563 B1	7/2006	Yamanaka et al.	7,953,286 B2	5/2011	Chiang et al.
7,084,905 B1	8/2006	Nayar et al.	7,962,030 B2	6/2011	Trevelyan
7,085,590 B2	8/2006	Kennedy et al.	7,966,661 B2	6/2011	Gunawardena
7,088,351 B2	8/2006	Wang	7,978,182 B2	7/2011	Ording et al.
7,095,879 B2	8/2006	Yan et al.	7,984,177 B2	7/2011	Girardeau, Jr. et al.
7,098,952 B2	8/2006	Morris et al.	7,991,887 B2	8/2011	Sutardja
7,113,497 B2	9/2006	Cromer et al.	7,999,858 B2	8/2011	Nayar et al.
7,113,648 B1	9/2006	Aihara	8,018,525 B2	9/2011	Trevelyan et al.
7,127,081 B1	10/2006	Erdem	8,063,967 B2	11/2011	Itano et al.
7,133,070 B2	11/2006	Wheeler et al.	8,068,121 B2	11/2011	Williamson et al.
7,138,639 B2	11/2006	Kameshima	8,081,606 B2	12/2011	Cai et al.
7,142,697 B2	11/2006	Huang et al.	8,107,497 B2	1/2012	Zerillo et al.
7,145,966 B2	12/2006	Lai et al.	8,114,172 B2	2/2012	Givon
7,164,423 B1	1/2007	Westen	8,120,625 B2	2/2012	Hinckley
7,206,449 B2	4/2007	Raskar et al.	8,125,499 B2	2/2012	Yamada
7,256,381 B2	8/2007	Asaba	8,125,526 B2	2/2012	Maruyama et al.
7,265,784 B1	9/2007	Frank	8,129,760 B2	3/2012	Hiromatsu
7,352,361 B2	4/2008	Yi	8,144,253 B2	3/2012	Su et al.
7,362,886 B2	4/2008	Rowe et al.	8,155,391 B1	4/2012	Tang et al.
7,381,963 B2	6/2008	Endo et al.	8,155,397 B2	4/2012	Bigioi et al.
7,415,152 B2	8/2008	Jiang et al.	8,170,576 B2	5/2012	Wu
7,421,063 B2	9/2008	Takenaka et al.	8,189,097 B2	5/2012	Chou et al.
7,428,378 B1	9/2008	Warpakowski Furlan	8,189,944 B1	5/2012	Lim
7,433,547 B2	10/2008	Yamamoto	8,194,993 B1	6/2012	Chen et al.
7,443,443 B2	10/2008	Raskar et al.	8,199,203 B2	6/2012	Sugimoto
7,470,911 B2	12/2008	Yagi	8,208,051 B2	6/2012	Kitani
7,514,690 B2	4/2009	Endo et al.	8,217,964 B2	7/2012	Laine et al.
7,518,645 B2	4/2009	Farrier	8,218,070 B2	7/2012	Kameshima
7,522,210 B2	4/2009	Shimada	8,218,625 B2	7/2012	Ward et al.
7,535,486 B2	5/2009	Motomura et al.	8,224,176 B1	7/2012	Pillman et al.
7,564,037 B2	7/2009	Tashiro et al.	8,228,560 B2	7/2012	Hooper
7,573,037 B1	8/2009	Kameshima et al.	8,233,003 B2	7/2012	Obinata
7,587,099 B2	9/2009	Szeliski et al.	8,237,813 B2	8/2012	Garten
7,590,775 B2	9/2009	Gildfind et al.	8,247,779 B2	8/2012	Kameshima et al.
7,592,599 B2	9/2009	Kameshima	8,270,764 B1	9/2012	Agarwala et al.
7,595,831 B2	9/2009	Kameshima et al.	8,276,085 B2	9/2012	Sherwani
7,599,541 B2	10/2009	Hayashida	8,291,446 B2	10/2012	Kennedy
7,599,569 B2	10/2009	Smirnov et al.	8,294,723 B2	10/2012	Labour et al.
7,609,860 B2	10/2009	Lee et al.	8,300,900 B2	10/2012	Lai et al.
7,622,698 B2	11/2009	Igaki et al.	8,314,817 B2	11/2012	Williamson et al.
7,626,598 B2	12/2009	Manchester	8,330,811 B2	12/2012	Macguire, Jr.
7,633,528 B2	12/2009	Fukushima et al.	8,335,021 B2	12/2012	Tanaka et al.
7,646,417 B2	1/2010	Goto et al.	8,339,508 B2	12/2012	Levy
7,646,909 B2	1/2010	Jiang et al.	8,345,327 B2	1/2013	Mikami
7,656,456 B2	2/2010	Zhang	8,346,923 B2	1/2013	Rowles et al.
7,660,464 B1	2/2010	Peterson	8,351,711 B2	1/2013	Takano et al.
7,676,170 B2	3/2010	Hata	8,363,122 B2	1/2013	Onozawa
7,705,911 B2	4/2010	Kameshima	8,363,145 B2	1/2013	Iwamoto
7,715,598 B2	5/2010	Li et al.	8,363,951 B2	1/2013	Bigioi et al.
7,724,292 B2	5/2010	Ueno et al.	8,363,952 B2	1/2013	Bigioi et al.
7,724,735 B2	5/2010	Locatelli et al.	8,369,586 B2	2/2013	Corcoran et al.
			8,369,893 B2	2/2013	Kirch et al.
			8,406,482 B1	3/2013	Chien et al.
			8,412,277 B2	4/2013	Fujiwara
			8,441,560 B2	5/2013	Yanai

(56)

References Cited

U.S. PATENT DOCUMENTS

8,451,296 B2	5/2013	Ono	8,928,654 B2	1/2015	Givon
8,488,219 B2	7/2013	Mikami	8,933,960 B2	1/2015	Lindahl et al.
8,493,413 B2	7/2013	Mo	8,934,029 B2	1/2015	Nayar et al.
8,508,600 B2	8/2013	Ibi	8,935,767 B2	1/2015	Li et al.
8,521,883 B1	8/2013	Wartnick et al.	8,937,735 B2	1/2015	Mori
RE44,499 E	9/2013	Inoue et al.	8,942,436 B2	1/2015	Mori et al.
8,531,465 B2	9/2013	Platzer et al.	8,947,382 B2	2/2015	Winkler et al.
8,532,395 B2	9/2013	Mitsui	8,948,584 B2	2/2015	Inoue et al.
8,543,946 B2	9/2013	Kethireddy	8,953,094 B2	2/2015	Baer
8,548,257 B2	10/2013	Reid et al.	8,953,882 B2	2/2015	Lim et al.
8,563,915 B2	10/2013	Takenaka et al.	8,965,390 B2	2/2015	Morrison
8,570,415 B2	10/2013	Takeda	8,966,401 B2	2/2015	Kang et al.
8,576,294 B2	11/2013	Kameshima et al.	8,970,770 B2	3/2015	Nanu et al.
8,581,935 B2	11/2013	Handa	8,976,264 B2	3/2015	Rivard et al.
8,605,142 B2	12/2013	Hayashi	8,977,073 B2	3/2015	Kwon et al.
8,610,724 B2	12/2013	Garg	8,988,349 B2	3/2015	Alberth et al.
8,610,789 B1	12/2013	Nayar et al.	8,988,548 B2	3/2015	Manabe
8,619,728 B2	12/2013	Frank et al.	8,988,559 B2	3/2015	Chao et al.
8,633,978 B2	1/2014	Yang et al.	8,988,561 B2	3/2015	Yamauchi
8,644,644 B2	2/2014	Yadav	9,001,232 B2	4/2015	Tatsumi
8,675,086 B1	3/2014	Linzer	9,001,249 B2	4/2015	Iwane
8,675,960 B2	3/2014	Reid et al.	9,003,293 B2	4/2015	Grosz et al.
8,682,029 B2	3/2014	Piramuthu	9,007,505 B2	4/2015	Ichikawa
8,692,851 B2	4/2014	Ordning et al.	9,014,459 B2	4/2015	Xiang et al.
8,692,917 B2	4/2014	Takeda	9,015,640 B2	4/2015	de Leon
8,698,062 B2	4/2014	Yoshida	9,030,566 B2	5/2015	Miyasako
8,699,822 B2	4/2014	Park et al.	9,030,587 B2	5/2015	Suzuki et al.
8,712,160 B2	4/2014	Bigioi et al.	9,036,056 B2	5/2015	Ohtsuka
8,712,189 B2	4/2014	Bitouk et al.	9,055,003 B2	6/2015	Baratakke et al.
8,717,293 B2	5/2014	Wong et al.	9,058,655 B2	6/2015	Doepke et al.
8,723,284 B1	5/2014	Hynecek	9,070,185 B2	6/2015	Lee et al.
8,754,978 B2	6/2014	Hayashi et al.	9,070,229 B2	6/2015	Williamson et al.
8,761,245 B2	6/2014	Puri et al.	9,077,943 B2	7/2015	Lim et al.
8,763,056 B2	6/2014	Bagasra	9,080,914 B2	7/2015	Kinugasa
8,764,560 B2	7/2014	Sitrick	9,081,257 B2	7/2015	Miyazaki
8,774,553 B1	7/2014	Winn	9,083,905 B2	7/2015	Wan et al.
8,774,554 B1	7/2014	Winn	9,083,935 B2	7/2015	Demandolx et al.
8,780,420 B1	7/2014	Bluzer et al.	9,098,069 B2	8/2015	Dickinson et al.
8,783,816 B2	7/2014	Masuda	9,100,514 B2	8/2015	Gu et al.
8,785,870 B2	7/2014	Matsumoto et al.	9,104,410 B2	8/2015	Wong et al.
8,786,725 B2	7/2014	Maruyama et al.	9,106,888 B2	8/2015	Chou
8,786,920 B2	7/2014	Mikami	9,118,876 B2	8/2015	Felt
8,792,679 B2	7/2014	Sengupta et al.	9,123,621 B2	9/2015	Watanabe
8,793,714 B2	7/2014	Kelsen et al.	9,124,814 B2	9/2015	Kim et al.
8,794,733 B2	8/2014	Kanno et al.	9,129,550 B2	9/2015	Doll
8,797,337 B1	8/2014	Labour et al.	9,137,455 B1	9/2015	Rivard et al.
8,797,445 B2	8/2014	Kang	9,144,714 B2	9/2015	Hollinger
8,805,091 B1	8/2014	Hensel et al.	9,154,708 B1	10/2015	Rivard et al.
8,809,760 B2	8/2014	Takenaka et al.	9,158,492 B2	10/2015	Miyata
8,811,757 B2	8/2014	Batur	9,160,936 B1	10/2015	Rivard et al.
8,817,048 B2	8/2014	Kerr et al.	9,165,533 B2	10/2015	Paulson
8,824,747 B2	9/2014	Free	9,167,169 B1	10/2015	Rivard et al.
8,830,177 B2	9/2014	Woo	9,167,174 B1	10/2015	Rivard et al.
8,830,261 B2	9/2014	Asai	9,177,362 B2	11/2015	Restrepo et al.
8,836,837 B2	9/2014	Kinugasa	9,179,062 B1	11/2015	Rivard et al.
8,849,984 B2	9/2014	Yamagishi	9,179,085 B1	11/2015	Rivard et al.
8,852,003 B2	10/2014	Oku	9,189,069 B2	11/2015	Hinckley
8,854,325 B2	10/2014	Byrd et al.	9,195,880 B1	11/2015	Levoy et al.
8,854,421 B2	10/2014	Kasahara	9,196,076 B1	11/2015	MacLeod
8,861,805 B2	10/2014	Hwang et al.	9,208,548 B1	12/2015	Noble et al.
8,861,847 B2	10/2014	Srinivasan et al.	9,215,405 B2	12/2015	Atkinson
8,861,886 B2	10/2014	Huo et al.	9,215,433 B2	12/2015	Rivard et al.
8,867,105 B2	10/2014	Mikami	9,218,662 B1	12/2015	Feder et al.
8,872,855 B2	10/2014	Doll	9,219,825 B2	12/2015	Sheikh Naziruddin et al.
8,878,117 B2	11/2014	Ogushi	9,223,467 B1	12/2015	Landry
8,878,963 B2	11/2014	Prabhudesai et al.	9,230,343 B2	1/2016	Ozawa
8,881,057 B2	11/2014	Mori et al.	9,232,124 B2	1/2016	Song
8,890,897 B2	11/2014	Homma et al.	9,232,164 B2	1/2016	Minowa et al.
8,896,632 B2	11/2014	MacDougall et al.	9,232,173 B1	1/2016	Duggal et al.
8,897,501 B2	11/2014	Okada et al.	9,239,947 B2	1/2016	Auberger et al.
8,897,504 B2	11/2014	Steinberg et al.	9,256,974 B1	2/2016	Hines
8,902,342 B2	12/2014	Araoka et al.	9,261,909 B2	2/2016	Lam
8,908,932 B2	12/2014	Corcoran et al.	9,277,135 B2	3/2016	Ichikawa
8,914,744 B2	12/2014	Spencer et al.	9,282,167 B2	3/2016	Jo et al.
8,915,437 B2	12/2014	Hoshino et al.	9,285,729 B2	3/2016	Saito et al.
			9,292,154 B2	3/2016	Ji et al.
			9,292,911 B2	3/2016	Paris et al.
			9,294,744 B2	3/2016	Takeda
			9,298,745 B2	3/2016	Lee et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,313,413 B2	4/2016	Cho et al.	9,942,504 B2	4/2018	Nishizawa
9,336,574 B2	5/2016	Zhang et al.	9,953,454 B1	4/2018	Rivard et al.
9,342,138 B2	5/2016	Ding	9,979,910 B2	5/2018	Ikeda
9,354,184 B2	5/2016	Dowaki	9,998,721 B2	6/2018	Rivard et al.
9,361,319 B2	6/2016	Feder et al.	9,998,935 B2	6/2018	Rivard et al.
9,383,202 B2	7/2016	Zhou et al.	10,007,842 B2	6/2018	Hu
9,384,384 B1	7/2016	Tyagi	10,025,972 B2	7/2018	Matas et al.
9,406,147 B2	8/2016	Rivard et al.	10,027,726 B1	7/2018	Ozog
9,407,675 B1	8/2016	Ludwig et al.	10,033,917 B1	7/2018	Silverstein et al.
9,417,836 B2	8/2016	Postal et al.	10,043,058 B2	8/2018	Ahmed
9,418,408 B1	8/2016	Thompson	10,049,425 B2	8/2018	Cornell
9,421,462 B2	8/2016	Oku	10,055,646 B2	8/2018	Bataller et al.
9,424,798 B2	8/2016	Park	10,088,866 B2	10/2018	Braun et al.
9,426,398 B2	8/2016	Ohshitanai	10,088,989 B2	10/2018	Rivard et al.
9,438,809 B2	9/2016	Sheikh et al.	10,097,765 B2	10/2018	Sheikh et al.
9,438,836 B2	9/2016	Tashiro et al.	10,102,829 B2	10/2018	Paulson
9,443,132 B2	9/2016	Linguraru et al.	10,109,098 B2	10/2018	Feder et al.
9,445,029 B2	9/2016	Shimizu	10,109,107 B2	10/2018	Knorr et al.
9,448,771 B2	9/2016	Feder et al.	10,110,867 B2	10/2018	Rivard et al.
9,456,144 B2	9/2016	Miyazaki	10,110,870 B2	10/2018	Rivard et al.
9,459,781 B2	10/2016	Wilson et al.	10,129,476 B1	11/2018	Hushchyn et al.
9,460,118 B2	10/2016	Feder et al.	10,129,514 B2	11/2018	Rivard et al.
9,460,125 B2	10/2016	Feder et al.	10,148,898 B2	12/2018	Dowaki et al.
9,467,607 B2	10/2016	Ng et al.	10,178,300 B2	1/2019	Rivard et al.
9,478,012 B2	10/2016	Uratani et al.	10,178,323 B2	1/2019	Rivard et al.
9,489,927 B2	11/2016	Aizawa	10,182,197 B2	1/2019	Feder et al.
9,495,025 B2	11/2016	Ishikawa et al.	10,186,019 B2	1/2019	Homma et al.
9,507,379 B2	11/2016	Kamei et al.	10,225,479 B2	3/2019	Shabtay et al.
9,507,445 B2	11/2016	Sutton et al.	10,270,958 B2	4/2019	Rivard et al.
9,508,133 B2	11/2016	Kindle et al.	10,271,001 B2	4/2019	Uchida et al.
9,509,919 B2	11/2016	Rivard et al.	10,284,834 B2	5/2019	Rivard et al.
9,516,217 B2	12/2016	Corcoran et al.	10,346,676 B2	7/2019	Ahmed
9,531,961 B2	12/2016	Rivard et al.	10,346,677 B2	7/2019	Steinberg et al.
9,538,112 B2	1/2017	Wada et al.	10,360,716 B1	7/2019	van der Meulen et al.
9,552,076 B2	1/2017	Homma et al.	10,365,820 B2	7/2019	Lee et al.
9,560,269 B2	1/2017	Baldwin	10,366,526 B2	7/2019	Rivard et al.
9,571,745 B2	2/2017	Mahowald	10,372,971 B2	8/2019	Rivard et al.
9,578,211 B2	2/2017	Kong et al.	10,375,367 B2	8/2019	Rivard et al.
9,591,225 B2	3/2017	Jung et al.	10,375,369 B2	8/2019	Rivard et al.
9,600,735 B2	3/2017	Nakagome et al.	10,382,702 B2	8/2019	Rivard et al.
9,600,741 B1	3/2017	Su et al.	10,387,963 B1	8/2019	Leise et al.
9,621,796 B2	4/2017	Muukki et al.	10,410,605 B2	9/2019	Gardenfors et al.
9,628,647 B2	4/2017	Tomono et al.	10,469,714 B2	11/2019	Rivard et al.
9,639,742 B2	5/2017	Lee et al.	10,477,077 B2	11/2019	Rivard et al.
9,646,576 B2	5/2017	Masuko	10,477,087 B2	11/2019	Rivard et al.
9,661,327 B2	5/2017	Nilsson	10,491,834 B2	11/2019	Rivard et al.
9,667,892 B2	5/2017	Sakuragi	10,498,982 B2	12/2019	Feder et al.
9,684,434 B2	6/2017	Lewin et al.	10,506,463 B2	12/2019	Rivard et al.
9,704,250 B1	7/2017	Shah et al.	10,521,946 B1	12/2019	Roche et al.
9,716,880 B2	7/2017	Chen et al.	10,552,016 B2	2/2020	Cherna et al.
9,721,375 B1	8/2017	Rivard et al.	10,552,946 B2	2/2020	Furukawa
9,741,150 B2	8/2017	Feder et al.	10,554,890 B1	2/2020	Le et al.
9,756,267 B2	9/2017	Dowaki et al.	10,554,943 B2	2/2020	Rivard et al.
9,760,764 B2	9/2017	Mishra et al.	10,558,848 B2	2/2020	Rivard et al.
9,761,033 B2	9/2017	Flider	10,586,097 B2	3/2020	Rivard et al.
9,769,404 B2	9/2017	Ohshitanai	10,586,369 B1	3/2020	Roche et al.
9,773,827 B2	9/2017	Minowa et al.	10,587,864 B2	3/2020	Hayasaka et al.
9,779,287 B2	10/2017	Steinberg et al.	10,621,771 B2	4/2020	Matts et al.
9,779,481 B2	10/2017	Yuasa	10,627,854 B2	4/2020	Gurr et al.
9,794,607 B2	10/2017	Joy et al.	10,628,666 B2	4/2020	Sareen et al.
9,798,395 B2	10/2017	Ye et al.	10,628,729 B2	4/2020	Sareen
9,800,638 B2	10/2017	Munagala et al.	10,630,903 B2	4/2020	Srivastava et al.
9,807,322 B2	10/2017	Feder et al.	10,652,478 B2	5/2020	Rivard et al.
9,819,849 B1	11/2017	Rivard et al.	10,708,545 B2	7/2020	Rivard et al.
9,846,804 B2	12/2017	Lim et al.	10,742,892 B1	8/2020	Le et al.
9,858,648 B2	1/2018	Li et al.	10,785,401 B2	9/2020	Rivard et al.
9,860,461 B2	1/2018	Feder et al.	10,810,781 B2	10/2020	Feder et al.
9,886,192 B2	2/2018	Masuko	10,832,388 B2	11/2020	Liu et al.
9,894,289 B2	2/2018	Rivard et al.	10,834,329 B2	11/2020	Gatt et al.
9,898,674 B2	2/2018	Connell, II et al.	10,880,521 B2	12/2020	Rivard et al.
9,912,928 B2	3/2018	Rivard et al.	10,904,505 B2	1/2021	Rivard et al.
9,918,017 B2	3/2018	Rivard et al.	10,917,571 B2	2/2021	Shanmugam et al.
9,924,147 B2	3/2018	Rivard et al.	10,924,688 B2	2/2021	Rivard et al.
9,934,561 B2	4/2018	Feder et al.	10,929,945 B2	2/2021	Lim et al.
9,942,464 B2	4/2018	Voss	10,931,897 B2	2/2021	Feder et al.
			10,937,222 B2	3/2021	Rivard et al.
			11,025,831 B2	6/2021	Rivard et al.
			11,113,821 B2	9/2021	Rivard et al.
			11,113,859 B1	9/2021	Xiao et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

11,202,047 B2	12/2021	Rivard et al.	2004/0181375 A1	9/2004	Szu et al.
11,252,589 B2	2/2022	Rivard et al.	2004/0184115 A1	9/2004	Suzuki
11,308,626 B2	4/2022	Rivard et al.	2004/0184677 A1	9/2004	Raskar et al.
11,356,647 B2	6/2022	Rivard	2004/0201589 A1	10/2004	Ekstrom
11,363,179 B2	6/2022	Rivard et al.	2004/0228528 A1	11/2004	Lao
11,375,085 B2	6/2022	Rivard et al.	2004/0234259 A1	11/2004	Muramatsu
11,394,894 B2	7/2022	Rivard et al.	2004/0239792 A1	12/2004	Shibutani et al.
11,394,895 B2	7/2022	Rivard et al.	2004/0247177 A1	12/2004	Rowe et al.
11,455,829 B2	9/2022	Rivard et al.	2004/0252199 A1	12/2004	Cheung et al.
11,463,630 B2	10/2022	Rivard et al.	2004/0263510 A1	12/2004	Marschner et al.
11,620,754 B2	4/2023	Rivard et al.	2005/0010697 A1	1/2005	Kinawi et al.
11,683,448 B2	6/2023	Rivard et al.	2005/0022131 A1	1/2005	Saint-Hilaire et al.
11,699,219 B2	7/2023	Rivard et al.	2005/0047333 A1	3/2005	Todd et al.
11,729,518 B2	8/2023	Rivard et al.	2005/0068432 A1	3/2005	Kozuka
12,003,853 B2	6/2024	Rivard et al.	2005/0088570 A1	4/2005	Seo
12,003,864 B2	6/2024	Rivard et al.	2005/0089212 A1	4/2005	Mashitani et al.
12,143,842 B2	11/2024	Rivard et al.	2005/0093997 A1	5/2005	Dalton et al.
12,175,632 B2	12/2024	Gao et al.	2005/0104848 A1	5/2005	Yamaguchi et al.
2001/0009437 A1	7/2001	Klein et al.	2005/0128438 A1	6/2005	Kang et al.
2001/0030770 A1	10/2001	Ohashi	2005/0134723 A1	6/2005	Lee et al.
2001/0033284 A1	10/2001	Chan	2005/0143124 A1	6/2005	Kennedy et al.
2001/0033336 A1	10/2001	Kameshima et al.	2005/0147292 A1	7/2005	Huang et al.
2001/0033675 A1	10/2001	Maurer et al.	2005/0154798 A1	7/2005	Nurmi
2001/0039582 A1	11/2001	McKinnon et al.	2005/0155043 A1	7/2005	Schulz et al.
2002/0003545 A1	1/2002	Nakamura	2005/0173646 A1	8/2005	Tashiro et al.
2002/0006232 A1	1/2002	Inui	2005/0180657 A1	8/2005	Zhang et al.
2002/0012450 A1	1/2002	Tsujii	2005/0182808 A1	8/2005	Kanai
2002/0059432 A1	5/2002	Masuda et al.	2005/0196069 A1	9/2005	Yonaha
2002/0060566 A1	5/2002	Debbins et al.	2005/0218333 A1	10/2005	Kameshima
2002/0063714 A1	5/2002	Haas et al.	2005/0237587 A1	10/2005	Nakamura
2002/0070945 A1	6/2002	Kage	2005/0237588 A1	10/2005	Gohara et al.
2002/0087300 A1	7/2002	Patwari	2005/0253945 A1	11/2005	Shinohara
2002/0106114 A1	8/2002	Yan et al.	2005/0253946 A1	11/2005	Shinohara
2002/0107750 A1	8/2002	Kanevsky et al.	2005/0264688 A1	12/2005	Ouchi
2002/0113882 A1	8/2002	Pollard et al.	2005/0286559 A1	12/2005	Miernik et al.
2002/0114296 A1	8/2002	Hardy	2006/0007346 A1	1/2006	Nakamura et al.
2002/0122014 A1	9/2002	Rajasingham	2006/0008171 A1	1/2006	Petschnigg et al.
2002/0141256 A1	10/2002	Barri et al.	2006/0012689 A1	1/2006	Dalton et al.
2002/0146074 A1	10/2002	Ariel et al.	2006/0015308 A1	1/2006	Marschner et al.
2002/0196472 A1	12/2002	Enomoto	2006/0017837 A1	1/2006	Sorek et al.
2003/0015645 A1	1/2003	Brickell et al.	2006/0026535 A1	2/2006	Hotelling et al.
2003/0025816 A1	2/2003	Sakuragi	2006/0028987 A1	2/2006	Alexander Gildfind et al.
2003/0039211 A1	2/2003	Hvostov et al.	2006/0029292 A1	2/2006	Hagiwara
2003/0042425 A1	3/2003	Tashiro et al.	2006/0033724 A1	2/2006	Chaudhri et al.
2003/0046477 A1	3/2003	Jeddeloh	2006/0033760 A1	2/2006	Koh
2003/0086002 A1	5/2003	Cahill et al.	2006/0038899 A1	2/2006	Tamaru et al.
2003/0090577 A1	5/2003	Shirakawa	2006/0039630 A1	2/2006	Kotani
2003/0097531 A1	5/2003	Arimilli et al.	2006/0050165 A1	3/2006	Amano
2003/0103523 A1	6/2003	Frossard et al.	2006/0050335 A1	3/2006	Dorrell et al.
2003/0133599 A1	7/2003	Tian et al.	2006/0053371 A1	3/2006	Anderson
2003/0142745 A1	7/2003	Osawa	2006/0054834 A1	3/2006	Kameshima
2003/0148811 A1	8/2003	Sitrick	2006/0067668 A1*	3/2006	Kita
2003/0152096 A1	8/2003	Chapman	2006/0072810 A1	4/2006	Scharlack et al.
2003/0179911 A1	9/2003	Ho et al.	2006/0085541 A1	4/2006	Cuomo et al.
2003/0184660 A1	10/2003	Skow	2006/0087702 A1	4/2006	Satoh et al.
2003/0206654 A1	11/2003	Teng	2006/0115157 A1	6/2006	Mori et al.
2003/0210672 A1	11/2003	Cromer et al.	2006/0120571 A1	6/2006	Tu et al.
2003/0212792 A1	11/2003	Raymond	2006/0133375 A1	6/2006	Napierala
2003/0215153 A1	11/2003	Gindele et al.	2006/0133695 A1	6/2006	Obinata
2003/0222978 A9	12/2003	Thomason	2006/0139460 A1	6/2006	Ozaki
2003/0223622 A1	12/2003	Simon et al.	2006/0153127 A1	7/2006	Lee
2004/0021701 A1	2/2004	Iwema et al.	2006/0177150 A1	8/2006	Uyttendaele et al.
2004/0027471 A1	2/2004	Koseki et al.	2006/0181614 A1	8/2006	Yen et al.
2004/0042807 A1	3/2004	Nakayama	2006/0188132 A1	8/2006	Shigeta
2004/0045006 A1	3/2004	Simonson et al.	2006/0192130 A1	8/2006	Yagi
2004/0052248 A1	3/2004	Frank et al.	2006/0192785 A1	8/2006	Marschner et al.
2004/0059783 A1	3/2004	Kazui et al.	2006/0195252 A1	8/2006	Orr et al.
2004/0070676 A1	4/2004	Lawther et al.	2006/0203100 A1	9/2006	Ajito et al.
2004/0071465 A1	4/2004	Smart et al.	2006/0212538 A1	9/2006	Sutardja
2004/0085259 A1	5/2004	Tarleton et al.	2006/0215016 A1	9/2006	Cohen et al.
2004/0135912 A1	7/2004	Hofflinger et al.	2006/0222260 A1	10/2006	Sambongi et al.
2004/0145674 A1	7/2004	Hoppe et al.	2006/0225034 A1	10/2006	Peck et al.
2004/0150622 A1	8/2004	Bohn	2006/0231875 A1	10/2006	Patrick et al.
2004/0158582 A1	8/2004	Takagi et al.	2006/0245014 A1	11/2006	Haneda
2004/0178349 A1	9/2004	Kameshima	2006/0245639 A1	11/2006	Jiang et al.
			2006/0262363 A1	11/2006	Henley
			2006/0280343 A1	12/2006	Lee et al.
			2007/0002897 A1	1/2007	Goshen et al.

H04N 23/56
348/E.029

(56)

References Cited

U.S. PATENT DOCUMENTS

2007/0004451 A1	1/2007	C. Anderson	2008/0158403 A1	7/2008	Itano et al.
2007/0019000 A1	1/2007	Motomura et al.	2008/0165144 A1	7/2008	Forstall et al.
2007/0023798 A1	2/2007	McKee	2008/0170160 A1	7/2008	Lukac
2007/0025714 A1	2/2007	Shiraki	2008/0174570 A1	7/2008	Jobs et al.
2007/0025717 A1	2/2007	Raskar et al.	2008/0181597 A1	7/2008	Tamura
2007/0025720 A1	2/2007	Raskar et al.	2008/0192064 A1	8/2008	Hong et al.
2007/0030357 A1	2/2007	Levien et al.	2008/0192773 A1	8/2008	Ou et al.
2007/0045433 A1	3/2007	Chapman et al.	2008/0192819 A1	8/2008	Ward et al.
2007/0052838 A1	3/2007	Zhang	2008/0192991 A1	8/2008	Gremse et al.
2007/0060135 A1	3/2007	Lin	2008/0193119 A1	8/2008	Miyazaki
2007/0069144 A1	3/2007	Kameshima	2008/0198219 A1	8/2008	Yoshida et al.
2007/0070364 A1	3/2007	Henley	2008/0199172 A1	8/2008	Hamada
2007/0080299 A1	4/2007	Endo et al.	2008/0207322 A1	8/2008	Mizrahi
2007/0092137 A1	4/2007	Zhao et al.	2008/0212591 A1	9/2008	Wu et al.
2007/0101067 A1	5/2007	Shafi et al.	2008/0218611 A1	9/2008	Parulski et al.
2007/0101251 A1	5/2007	Lee et al.	2008/0225057 A1	9/2008	Hertzfeld et al.
2007/0110305 A1	5/2007	Corcoran et al.	2008/0266326 A1	10/2008	Porwal
2007/0121182 A1	5/2007	Fukushima et al.	2008/0269958 A1	10/2008	Filev et al.
2007/0122034 A1	5/2007	Maor	2008/0275881 A1	11/2008	Conn et al.
2007/0133988 A1	6/2007	Kim et al.	2008/0297596 A1	12/2008	Inomata et al.
2007/0136208 A1	6/2007	Hamashima et al.	2008/0298794 A1	12/2008	Subbotin et al.
2007/0146529 A1	6/2007	Suzuki	2008/0303813 A1	12/2008	Joung et al.
2007/0146538 A1	6/2007	Kakinuma et al.	2008/0307363 A1	12/2008	Jalon et al.
2007/0165960 A1	7/2007	Yamada	2008/0309788 A1	12/2008	Onozawa
2007/0171210 A1	7/2007	Chaudhri et al.	2008/0309810 A1	12/2008	Smith et al.
2007/0182823 A1	8/2007	Maruyama et al.	2008/0310753 A1	12/2008	Edgar
2007/0182936 A1	8/2007	Suzuki	2008/0316226 A1	12/2008	Weibrecht et al.
2007/0189748 A1	8/2007	Drimbarean et al.	2008/0317376 A1	12/2008	Kasperkiewicz
2007/0198931 A1	8/2007	Ono et al.	2009/0002335 A1	1/2009	Chaudhri
2007/0200663 A1	8/2007	White et al.	2009/0002391 A1	1/2009	Williamson et al.
2007/0200925 A1	8/2007	Kim	2009/0002395 A1	1/2009	Yamada
2007/0201853 A1	8/2007	Petschnigg	2009/0002475 A1	1/2009	Jelley et al.
2007/0206885 A1	9/2007	Wen	2009/0006626 A1	1/2009	Yamagishi
2007/0223062 A1	9/2007	Tanaka et al.	2009/0009636 A1	1/2009	Endo
2007/0223954 A1	9/2007	Hata	2009/0015581 A1	1/2009	Hirohara
2007/0236515 A1	10/2007	Montague	2009/0021937 A1	1/2009	Sloan et al.
2007/0236709 A1	10/2007	Mitani	2009/0027516 A1	1/2009	Suzuki
2007/0242900 A1	10/2007	Chen et al.	2009/0034460 A1	2/2009	Moratt et al.
2007/0244925 A1	10/2007	Albouze	2009/0040364 A1	2/2009	Rubner
2007/0248342 A1	10/2007	Tamminen et al.	2009/0041376 A1	2/2009	Carletta et al.
2007/0258008 A1	11/2007	Kameshima et al.	2009/0052748 A1	2/2009	Jiang et al.
2007/0260979 A1	11/2007	Hertzfeld et al.	2009/0058845 A1	3/2009	Fukuda et al.
2007/0263099 A1	11/2007	Motta et al.	2009/0058873 A1	3/2009	Brown Elliott et al.
2007/0263106 A1	11/2007	Tanaka et al.	2009/0058882 A1	3/2009	Adachi et al.
2007/0264000 A1	11/2007	Hsieh et al.	2009/0060379 A1	3/2009	Manabe
2007/0274331 A1	11/2007	Locatelli et al.	2009/0066782 A1	3/2009	Choi et al.
2007/0280505 A1	12/2007	Breed	2009/0080713 A1	3/2009	Bigioi et al.
2007/0291081 A1	12/2007	Kanno et al.	2009/0085919 A1	4/2009	Chen et al.
2007/0296820 A1	12/2007	Lonn	2009/0110274 A1	4/2009	Atanassov et al.
2007/0297567 A1	12/2007	Takenaka et al.	2009/0141149 A1	6/2009	Zhang et al.
2008/0001945 A1	1/2008	Kashito et al.	2009/0153245 A1	6/2009	Lee
2008/0001950 A1	1/2008	Lin et al.	2009/0160944 A1	6/2009	Trevelyan et al.
2008/0004946 A1	1/2008	Schwarz	2009/0160992 A1	6/2009	Inaba et al.
2008/0018763 A1	1/2008	Sato	2009/0166547 A1	7/2009	Endo et al.
2008/0018911 A1	1/2008	Igaki et al.	2009/0172516 A1	7/2009	Gill
2008/0019575 A1	1/2008	Scalise et al.	2009/0175551 A1	7/2009	Thorn
2008/0019680 A1	1/2008	Kasahara et al.	2009/0175555 A1	7/2009	Mahowald
2008/0025576 A1	1/2008	Li et al.	2009/0181770 A1	7/2009	Viner et al.
2008/0030592 A1	2/2008	Border et al.	2009/0185659 A1	7/2009	Ishii et al.
2008/0037829 A1	2/2008	Givon	2009/0196236 A1	8/2009	Cai et al.
2008/0043032 A1	2/2008	Mamona et al.	2009/0218476 A1	9/2009	Kameshima et al.
2008/0050022 A1	2/2008	Okada et al.	2009/0219387 A1	9/2009	Marman et al.
2008/0052945 A1	3/2008	Matas et al.	2009/0230290 A1	9/2009	Tashiro et al.
2008/0076481 A1	3/2008	Iwasaki et al.	2009/0230312 A1	9/2009	Tashiro et al.
2008/0079842 A1	4/2008	Aoki et al.	2009/0237420 A1	9/2009	Lawrenz
2008/0092051 A1	4/2008	Sidon et al.	2009/0238419 A1	9/2009	Steinberg et al.
2008/0094419 A1	4/2008	Leigh et al.	2009/0257683 A1	10/2009	Cloud et al.
2008/0106636 A1	5/2008	Wernersson	2009/0262074 A1	10/2009	Nasiri et al.
2008/0107411 A1	5/2008	Hope	2009/0262087 A1	10/2009	Kim
2008/0112361 A1	5/2008	Wu	2009/0268055 A1	10/2009	Hamilton, Jr. et al.
2008/0122796 A1	5/2008	Jobs et al.	2009/0278922 A1	11/2009	Tinker et al.
2008/0122933 A1	5/2008	Murayama	2009/0295941 A1	12/2009	Nakajima et al.
2008/0129848 A1	6/2008	Yamauchi	2009/0303373 A1	12/2009	Yamada
2008/0151097 A1	6/2008	Chen et al.	2009/0304070 A1	12/2009	Lamy-Bergot et al.
2008/0158398 A1	7/2008	Yaffe et al.	2009/0309035 A1	12/2009	Kameshima
			2009/0309985 A1	12/2009	Ibi
			2009/0309990 A1	12/2009	Levoy et al.
			2009/0309994 A1	12/2009	Inoue
			2009/0322903 A1	12/2009	Hashimoto et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2009/0323897	A1	12/2009	Kameshima et al.	2011/0058237	A1	3/2011	Mikami
2010/0007603	A1	1/2010	Kirkup	2011/0071911	A1	3/2011	Tung et al.
2010/0007665	A1	1/2010	Smith et al.	2011/0074973	A1	3/2011	Hayashi
2010/0010986	A1	1/2010	Ichio et al.	2011/0090256	A1	4/2011	Manchester
2010/0013842	A1	1/2010	Green et al.	2011/0090385	A1	4/2011	Aoyama et al.
2010/0017545	A1	1/2010	Gildfind et al.	2011/0095169	A1	4/2011	Takenaka et al.
2010/0020822	A1	1/2010	Zerillo et al.	2011/0096192	A1	4/2011	Niikura
2010/0026836	A1	2/2010	Sugimoto	2011/0096375	A1	4/2011	Mikami
2010/0066763	A1	3/2010	MacDougall et al.	2011/0105229	A1	5/2011	Sitrick
2010/0066822	A1	3/2010	Steinberg et al.	2011/0115893	A1	5/2011	Hayashi
2010/0073499	A1	3/2010	Gere	2011/0115971	A1	5/2011	Furuya et al.
2010/0079491	A1	4/2010	Nonaka	2011/0123118	A1	5/2011	Nayar et al.
2010/0079494	A1	4/2010	Sung et al.	2011/0131041	A1	6/2011	Cortez et al.
2010/0092036	A1	4/2010	Das et al.	2011/0131331	A1	6/2011	Brunson et al.
2010/0115462	A1	5/2010	Spencer et al.	2011/0134267	A1	6/2011	Ohya
2010/0115605	A1	5/2010	Beattie et al.	2011/0134297	A1	6/2011	Takeda
2010/0118038	A1	5/2010	Labour et al.	2011/0145694	A1	6/2011	Graves et al.
2010/0118115	A1	5/2010	Takahashi et al.	2011/0150332	A1	6/2011	Sibiryakov et al.
2010/0118204	A1	5/2010	Proca et al.	2011/0150357	A1	6/2011	Prentice
2010/0121964	A1	5/2010	Rowles et al.	2011/0157412	A1	6/2011	Yoshida
2010/0123737	A1	5/2010	Williamson et al.	2011/0158473	A1	6/2011	Sun et al.
2010/0123805	A1	5/2010	Craig et al.	2011/0167382	A1	7/2011	van Os
2010/0123929	A1	5/2010	Yoshimoto	2011/0185296	A1	7/2011	Lanier et al.
2010/0128145	A1	5/2010	Pitts et al.	2011/0187749	A1	8/2011	Dehmann et al.
2010/0135541	A1	6/2010	Lai et al.	2011/0193982	A1	8/2011	Kook et al.
2010/0146446	A1	6/2010	Ahn et al.	2011/0194618	A1	8/2011	Gish et al.
2010/0149377	A1	6/2010	Shintani et al.	2011/0199519	A1	8/2011	Yamauchi
2010/0157079	A1	6/2010	Atanassov et al.	2011/0205395	A1	8/2011	Levy
2010/0157139	A1	6/2010	Velarde et al.	2011/0212717	A1	9/2011	Rhoads et al.
2010/0160049	A1	6/2010	Oku	2011/0221758	A1	9/2011	Livingston
2010/0165178	A1	7/2010	Chou et al.	2011/0221911	A1	9/2011	Kang
2010/0165181	A1	7/2010	Murakami et al.	2011/0221935	A1	9/2011	Kameshima
2010/0172578	A1	7/2010	Reid et al.	2011/0227923	A1	9/2011	Mariani et al.
2010/0172579	A1	7/2010	Reid et al.	2011/0242334	A1	10/2011	Wilburn et al.
2010/0182465	A1	7/2010	Okita	2011/0249086	A1	10/2011	Guo et al.
2010/0183071	A1	7/2010	Segall et al.	2011/0254792	A1	10/2011	Waters et al.
2010/0194851	A1	8/2010	Pasupaleti et al.	2011/0254860	A1	10/2011	Zontrop et al.
2010/0194963	A1	8/2010	Terashima	2011/0256886	A1	10/2011	Velusamy
2010/0201831	A1	8/2010	Weinstein	2011/0261075	A1	10/2011	Tanaka
2010/0201846	A1	8/2010	Silverbrook	2011/0279698	A1	11/2011	Yoshikawa
2010/0208099	A1	8/2010	Nomura	2011/0279699	A1	11/2011	Matsui
2010/0214320	A1	8/2010	Mo	2011/0280541	A1	11/2011	Lee
2010/0215259	A1	8/2010	Scalise et al.	2011/0283346	A1	11/2011	Li et al.
2010/0218113	A1	8/2010	White et al.	2011/0286658	A1	11/2011	Mitsui
2010/0220933	A1	9/2010	Takano et al.	2011/0292242	A1	12/2011	Imai
2010/0231747	A1	9/2010	Yim	2011/0298828	A1	12/2011	Suzuki
2010/0231761	A1	9/2010	Yanai	2011/0298982	A1	12/2011	Kobayashi
2010/0257239	A1	10/2010	Roberts	2011/0299741	A1	12/2011	Zhang et al.
2010/0265079	A1	10/2010	Yin	2011/0304613	A1	12/2011	Thoreson
2010/0278452	A1	11/2010	Sarkijarvi et al.	2011/0310094	A1	12/2011	Park et al.
2010/0299630	A1	11/2010	McCutchen et al.	2011/0311150	A1	12/2011	Okamoto
2010/0302278	A1	12/2010	Shaffer et al.	2011/0312376	A1	12/2011	Woo et al.
2010/0302407	A1	12/2010	Ayers et al.	2011/0316859	A1	12/2011	Nikula et al.
2010/0302408	A1	12/2010	Ito	2011/0316888	A1	12/2011	Sachs et al.
2010/0315656	A1	12/2010	Agata	2011/0317005	A1	12/2011	Atkinson
2010/0317332	A1	12/2010	Bathiche et al.	2011/0317917	A1	12/2011	Free
2010/0321439	A1	12/2010	Kanno et al.	2012/0001943	A1	1/2012	Ishidera
2010/0322074	A1	12/2010	Nakahira	2012/0002082	A1	1/2012	Johnson et al.
2010/0328442	A1	12/2010	Yang et al.	2012/0002089	A1	1/2012	Wang et al.
2010/0328486	A1	12/2010	Steinberg et al.	2012/0007859	A1	1/2012	Lee et al.
2010/0329564	A1	12/2010	Hervas et al.	2012/0007908	A1	1/2012	Masuda
2010/0333044	A1	12/2010	Kethireddy	2012/0008011	A1	1/2012	Garcia Manchado
2011/0012914	A1	1/2011	Nakamura et al.	2012/0026359	A1	2/2012	Fukushima et al.
2011/0013042	A1	1/2011	Itano et al.	2012/0033118	A1	2/2012	Lee et al.
2011/0013043	A1	1/2011	Corcoran et al.	2012/0033262	A1	2/2012	Sakurai
2011/0013052	A1	1/2011	Yanai	2012/0036051	A1	2/2012	Sachson
2011/0019051	A1	1/2011	Yin et al.	2012/0038635	A1	2/2012	Stamate et al.
2011/0019094	A1	1/2011	Rossignol et al.	2012/0044266	A1	2/2012	Mori
2011/0029635	A1	2/2011	Shkurko et al.	2012/0044543	A1	2/2012	Mikami
2011/0032384	A1	2/2011	Ono	2012/0056889	A1	3/2012	Carter et al.
2011/0036986	A1	2/2011	Tashiro et al.	2012/0057051	A1	3/2012	Ito et al.
2011/0037712	A1	2/2011	Kim et al.	2012/0057064	A1	3/2012	Gardiner et al.
2011/0037777	A1	2/2011	Lindahl et al.	2012/0057786	A1	3/2012	Yano
2011/0057880	A1	3/2011	Kasahara	2012/0066355	A1	3/2012	Tiwari et al.
2011/0058060	A1	3/2011	Bigioi et al.	2012/0069213	A1	3/2012	Jannard et al.
				2012/0075285	A1	3/2012	Oyagi et al.
				2012/0075492	A1	3/2012	Nanu et al.
				2012/0081382	A1	4/2012	Lindahl et al.
				2012/0104267	A1	5/2012	Matsumoto et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2012/0105579 A1	5/2012	Jeon et al.	2013/0027580 A1	1/2013	Olsen et al.
2012/0105584 A1	5/2012	Gallagher et al.	2013/0038634 A1	2/2013	Yamada et al.
2012/0113092 A1	5/2012	Bar-Zeev et al.	2013/0044237 A1	2/2013	Ikizyan et al.
2012/0113106 A1	5/2012	Choi et al.	2013/0050460 A1	2/2013	Steinberg et al.
2012/0117499 A1	5/2012	Mori et al.	2013/0050520 A1	2/2013	Takeuchi
2012/0120304 A1	5/2012	Corcoran et al.	2013/0050551 A1	2/2013	Ichikawa
2012/0127072 A1	5/2012	Kim	2013/0057713 A1	3/2013	Khawand
2012/0127333 A1	5/2012	Maruyama et al.	2013/0063571 A1	3/2013	Ishii
2012/0139904 A1	6/2012	Lee et al.	2013/0063633 A1	3/2013	Hayashi et al.
2012/0144347 A1	6/2012	Jo et al.	2013/0067093 A1	3/2013	Moreno et al.
2012/0154276 A1	6/2012	Shin et al.	2013/0069988 A1	3/2013	Kamei et al.
2012/0154541 A1	6/2012	Scott	2013/0069989 A1	3/2013	Nagata et al.
2012/0154627 A1	6/2012	Rivard et al.	2013/0070111 A1	3/2013	Ohtsuka
2012/0159372 A1	6/2012	Stallings et al.	2013/0070145 A1	3/2013	Matsuyama
2012/0162251 A1	6/2012	Minamino et al.	2013/0101219 A1	4/2013	Bosworth et al.
2012/0162263 A1	6/2012	Griffin et al.	2013/0101220 A1	4/2013	Bosworth et al.
2012/0162465 A1	6/2012	Culbert et al.	2013/0107062 A1	5/2013	Okazaki
2012/0172088 A1	7/2012	Kirch et al.	2013/0108123 A1	5/2013	Hwang et al.
2012/0173889 A1	7/2012	Wong et al.	2013/0111337 A1	5/2013	Deng et al.
2012/0176413 A1	7/2012	Kulik et al.	2013/0111369 A1	5/2013	Pasquero et al.
2012/0176474 A1	7/2012	Border	2013/0114853 A1	5/2013	Sengupta et al.
2012/0176477 A1	7/2012	Givon	2013/0114894 A1	5/2013	Yadav et al.
2012/0177352 A1	7/2012	Pillman et al.	2013/0120011 A1	5/2013	Yamazaki et al.
2012/0188386 A1	7/2012	Kulkarni et al.	2013/0120256 A1	5/2013	Ishidera et al.
2012/0188392 A1	7/2012	Smith	2013/0120607 A1	5/2013	Manabe
2012/0194545 A1	8/2012	Shibata et al.	2013/0129142 A1	5/2013	Miranda-Steiner
2012/0194905 A1	8/2012	Ushio et al.	2013/0129209 A1	5/2013	Reid et al.
2012/0206488 A1	8/2012	Wong et al.	2013/0132903 A1	5/2013	Krishnaswamy
2012/0206582 A1	8/2012	DiCarlo et al.	2013/0140435 A1	6/2013	Kikuchi
2012/0210275 A1	8/2012	Park et al.	2013/0141456 A1	6/2013	Sokolov et al.
2012/0212661 A1	8/2012	Yamaguchi et al.	2013/0141464 A1	6/2013	Hunt et al.
2012/0213407 A1	8/2012	Haikin et al.	2013/0147979 A1	6/2013	McMahon et al.
2012/0218290 A1	8/2012	Waschbuesch et al.	2013/0148013 A1	6/2013	Shiohara
2012/0224042 A1	9/2012	Saijo	2013/0154978 A1	6/2013	Kim et al.
2012/0224788 A1	9/2012	Jia et al.	2013/0162542 A1	6/2013	Badali et al.
2012/0226800 A1	9/2012	Baratakke et al.	2013/0162848 A1	6/2013	Miyasako
2012/0229370 A1	9/2012	Stroffolino et al.	2013/0162874 A1	6/2013	Hashimoto et al.
2012/0242683 A1	9/2012	Asai	2013/0176222 A1	7/2013	Tanaka
2012/0242844 A1	9/2012	Walker et al.	2013/0176442 A1	7/2013	Shuster et al.
2012/0242886 A1	9/2012	Kawarada	2013/0176458 A1	7/2013	Van Dalen et al.
2012/0250082 A1	10/2012	Mori	2013/0179308 A1	7/2013	Agustin et al.
2012/0250952 A1	10/2012	Kveton et al.	2013/0179831 A1	7/2013	Izaki
2012/0254808 A1	10/2012	Gildfind	2013/0188886 A1	7/2013	Petrou et al.
2012/0256959 A1	10/2012	Ye et al.	2013/0194963 A1	8/2013	Hampel
2012/0262450 A1	10/2012	Aiba	2013/0196653 A1	8/2013	Morrison
2012/0262600 A1	10/2012	Velarde et al.	2013/0201217 A1	8/2013	Morinaga et al.
2012/0273651 A1	11/2012	Willassen	2013/0205244 A1	8/2013	Decker et al.
2012/0274661 A1	11/2012	Ye et al.	2013/0215113 A1	8/2013	Corazza et al.
2012/0274806 A1	11/2012	Mori	2013/0222231 A1	8/2013	Gardenfors et al.
2012/0277914 A1	11/2012	Crow et al.	2013/0222275 A1	8/2013	Byrd et al.
2012/0278727 A1	11/2012	Ananthakrishnan et al.	2013/0222516 A1	8/2013	Do et al.
2012/0287223 A1	11/2012	Zhang et al.	2013/0222646 A1	8/2013	Tsubota et al.
2012/0293556 A1	11/2012	Kim et al.	2013/0223530 A1	8/2013	Demos
2012/0294533 A1	11/2012	Ikenoue	2013/0227469 A1	8/2013	Park
2012/0299964 A1	11/2012	Homma et al.	2013/0228673 A1	9/2013	Hashimoto et al.
2012/0307096 A1	12/2012	Ford et al.	2013/0235068 A1	9/2013	Ubillos et al.
2012/0307097 A1	12/2012	Araoka et al.	2013/0235069 A1	9/2013	Ubillos et al.
2012/0307100 A1	12/2012	Iwane	2013/0235071 A1	9/2013	Ubillos et al.
2012/0314100 A1	12/2012	Frank	2013/0235081 A1	9/2013	Ishibashi
2012/0314124 A1	12/2012	Kaizu et al.	2013/0239057 A1	9/2013	Ubillos et al.
2012/0324400 A1	12/2012	Caliendo, Jr. et al.	2013/0239154 A1	9/2013	Bagasra
2013/0001402 A1	1/2013	Ogushi	2013/0241442 A1	9/2013	Trattler
2013/0001429 A1	1/2013	Dowaki et al.	2013/0243271 A1	9/2013	Kawahara et al.
2013/0002932 A1	1/2013	Guenter et al.	2013/0251202 A1	9/2013	Auberger et al.
2013/0006953 A1	1/2013	Epshtein et al.	2013/0257877 A1	10/2013	Davis
2013/0010075 A1	1/2013	Gallagher et al.	2013/0258040 A1	10/2013	Kaytaz et al.
2013/0010138 A1	1/2013	Bigioi et al.	2013/0258044 A1	10/2013	Betts-Lacroix
2013/0011039 A1	1/2013	Kadir et al.	2013/0258118 A1	10/2013	Felt
2013/0016122 A1	1/2013	Bhatt et al.	2013/0262258 A1	10/2013	Jennings
2013/0016222 A1	1/2013	Jiang et al.	2013/0262486 A1	10/2013	O'Dell et al.
2013/0019196 A1	1/2013	Bhatt	2013/0271622 A1	10/2013	Tatsumi
2013/0021358 A1	1/2013	Nordlund et al.	2013/0271631 A1	10/2013	Tatsuzawa et al.
2013/0021377 A1	1/2013	Doll	2013/0277534 A1	10/2013	Watanabe
2013/0021447 A1	1/2013	Brisedoux et al.	2013/0278482 A1	10/2013	Hsu
2013/0026349 A1	1/2013	Kinugasa	2013/0278798 A1	10/2013	Hattori
			2013/0278819 A1	10/2013	Liaw
			2013/0278979 A1	10/2013	Mikami
			2013/0279584 A1	10/2013	Demos
			2013/0286876 A1	10/2013	Moratt et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2013/0287265	A1	10/2013	Nepomniachtchi et al.	2014/0210754	A1	7/2014	Ryu et al.
2013/0293502	A1	11/2013	Kitatani	2014/0210847	A1	7/2014	Knibbeler et al.
2013/0293744	A1	11/2013	Attar et al.	2014/0211852	A1	7/2014	Demos
2013/0294688	A1	11/2013	Auberger et al.	2014/0215365	A1	7/2014	Hiraga et al.
2013/0301729	A1	11/2013	Demos	2014/0219517	A1	8/2014	Mishra et al.
2013/0301885	A1	11/2013	Mori et al.	2014/0219526	A1	8/2014	Linguraru et al.
2013/0303247	A1	11/2013	Wu et al.	2014/0240453	A1	8/2014	Kim et al.
2013/0307999	A1	11/2013	Motta	2014/0240543	A1	8/2014	Kim et al.
2013/0314576	A1	11/2013	Suzuki et al.	2014/0244858	A1	8/2014	Okazaki
2013/0321671	A1	12/2013	Cote et al.	2014/0247342	A1	9/2014	Ellenby et al.
2013/0328864	A1	12/2013	Lee et al.	2014/0247870	A1	9/2014	Mertens
2013/0328935	A1	12/2013	Tu	2014/0247979	A1	9/2014	Roffet et al.
2013/0329015	A1	12/2013	Pulli et al.	2014/0253752	A1	9/2014	Kawano et al.
2013/0335317	A1	12/2013	Liu et al.	2014/0258674	A1	9/2014	Kim et al.
2013/0335596	A1	12/2013	Demandolx et al.	2014/0267833	A1	9/2014	Chen et al.
2013/0336600	A1	12/2013	Bitouk et al.	2014/0267869	A1	9/2014	Sawa
2013/0342526	A1	12/2013	Ng et al.	2014/0269550	A1	9/2014	Webb
2013/0342739	A1	12/2013	Yanowitz et al.	2014/0270543	A1	9/2014	Zhang et al.
2013/0342740	A1	12/2013	Govindarao	2014/0279181	A1	9/2014	Wills
2014/0001340	A1	1/2014	Takenaka et al.	2014/0283113	A1	9/2014	Hanna
2014/0002606	A1	1/2014	Blayvas et al.	2014/0289360	A1	9/2014	Mahkovec et al.
2014/0002718	A1	1/2014	Spielberg	2014/0298323	A1	10/2014	Stolberg et al.
2014/0009636	A1	1/2014	Lee et al.	2014/0300795	A1	10/2014	Bilcu et al.
2014/0009664	A1	1/2014	Kinugasa	2014/0301642	A1	10/2014	Muninder
2014/0016001	A1	1/2014	Ichikawa	2014/0306899	A1	10/2014	Hicks
2014/0036118	A1	2/2014	Dowaki et al.	2014/0307001	A1	10/2014	Aizawa
2014/0036121	A1	2/2014	Minowa et al.	2014/0307117	A1	10/2014	Feng et al.
2014/0043628	A1	2/2014	Kishino et al.	2014/0310788	A1	10/2014	Ricci
2014/0055494	A1	2/2014	Mikawa	2014/0317295	A1	10/2014	Martini
2014/0063287	A1	3/2014	Yamada	2014/0320720	A1	10/2014	Ohshitanai
2014/0063301	A1	3/2014	Solhusvik	2014/0337791	A1	11/2014	Agnetta et al.
2014/0063611	A1	3/2014	Raymond et al.	2014/0340428	A1	11/2014	Shibayama
2014/0070965	A1	3/2014	Letsu-Dake	2014/0351687	A1	11/2014	Hall
2014/0075286	A1	3/2014	Harada	2014/0354781	A1	12/2014	Matsuyama
2014/0075372	A1	3/2014	Wu et al.	2014/0359517	A1	12/2014	Elings et al.
2014/0078171	A1	3/2014	Miyatake et al.	2014/0359656	A1	12/2014	Banica et al.
2014/0079279	A1	3/2014	Pulli et al.	2014/0362117	A1	12/2014	Paulson
2014/0085339	A1	3/2014	Brady et al.	2014/0364241	A1	12/2014	Oku
2014/0085422	A1	3/2014	Aronsson et al.	2014/0365977	A1	12/2014	Elyada et al.
2014/0085430	A1	3/2014	Komori et al.	2014/0372914	A1	12/2014	Byrd et al.
2014/0085508	A1	3/2014	Ichikawa	2014/0375837	A1	12/2014	Ichihara
2014/0096064	A1	4/2014	Suzuki	2015/0005637	A1	1/2015	Stegman et al.
2014/0098118	A1	4/2014	Liu et al.	2015/0015740	A1	1/2015	Cho et al.
2014/0098248	A1	4/2014	Okazaki	2015/0015774	A1	1/2015	Sugie
2014/0098259	A1	4/2014	Song et al.	2015/0016693	A1	1/2015	Gattuso
2014/0111548	A1	4/2014	Shin	2015/0016735	A1	1/2015	Kikuchi
2014/0111657	A1	4/2014	Weatherford et al.	2015/0025359	A1	1/2015	Fenchel et al.
2014/0112299	A1	4/2014	Frank et al.	2015/0026101	A1	1/2015	Lin et al.
2014/0118256	A1	5/2014	Sonoda et al.	2015/0029226	A1	1/2015	Feder et al.
2014/0119290	A1	5/2014	Grewal et al.	2015/0030242	A1	1/2015	Shen
2014/0125856	A1	5/2014	Kim et al.	2015/0030246	A1 *	1/2015	Wilensky G06T 5/70 382/167
2014/0129966	A1	5/2014	Kolesnikov et al.	2015/0035991	A1	2/2015	Sachs et al.
2014/0132735	A1	5/2014	Lee et al.	2015/0042669	A1	2/2015	Van Nostrand et al.
2014/0140630	A1	5/2014	Hwang et al.	2015/0042743	A1	2/2015	Cullen
2014/0160333	A1	6/2014	Takeda	2015/0049119	A1	2/2015	Homma et al.
2014/0161351	A1	6/2014	Yagnik	2015/0055835	A1	2/2015	Ozawa
2014/0168271	A1	6/2014	Yu et al.	2015/0062038	A1	3/2015	Taniuchi
2014/0168468	A1	6/2014	Levoy et al.	2015/0062044	A1	3/2015	Chen et al.
2014/0176458	A1	6/2014	Matsuda	2015/0063694	A1	3/2015	Shroff et al.
2014/0176750	A1	6/2014	Pajak et al.	2015/0067600	A1	3/2015	Steinberg et al.
2014/0176757	A1	6/2014	Rivard et al.	2015/0070458	A1	3/2015	Kim et al.
2014/0176759	A1	6/2014	Goto	2015/0077326	A1	3/2015	Kramer et al.
2014/0177008	A1	6/2014	Raymond et al.	2015/0077581	A1	3/2015	Baltz et al.
2014/0181089	A1	6/2014	Desmond et al.	2015/0077603	A1	3/2015	Matsuzawa et al.
2014/0184865	A1	7/2014	Muto et al.	2015/0078661	A1	3/2015	Granados et al.
2014/0184894	A1	7/2014	Motta	2015/0084885	A1	3/2015	Kawamoto
2014/0186050	A1	7/2014	Oshima et al.	2015/0085184	A1	3/2015	Vidal et al.
2014/0189181	A1	7/2014	Cheng et al.	2015/0091945	A1	4/2015	Uratani et al.
2014/0192216	A1	7/2014	Matsumoto	2015/0092019	A1	4/2015	Asano et al.
2014/0192267	A1	7/2014	Biswas et al.	2015/0092077	A1	4/2015	Feder et al.
2014/0193088	A1	7/2014	Capata et al.	2015/0092852	A1	4/2015	Demos
2014/0197302	A1	7/2014	Araoka et al.	2015/0093044	A1	4/2015	Feder et al.
2014/0198242	A1	7/2014	Weng et al.	2015/0095775	A1	4/2015	Lewis et al.
2014/0204084	A1	7/2014	Corazza et al.	2015/0098014	A1	4/2015	Raskar et al.
2014/0208208	A1	7/2014	Chevallier et al.	2015/0098651	A1	4/2015	Rivard et al.
				2015/0103192	A1	4/2015	Venkatraman et al.
				2015/0109505	A1	4/2015	Sakuragi
				2015/0113368	A1	4/2015	Flider

(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0113370	A1	4/2015	Flider	2016/0092472	A1	3/2016	Feder et al.
2015/0113371	A1	4/2015	Flider	2016/0110168	A1	4/2016	Feder et al.
2015/0116365	A1	4/2015	Ding	2016/0132130	A1	5/2016	Lu
2015/0116523	A1	4/2015	Sinha et al.	2016/0133037	A1	5/2016	Vemulapalli et al.
2015/0117786	A1	4/2015	James et al.	2016/0139774	A1	5/2016	Rivard et al.
2015/0127775	A1	5/2015	Munagala et al.	2016/0140702	A1	5/2016	Kindle et al.
2015/0130978	A1	5/2015	Tashiro et al.	2016/0142610	A1	5/2016	Rivard et al.
2015/0138366	A1	5/2015	Keelan et al.	2016/0143040	A1	5/2016	Rivard et al.
2015/0142182	A1	5/2015	Kimura et al.	2016/0148551	A1	5/2016	Jian
2015/0146079	A1	5/2015	Kim	2016/0148648	A1	5/2016	Dimson et al.
2015/0146979	A1	5/2015	Paliy et al.	2016/0150147	A1	5/2016	Shioya
2015/0169166	A1	6/2015	Kim et al.	2016/0150175	A1	5/2016	Hynecek
2015/0169940	A1	6/2015	Kang et al.	2016/0154994	A1	6/2016	Kim et al.
2015/0178977	A1	6/2015	Kontkanen	2016/0157587	A1	6/2016	Yamanashi et al.
2015/0189161	A1	7/2015	Chun et al.	2016/0163289	A1	6/2016	Masuko
2015/0193912	A1	7/2015	Yuasa	2016/0170608	A1	6/2016	Zambetti et al.
2015/0195330	A1	7/2015	Lee	2016/0173782	A1	6/2016	Dimson et al.
2015/0199022	A1	7/2015	Gottesman et al.	2016/0179387	A1	6/2016	Gaur et al.
2015/0205236	A1	7/2015	Saito et al.	2016/0180567	A1	6/2016	Lee
2015/0207920	A1	7/2015	Choi et al.	2016/0182874	A1	6/2016	Richards et al.
2015/0213784	A1	7/2015	Jafarzadeh et al.	2016/0191973	A1	6/2016	Joy et al.
2015/0215526	A1	7/2015	Jafarzadeh et al.	2016/0196042	A1	7/2016	Laute et al.
2015/0215532	A1	7/2015	Jafarzadeh et al.	2016/0202866	A1	7/2016	Zambetti et al.
2015/0222809	A1	8/2015	Osuka et al.	2016/0202872	A1	7/2016	Jang et al.
2015/0222836	A1	8/2015	Wada et al.	2016/0210716	A1	7/2016	Ghosh et al.
2015/0229819	A1	8/2015	Rivard et al.	2016/0219211	A1	7/2016	Katayama
2015/0229898	A1	8/2015	Rivard et al.	2016/0231883	A1	8/2016	Zambetti et al.
2015/0235073	A1	8/2015	Hua et al.	2016/0232419	A1	8/2016	Bai et al.
2015/0244980	A1	8/2015	Matthews	2016/0240168	A1	8/2016	Keal et al.
2015/0249810	A1	9/2015	Sasaki	2016/0248968	A1	8/2016	Baldwin
2015/0254502	A1	9/2015	Lim et al.	2016/0274622	A1	9/2016	Braun et al.
2015/0256752	A1	9/2015	Ninan et al.	2016/0274768	A1	9/2016	Tian et al.
2015/0264273	A1	9/2015	Feder et al.	2016/0275650	A1	9/2016	Case et al.
2015/0269423	A1	9/2015	Nakagome et al.	2016/0277656	A1	9/2016	Tsunoda
2015/0278853	A1	10/2015	McLaughlin et al.	2016/0284065	A1	9/2016	Cohen
2015/0278999	A1	10/2015	Summers et al.	2016/0292905	A1	10/2016	Nehmadi et al.
2015/0279113	A1	10/2015	Knorr et al.	2016/0313781	A1	10/2016	Jeon et al.
2015/0287189	A1	10/2015	Hirai et al.	2016/0315830	A1	10/2016	Cote et al.
2015/0288870	A1	10/2015	Nagaraja et al.	2016/0316154	A1	10/2016	Elmfors et al.
2015/0296145	A1	10/2015	Hwang et al.	2016/0316156	A1	10/2016	Ohshitanai
2015/0302587	A1	10/2015	Hirano et al.	2016/0323518	A1	11/2016	Rivard et al.
2015/0310261	A1	10/2015	Lee et al.	2016/0330383	A1	11/2016	Oyama
2015/0312397	A1	10/2015	Chiang	2016/0342863	A1	11/2016	Kwon et al.
2015/0324983	A1	11/2015	Takasugi et al.	2016/0344927	A1	11/2016	Brasket et al.
2015/0334318	A1	11/2015	Georgiev et al.	2016/0350587	A1	12/2016	Bataller et al.
2015/0339002	A1	11/2015	Arnold et al.	2016/0352996	A1	12/2016	Qian et al.
2015/0339006	A1	11/2015	Chaland et al.	2016/0357420	A1	12/2016	Wilson et al.
2015/0341536	A1	11/2015	Huang et al.	2016/0366331	A1	12/2016	Barron et al.
2015/0341593	A1	11/2015	Zhang et al.	2016/0371824	A1	12/2016	Feder et al.
2015/0350516	A1	12/2015	Rivard et al.	2016/0373362	A1	12/2016	Cheng et al.
2015/0350562	A1	12/2015	Hubel et al.	2016/0381304	A9	12/2016	Feder et al.
2015/0363915	A1	12/2015	Li et al.	2016/0381348	A1	12/2016	Hayasaka et al.
2015/0370399	A1	12/2015	Kim et al.	2017/0006322	A1	1/2017	Dury et al.
2015/0373414	A1	12/2015	Kinoshita	2017/0011745	A1	1/2017	Navaratnam
2015/0379690	A1	12/2015	Stepanov et al.	2017/0019617	A1	1/2017	Dowaki et al.
2016/0006949	A1	1/2016	Kim et al.	2017/0026562	A1	1/2017	Rivard et al.
2016/0006987	A1	1/2016	Li et al.	2017/0032181	A1	2/2017	Hu
2016/0014312	A1	1/2016	Nikkanen et al.	2017/0039750	A1	2/2017	Tong et al.
2016/0014421	A1	1/2016	Cote et al.	2017/0048442	A1	2/2017	Cote et al.
2016/0026658	A1	1/2016	Krishnaraj et al.	2017/0048449	A1	2/2017	Chen et al.
2016/0027150	A1	1/2016	Lee et al.	2017/0052604	A1	2/2017	Hunt et al.
2016/0028948	A1	1/2016	Omori et al.	2017/0054966	A1	2/2017	Zhou et al.
2016/0034166	A1	2/2016	Wilson et al.	2017/0055101	A1	2/2017	Studerus et al.
2016/0034167	A1	2/2016	Wilson et al.	2017/0061234	A1	3/2017	Lim et al.
2016/0044293	A1	2/2016	Rivard et al.	2017/0061236	A1	3/2017	Pope
2016/0054851	A1	2/2016	Kim et al.	2017/0061567	A1	3/2017	Lim et al.
2016/0057348	A1	2/2016	Liang et al.	2017/0061669	A1	3/2017	Hirano et al.
2016/0062515	A1	3/2016	Bae et al.	2017/0064192	A1	3/2017	Mori
2016/0062615	A1	3/2016	Price et al.	2017/0064204	A1	3/2017	Sapiro et al.
2016/0062645	A1	3/2016	Masuko	2017/0064227	A1	3/2017	Lin et al.
2016/0065926	A1	3/2016	Nonaka et al.	2017/0064276	A1	3/2017	Rivard et al.
2016/0070963	A1	3/2016	Chakraborty et al.	2017/0068846	A1	3/2017	Linguraru et al.
2016/0071241	A1	3/2016	Karunamuni et al.	2017/0070690	A1	3/2017	Feder et al.
2016/0071289	A1	3/2016	Kobayashi et al.	2017/0076430	A1	3/2017	Xu
2016/0086318	A1	3/2016	Hannuksela et al.	2017/0085785	A1	3/2017	Corcoran et al.
				2017/0109807	A1	4/2017	Krishnan et al.
				2017/0109931	A1	4/2017	Knorr et al.
				2017/0115749	A1	4/2017	Li
				2017/0118394	A1	4/2017	Van Hoeckel et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0140214	A1	5/2017	Matas et al.	2019/0012525	A1	1/2019	Wang et al.
2017/0150080	A1	5/2017	Nishizawa	2019/0026010	A1	1/2019	Rivard et al.
2017/0150118	A1	5/2017	Pacheco et al.	2019/0031145	A1	1/2019	Trelin
2017/0154211	A1	6/2017	Shaburov et al.	2019/0035047	A1	1/2019	Lim et al.
2017/0154457	A1	6/2017	Theobald et al.	2019/0035135	A1	1/2019	Feder et al.
2017/0169303	A1	6/2017	Connell, II et al.	2019/0037192	A1	1/2019	Rivard et al.
2017/0187938	A1	6/2017	Ichihara	2019/0039570	A1	2/2019	Foster et al.
2017/0187953	A1	6/2017	Graham et al.	2019/0042833	A1	2/2019	Gernoth et al.
2017/0201677	A1	7/2017	Otani	2019/0043176	A1	2/2019	Li et al.
2017/0201692	A1	7/2017	Wu	2019/0045165	A1	2/2019	Rivard et al.
2017/0208292	A1	7/2017	Smits	2019/0057554	A1	2/2019	Knorr et al.
2017/0213076	A1	7/2017	Francisco et al.	2019/0080119	A1	3/2019	Wang et al.
2017/0228583	A1	8/2017	Lee et al.	2019/0102279	A1	4/2019	Awan et al.
2017/0236253	A1	8/2017	Restrepo et al.	2019/0108387	A1	4/2019	Rivard et al.
2017/0237786	A1	8/2017	Crowe et al.	2019/0108388	A1	4/2019	Rivard et al.
2017/0237925	A1	8/2017	Uchida et al.	2019/0109974	A1	4/2019	Rivard et al.
2017/0256086	A1	9/2017	Park et al.	2019/0116306	A1	4/2019	Rivard et al.
2017/0262695	A1	9/2017	Ahmed	2019/0122378	A1	4/2019	Aswin
2017/0274768	A1	9/2017	Hök et al.	2019/0124280	A1	4/2019	Feder et al.
2017/0277941	A1	9/2017	Smith	2019/0149706	A1	5/2019	Rivard et al.
2017/0278292	A1	9/2017	Feder et al.	2019/0174028	A1	6/2019	Rivard et al.
2017/0280069	A1	9/2017	Smith	2019/0179594	A1	6/2019	Alameh et al.
2017/0285743	A1	10/2017	Yu et al.	2019/0188857	A1	6/2019	Rivard et al.
2017/0286752	A1	10/2017	Gusarov et al.	2019/0197297	A1	6/2019	Rivard et al.
2017/0300778	A1	10/2017	Sato	2019/0197330	A1	6/2019	Mahmoud et al.
2017/0302903	A1	10/2017	Ng et al.	2019/0208172	A1	7/2019	Rivard et al.
2017/0323149	A1	11/2017	Harary et al.	2019/0215440	A1	7/2019	Rivard et al.
2017/0337440	A1	11/2017	Green et al.	2019/0222769	A1	7/2019	Srivastava et al.
2017/0337657	A1	11/2017	Cornell	2019/0222807	A1	7/2019	Rivard et al.
2017/0343887	A1	11/2017	Hoshino	2019/0244010	A1	8/2019	Kim et al.
2017/0364752	A1	12/2017	Zhou et al.	2019/0251682	A1	8/2019	Feder et al.
2017/0372108	A1	12/2017	Corcoran et al.	2019/0263415	A1	8/2019	Gong
2017/0374336	A1	12/2017	Rivard et al.	2019/0285881	A1	9/2019	Ilic et al.
2018/0005420	A1	1/2018	Bondich et al.	2019/0335151	A1	10/2019	Rivard et al.
2018/0007240	A1	1/2018	Rivard et al.	2019/0342534	A1	11/2019	Rivard et al.
2018/0020156	A1	1/2018	Zobel	2019/0347843	A1	11/2019	Rivard et al.
2018/0024661	A1	1/2018	Lin et al.	2019/0349510	A1	11/2019	Rivard et al.
2018/0025218	A1	1/2018	Steinberg et al.	2019/0379863	A1	12/2019	Rivard et al.
2018/0025244	A1	1/2018	Bohl et al.	2019/0385311	A1	12/2019	Rivard et al.
2018/0061126	A1	3/2018	Huang et al.	2020/0029008	A1	1/2020	Rivard et al.
2018/0063019	A1	3/2018	Martin	2020/0057654	A1	2/2020	Yang
2018/0063409	A1	3/2018	Rivard et al.	2020/0059575	A1	2/2020	Rivard et al.
2018/0063411	A1	3/2018	Rivard et al.	2020/0059582	A1	2/2020	Rivard et al.
2018/0067633	A1	3/2018	Wilson et al.	2020/0059806	A1	2/2020	Rivard et al.
2018/0070069	A1	3/2018	Rivard et al.	2020/0067715	A1	2/2020	Yi et al.
2018/0074495	A1	3/2018	Myers et al.	2020/0077013	A1	3/2020	Rivard et al.
2018/0075637	A1	3/2018	Henry et al.	2020/0084398	A1	3/2020	Feder et al.
2018/0077367	A1	3/2018	Feder et al.	2020/0092596	A1	3/2020	Moore, Jr. et al.
2018/0088775	A1	3/2018	Ye et al.	2020/0106956	A1	4/2020	Kimball et al.
2018/0109771	A1	4/2018	Rivard et al.	2020/0118315	A1	4/2020	Ranzinger
2018/0114025	A1	4/2018	Cui et al.	2020/0126283	A1	4/2020	van Vuuren et al.
2018/0114351	A1	4/2018	Rivard et al.	2020/0154089	A1	5/2020	Rivard et al.
2018/0114352	A1	4/2018	Rivard et al.	2020/0193144	A1	6/2020	Rivard et al.
2018/0115702	A1	4/2018	Brauer et al.	2020/0242774	A1	7/2020	Park et al.
2018/0121716	A1	5/2018	Sun et al.	2020/0249763	A1	8/2020	Moritani
2018/0130182	A1	5/2018	Bhatt et al.	2020/0259991	A1	8/2020	Rivard et al.
2018/0131855	A1	5/2018	Rivard et al.	2020/0267299	A1	8/2020	Le et al.
2018/0137375	A1	5/2018	Takemura et al.	2020/0351432	A1	11/2020	Rivard et al.
2018/0137678	A1	5/2018	Kaehler	2021/0001810	A1	1/2021	Rivard et al.
2018/0160092	A1	6/2018	Rivard et al.	2021/0037178	A1	2/2021	Rivard et al.
2018/0165862	A1	6/2018	Sawaki	2021/0049984	A1	2/2021	Cain
2018/0183986	A1	6/2018	Smith et al.	2021/0074051	A1	3/2021	Feder et al.
2018/0183989	A1	6/2018	Rivard et al.	2021/0110554	A1	4/2021	Rivard et al.
2018/0197281	A1	7/2018	Feder et al.	2021/0152779	A1	5/2021	Rivard et al.
2018/0204052	A1	7/2018	Li et al.	2021/0274142	A1	9/2021	Rivard et al.
2018/0211101	A1	7/2018	Ahmed	2021/0314507	A1	10/2021	Feder et al.
2018/0253881	A1	9/2018	Edwards et al.	2021/0319613	A1	10/2021	Rivard et al.
2018/0262934	A1	9/2018	Rivard et al.	2021/0337104	A1	10/2021	Rivard et al.
2018/0288311	A1	10/2018	Baghert et al.	2021/0360141	A1	11/2021	Rivard et al.
2018/0295292	A1	10/2018	Lee et al.	2021/0382969	A1	12/2021	Park
2018/0341383	A1	11/2018	Sully	2022/0107550	A1*	4/2022	Yeh G03B 7/16
2018/0342091	A1	11/2018	Seibert et al.	2022/0210313	A1	6/2022	Rivard et al.
2018/0352241	A1	12/2018	Gatt et al.	2022/0210386	A1	6/2022	Rivard et al.
2018/0367774	A1	12/2018	Barron et al.	2022/0215504	A1	7/2022	Moens
2019/0005632	A1	1/2019	Huang	2022/0224820	A1	7/2022	Liu et al.
				2022/0239866	A1	7/2022	Rivard et al.
				2022/0272553	A1	8/2022	Rivard et al.
				2022/0291758	A1	9/2022	Wang et al.
				2022/0343476	A1	10/2022	Feder et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2022/0343509 A1 10/2022 Rivard et al.
 2022/0343678 A1 10/2022 Rivard et al.
 2022/0345613 A1 10/2022 Rivard et al.
 2022/0353712 A1 11/2022 Rivard et al.
 2022/0368793 A1 11/2022 Xu et al.
 2023/0005294 A1 1/2023 Rivard et al.
 2023/0008242 A1 1/2023 Rivard et al.
 2023/0047124 A1 2/2023 Rivard et al.
 2023/0050695 A1 2/2023 Rivard et al.
 2023/0052018 A1 2/2023 Rivard et al.
 2023/0060489 A1 3/2023 Rivard et al.
 2023/0061404 A1 3/2023 Feder et al.
 2023/0079783 A1 3/2023 Rivard et al.
 2023/0081630 A1 3/2023 Rivard et al.
 2023/0093132 A1 3/2023 Rivard et al.
 2023/0154097 A1 5/2023 Feder et al.
 2023/0156344 A1 5/2023 Rivard et al.
 2023/0156350 A1 5/2023 Feder et al.
 2023/0188676 A1 6/2023 Rivard et al.
 2023/0325080 A1 10/2023 Huo et al.
 2023/0351608 A1 11/2023 Rivard et al.
 2024/0064419 A1 2/2024 Rivard et al.
 2024/0073543 A1 2/2024 Rivard et al.
 2025/0030945 A1 1/2025 Rivard et al.
 2025/0053287 A1 2/2025 Rivard et al.
 2025/0056130 A1 2/2025 Rivard et al.
 2025/0056131 A1 2/2025 Rivard et al.
 2025/0063246 A1 2/2025 Rivard et al.
 2025/0071430 A1 2/2025 Rivard et al.

FOREIGN PATENT DOCUMENTS

CN 101547306 A 9/2009
 CN 102053453 A 5/2011
 CN 102165783 A 8/2011
 CN 102521814 A 6/2012
 CN 102608036 A 7/2012
 CN 103152519 A 6/2013
 CN 103581556 A 2/2014
 CN 103813098 A 5/2014
 CN 104040292 A 9/2014
 CN 204316606 U 5/2015
 CN 105026955 A 11/2015
 DE 102011107844 A1 1/2013
 EP 1549080 A1 6/2005
 EP 2169946 A2 3/2010
 EP 2346079 A1 7/2011
 EP 2565843 A2 3/2013
 EP 2731326 A2 5/2014
 EP 2879375 A1 6/2015
 EP 3105713 A1 12/2016
 GB 2486878 A 7/2012
 GB 2487943 A 8/2012
 JP H09-200617 A 7/1997
 JP 2000278532 A 10/2000
 JP 2000308068 A 11/2000
 JP 2001245213 A 9/2001
 JP 2002112008 A 4/2002
 JP 2003101886 A 4/2003
 JP 2003299067 A 10/2003
 JP 2004247983 A 9/2004
 JP 2004248061 A 9/2004
 JP 2004326119 A 11/2004
 JP 2004328532 A 11/2004
 JP 2006080752 A 3/2006
 JP 2006121612 A 5/2006
 JP 2006311311 A 11/2006
 JP 2007035028 A 2/2007
 JP 2007228099 A 9/2007
 JP 2008162498 A 7/2008
 JP 2008177738 A 7/2008
 JP 2008187615 A 8/2008
 JP 2008236726 A 10/2008
 JP 2009267923 A 11/2009
 JP 2009303010 A 12/2009

JP 2010016416 A 1/2010
 JP 2010512049 A 4/2010
 JP 2010136224 A 6/2010
 JP 2010157925 A 7/2010
 JP 2010166281 A 7/2010
 JP 2010239317 A 10/2010
 JP 4649623 B2 3/2011
 JP 2011097141 A 5/2011
 JP 2011101180 A 5/2011
 JP 2011120087 A 6/2011
 JP 2011120094 A 6/2011
 JP 2011146957 A 7/2011
 JP 2012080196 A 4/2012
 JP 2012119840 A 6/2012
 JP 2012156885 A 8/2012
 JP 2012195660 A 10/2012
 JP 2012213137 A 11/2012
 JP 2013026734 A 2/2013
 JP 2013055610 A 3/2013
 JP 2013066142 A 4/2013
 JP 2013093875 A 5/2013
 JP 2013120254 A 6/2013
 JP 2013207327 A 10/2013
 JP 2013219708 A 10/2013
 JP 2013258444 A 12/2013
 JP 2013258510 A 12/2013
 JP 2014057256 A 3/2014
 JP 2014140246 A 7/2014
 JP 2014140247 A 7/2014
 JP 2014142836 A 8/2014
 JP 2014155033 A 8/2014
 JP 2016019196 A 2/2016
 JP 2016066015 A 4/2016
 JP 6333095 B2 5/2018
 KR 20100094200 A 8/2010
 KR 20150130186 A 11/2015
 KR 20160127606 A 11/2016
 TW 201630406 A 8/2016
 WO 9746001 A1 12/1997
 WO 0237830 A2 5/2002
 WO 2004064391 A1 7/2004
 WO 2008010559 A1 1/2008
 WO 2009074938 A2 6/2009
 WO 2009074938 A3 8/2009
 WO 2013184256 A1 12/2013
 WO 2014094199 A1 6/2014
 WO 2014172059 A2 10/2014
 WO 2015120873 A1 8/2015
 WO 2015123455 A1 8/2015
 WO 2015173565 A1 11/2015
 WO 2016073643 A1 5/2016
 WO 2018044314 A1 3/2018
 WO 2021003261 A1 1/2021

OTHER PUBLICATIONS

Lim et al., U.S. Appl. No. 15/663,005, filed Jul. 28, 2017.
 First Action Interview Pilot Program Pre-Interview Communication from U.S. Appl. No. 15/663,005, dated May 28, 2020.
 Notice of Allowance from U.S. Appl. No. 15/663,005, dated Oct. 29, 2020.
 Sogaard et al., "Video Quality Assessment and Machine Learning: Performance and Interpretability," In Proceedings of IEEE QoMEX, 2015, 7 pages.
 Aghazadeh et al., "Novelty Detection from an Ego-Centric Perspective," IEEE Conference on Computer Vision and Pattern Recognition, 2011, pp. 3297-3304.
 Bouguet, J., "Pyramidal Implementation of the Affine Lucas Kanade Feature Tracker Description of the algorithm," Intel Corporation, 2001, 10 pages, retrieved from [http://robots.stanford.edu/cs223b04/ algo_affine_tracking.pdf](http://robots.stanford.edu/cs223b04/algo_affine_tracking.pdf).
 Doherty et al., "Investigating Keyframe Selection Methods in the Novel Domain of Passively Captured Visual Lifelogs," ACM International Conference on Image and Video Retrieval, Jul. 2008, 10 pages.
 Gokalp et al., "Scene Classification Using Bag-of-Regions Representations," IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Jun. 2007, 9 pages.

(56)

References Cited**OTHER PUBLICATIONS**

- Gygli et al., "Creating Summaries from User Videos," European Conference on Computer Vision, 2014, pp. 505-520.
- Horn et al., "Determining Optical Flow," *Artificial Intelligence*, Aug. 1981, pp. 185-203.
- Jain et al., "Displacement Measurement and Its Application in Interframe Image Coding," *IEEE Transactions on Communications*, vol. COM-29, No. 12, Dec. 1981, pp. 1799-1808.
- Khosla et al., "Large-Scale Video Summarization using Web-Image Priors," *IEEE Conference on Computer Vision and Pattern Recognition*, 2013, 8 pages, retrieved from https://people.csail.mit.edu/khosla/papers/cvpr2013_khosla.pdf.
- Lee et al., "Discovering Important People and Objects for Egocentric Video Summarization," *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, 8 pages, retrieved from https://vision.cs.utexas.edu/projects/egocentric/egocentric_cvpr2012.pdf.
- Ling et al., "Method for Fast Shot Boundary Detection Based on SVM," *Congress on Image and Signal Processing*, 2008, pp. 445-449.
- Liu et al., "Brief and High-Interest Video Summary Generation: Evaluating the AT&T Labs Rushes Summarizations," *Proceedings of the 2nd ACM TRECVID Video Summarization Workshop*, Oct. 2008, pp. 21-25.
- Lu et al., "Fast Video Shot Boundary Detection Based on SVD and Pattern Matching," *IEEE Transactions on Image Processing*, vol. 22, Dec. 2013, pp. 5136-5145.
- Lu et al., "Story-Driven Summarization for Egocentric Video," *IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 2714-2721.
- Lucas et al., "An Iterative Image Registration Technique with an Application to Stereo Vision," *Proceedings of DARPA Image Understanding Workshop*, Apr. 1981, pp. 121-130.
- Lupatini et al., "Scene Break Detection: A Comparison," *Proceedings Eighth International Workshop on Research Issues in Data Engineering*, Feb. 1998, 8 pages, retrieved from <https://ieeexplore.ieee.org/document/658276>.
- Ma et al., "A Generic Framework of User Attention Model and its Application in Video Summarization," *IEEE Transactions on Multimedia*, vol. 7, Oct. 2005, pp. 907-919.
- Marat et al., "Video Summarization using a Visual Attention Mode," *European Signal Processing Conference*, Sep. 2007, pp. 1784-1788.
- Marpe et al., "The H.264/MPEG4 Advanced Video Coding Standard and its Applications," *IEEE Communications Magazine*, Aug. 2006, pp. 134-143.
- Mertens et al., "Exposure Fusion," *Pacific Conference on Computer Graphics and Applications (PG '07)*, 2007, 9 pages, retrieved from https://ericam.github.io/exposure_fusion/index.html.
- Ngo et al., "On Clustering and Retrieval of Video Shots through Temporal Slices Analysis," *IEEE Transactions on Multimedia*, vol. 4, No. 4, Dec. 2002, pp. 446-458.
- Sheikh et al., "Blind Quality Assessment of JPEG2000 Compressed Images," *IEEE Asilomar Conference on Signals, Systems and Computers*, Nov. 2002, 4 pages, retrieved from <https://ece.uwaterloo.ca/~z70wang/publications/asilomar02.html>.
- Swain et al., "Indexing via Color Histograms," *Proceedings Third International Conference on Computer Vision*, Dec. 1990, pp. 390-393.
- Wan et al., "A New Approach to Image Retrieval with Hierarchical Color Clustering," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 8, No. 5, Sep. 1998, pp. 628-643.
- Wang, Z., "Objective Image Quality Assessment: Facing the Real-World Challenges," *Electronic Imaging, Image Quality and System Performance*, Feb. 2016, 6 pages.
- Wang et al., "Reduced and No Reference Visual Quality Assessment," *IEEE Signal Processing Magazine, Special Issue on Multimedia Quality Assessment*, vol. 29, Nov. 2011, pp. 29-40.
- Wolf, W., "Key Frame Selection by Motion Analysis," *IEEE International Conference on Acoustics, Speech, and Signal Processing Conference*, May 1996, pp. 1228-1231.
- Zhu et al., "A New Diamond Search Algorithm for Fast Block-Matching Motion Estimation," *IEEE Transactions on Image Processing*, vol. 9, No. 2, Feb. 2000, pp. 287-290.
- T-Mobile, "T-Mobile Sets Your Music Free," T-Mobile Webpage, Newsroom, Jun. 19, 2014, 5 pages retrieved from <https://www.t-mobile.com/news/press/t-mobile-sets-your-music-free>.
- T-Mobile, "T-Mobile Brings iPhone 6 & iPhone 6 Plus to its Data Strong Network," T-Mobile Webpage, Newsroom, Sep. 8, 2014, 6 pages, retrieved from <https://www.t-mobile.com/news/press/apple-devices>.
- Gatt et al., U.S. Appl. No. 15/978,122, filed May 12, 2018.
- Non-Final Office Action from U.S. Appl. No. 15/978,122, dated Sep. 16, 2019.
- Final Office Action from U.S. Appl. No. 15/978,122, dated Mar. 2, 2020.
- Notice of Allowance from U.S. Appl. No. 15/978,122, dated Aug. 18, 2020.
- Easy Flex, Two Examples of Layout Animations, Apr. 11, 2010, pp. 1-11, <http://evtimmy.com/2010/04/two-examples-of-layout-animations/>.
- Tidwell, J., "Animated Transition," from *Designing Interfaces*, O'Reilly Media, Inc., Dec. 2010, pp. 127-129.
- International Search Report and Written Opinion from PCT Application No. PCT/US17/57704, dated Nov. 16, 2017.
- Extended European Search Report from European Application No. 17863122.2, dated Dec. 2, 2019.
- European Search Report from European Application No. 17863122.2, dated Dec. 10, 2020.
- Examination Report from Indian Application No. 201927013698, dated Jun. 9, 2021.
- Examination Report from European Application No. 17863122.2, dated Aug. 4, 2021.
- XP055828762, Google display tablet "iPad mini 2", device pictures, 2014, 5 pages.
- Office Action from Chinese Patent Application No. 201780064462.1, dated Nov. 2, 2022.
- Second Office Action from Chinese Patent Application No. 201780064462.1, dated Jun. 14, 2023.
- International Search Report and Written Opinion from PCT Application No. PCT/US2018/066293, dated Mar. 15, 2019.
- Yoon et al., "Image Contrast Enhancement based Sub-histogram Equalization Technique without Over-equalization Noise," *World Academy of Science, Engineering and Technology*, vol. 50, 2009, 7 pages.
- Notice of Allowance from U.S. Appl. No. 15/687,278, dated Aug. 24, 2018.
- Rivard et al., U.S. Appl. No. 15/836,655, filed Dec. 8, 2017.
- Non-Final Office Action from U.S. Appl. No. 15/836,655, dated Apr. 6, 2018.
- Notice of Allowance from U.S. Appl. No. 15/836,655, dated Apr. 30, 2018.
- Corrected Notice of Allowance from U.S. Appl. No. 15/836,655, dated May 14, 2018.
- Rivard et al., U.S. Appl. No. 15/885,296, filed Jan. 31, 2018.
- Non-Final Office Action for U.S. Appl. No. 15/885,296, dated Jun. 4, 2018.
- Notice of Allowance for U.S. Appl. No. 15/885,296 dated Sep. 21, 2018.
- Corrected Notice of Allowance for U.S. Appl. No. 15/885,296 dated Oct. 16, 2018.
- Rivard et al., U.S. Appl. No. 16/154,999, filed Oct. 9, 2018.
- Non-Final Office Action for U.S. Appl. No. 16/154,999, dated Dec. 20, 2018.
- Notice of Allowance from U.S. Appl. No. 16/154,999, dated Jun. 7, 2019.
- Rivard et al., U.S. Appl. No. 16/505,278, filed Jul. 8, 2019.
- Non-Final Office Action for U.S. Appl. No. 16/505,278, dated Jan. 10, 2020.
- Notice of Allowance from U.S. Appl. No. 16/505,278, dated Sep. 25, 2020.
- Corrected Notice of Allowance from U.S. Appl. No. 16/505,278, dated Oct. 22, 2020.

(56)

References Cited**OTHER PUBLICATIONS**

Corrected Notice of Allowance from U.S. Appl. No. 16/505,278, dated Nov. 18, 2020.
 Corrected Notice of Allowance from U.S. Appl. No. 16/505,278, dated Dec. 24, 2020.
 Rivard et al., U.S. Appl. No. 17/144,915, filed Jan. 8, 2021.
 Non-Final Office Action for U.S. Appl. No. 17/144,915, dated Aug. 13, 2021.
 Notice of Allowance from U.S. Appl. No. 17/144,915, dated Feb. 10, 2022.
 Rivard et al., U.S. Appl. No. 17/749,919, filed May 20, 2022.
 Non-Final Office Action for U.S. Appl. No. 17/749,919, dated Feb. 16, 2023.
 Final Office Action for U.S. Appl. No. 17/749,919, dated Sep. 11, 2023.
 Rivard et al., U.S. Appl. No. 14/823,993, filed Aug. 11, 2015.
 Non-Final Office Action from U.S. Appl. No. 14/823,993, dated Jul. 28, 2016.
 Final Office Action from U.S. Appl. No. 14/823,993, dated Feb. 10, 2017.
 Non-Final Office Action from U.S. Appl. No. 14/823,993, dated Jul. 14, 2017.
 Notice of Allowance from U.S. Appl. No. 14/823,993, dated Oct. 31, 2017.
 Corrected Notice of Allowance from U.S. Appl. No. 14/823,993, dated Dec. 27, 2017.
 Corrected Notice of Allowance from U.S. Appl. No. 14/823,993, dated Jan. 4, 2018.
 Corrected Notice of Allowance from U.S. Appl. No. 14/823,993, dated Feb. 8, 2018.
 Rivard et al., U.S. Appl. No. 15/891,251, filed Feb. 7, 2018.
 Non-Final Office Action for U.S. Appl. No. 15/891,251, dated May 31, 2018.
 Final Office Action for U.S. Appl. No. 15/891,251, dated Nov. 29, 2018.
 Notice of Allowance from U.S. Appl. No. 15/891,251, dated May 7, 2019.
 Corrected Notice of Allowance from U.S. Appl. No. 15/891,251, dated Jul. 3, 2019.
 Non-Final Office Action for U.S. Appl. No. 16/519,244, dated Sep. 23, 2019.
 Notice of Allowance from U.S. Appl. No. 16/519,244, dated Jan. 14, 2020.
 Corrected Notice of Allowance from U.S. Appl. No. 16/519,244, dated Feb. 20, 2020.
 Corrected Notice of Allowance from U.S. Appl. No. 16/519,244, dated Apr. 9, 2020.
 Rivard et al., U.S. Appl. No. 16/857,016, filed Apr. 23, 2020.
 Non-Final Office Action for U.S. Appl. No. 16/857,016, dated Aug. 5, 2020.
 Notice of Allowance from U.S. Appl. No. 16/857,016, dated Jan. 27, 2021.
 Corrected Notice of Allowance from U.S. Appl. No. 16/857,016, dated Feb. 16, 2021.
 Corrected Notice of Allowance from U.S. Appl. No. 16/857,016, dated Apr. 13, 2021.
 Rivard et al., U.S. Appl. No. 17/321,166, filed May 14, 2021.
 Non-Final Office Action for U.S. Appl. No. 17/321,166, dated Apr. 25, 2022.
 Final Office Action for U.S. Appl. No. 17/321,166, dated Dec. 9, 2022.
 Notice of Allowance from U.S. Appl. No. 17/321,166, dated Jun. 5, 2023.
 Corrected Notice of Allowance from U.S. Appl. No. 16/271,604, dated Sep. 19, 2019.
 Non-Final Office Action for U.S. Appl. No. 16/662,965, dated Mar. 22, 2021.
 Final Office Action for U.S. Appl. No. 16/662,965, dated Sep. 3, 2021.

Notice of Allowance from U.S. Appl. No. 16/662,965, dated Mar. 1, 2022.
 Rivard et al., U.S. Appl. No. 17/835,823, filed Jun. 8, 2022.
 Non-Final Office Action for U.S. Appl. No. 17/835,823, dated Apr. 18, 2023.
 Final Office Action for U.S. Appl. No. 17/835,823, dated Nov. 14, 2023.
 Rivard et al., U.S. Appl. No. 15/254,964, filed Apr. 3, 2019.
 Non-Final Office Action from U.S. Appl. No. 15/254,964, dated Jan. 3, 2018.
 Final Office Action for U.S. Appl. No. 15/254,964, dated Jul. 24, 2018.
 Notice of Allowance for U.S. Appl. No. 15/254,964, dated Dec. 21, 2018.
 Supplemental Notice of Allowance for U.S. Appl. No. 15/254,964, dated Feb. 1, 2019.
 Supplemental Notice of Allowance for U.S. Appl. No. 15/254,964, dated Mar. 11, 2019.
 Rivard et al., U.S. Appl. No. 15/643,311, filed Jul. 6, 2017.
 Non-Final Office Action from U.S. Appl. No. 15/643,311, dated Jan. 4, 2018.
 Final Office Action for U.S. Appl. No. 15/643,311, dated Jul. 24, 2018.
 Notice of Allowance for U.S. Appl. No. 15/643,311, dated Oct. 31, 2018.
 Supplemental Notice of Allowance for U.S. Appl. No. 15/643,311, dated Dec. 11, 2018.
 Rivard et al., U.S. Appl. No. 16/213,041, filed Dec. 7, 2018.
 Non-Final Office Action for U.S. Appl. No. 16/213,041, dated Oct. 30, 2019.
 Notice of Allowance from U.S. Appl. No. 16/213,041, dated May 29, 2020.
 Supplemental Notice of Allowance from U.S. Appl. No. 16/213,041, dated Jun. 17, 2020.
 Supplemental Notice of Allowance from U.S. Appl. No. 16/213,041, dated Aug. 31, 2020.
 Rivard et al., U.S. Appl. No. 17/000,098, filed Aug. 21, 2020.
 Non-Final Office Action for U.S. Appl. No. 17/000,098, dated Dec. 7, 2021.
 Final Office Action for U.S. Appl. No. 17/000,098, dated Aug. 25, 2022.
 Non-Final Office Action for U.S. Appl. No. 17/000,098, dated Apr. 11, 2023.
 Notice of Allowance for U.S. Appl. No. 17/000,098, dated Jan. 25, 2024.
 Corrected Notice of Allowance for U.S. Appl. No. 17/000,098, dated May 1, 2024.
 Rivard et al., U.S. Appl. No. 18/646,620, filed Apr. 25, 2024.
 Rivard et al., U.S. Appl. No. 15/331,733, filed Oct. 21, 2016.
 Notice of Allowance from U.S. Appl. No. 15/331,733, dated Dec. 7, 2016.
 Notice of Allowance from U.S. Appl. No. 15/331,733, dated Apr. 17, 2017.
 Rivard et al., U.S. Appl. No. 15/452,639, filed Apr. 4, 2018.
 Non-Final Office Action from U.S. Appl. No. 15/452,639, dated May 11, 2017.
 Notice of Allowance from U.S. Appl. No. 15/452,639, dated Nov. 30, 2017.
 Rivard et al., U.S. Appl. No. 15/642,074, filed Jul. 5, 2017.
 Non-Final Office Action from U.S. Appl. No. 15/642,074, dated Oct. 19, 2018.
 Notice of Allowance from U.S. Appl. No. 15/642,074, dated Apr. 10, 2019.
 Rivard et al., U.S. Appl. No. 16/518,786, filed Jul. 22, 2019.
 Non-Final Office Action from U.S. Appl. No. 16/518,786, dated Jun. 29, 2020.
 Notice of Allowance from U.S. Appl. No. 16/518,786, dated Nov. 2, 2020.
 Corrected Notice of Allowance from U.S. Appl. No. 16/518,786, dated Nov. 20, 2020.
 Rivard et al., U.S. Appl. No. 17/180,526, filed Feb. 19, 2021.
 Non-Final Office Action from U.S. Appl. No. 17/180,526, dated Jul. 25, 2022.

(56)

References Cited**OTHER PUBLICATIONS**

Final Office Action from U.S. Appl. No. 17/180,526, dated Jan. 23, 2023.

Non-Final Office Action from U.S. Appl. No. 17/180,526, dated Sep. 15, 2023.

Non-Final Office Action from U.S. Appl. No. 17/824,773, dated Feb. 16, 2023.

Final Office Action from U.S. Appl. No. 17/824,773, dated Sep. 27, 2023.

Rivard et al., U.S. Appl. No. 15/976,756, filed May 10, 2018.

Final Office Action from U.S. Appl. No. 17/874,086, dated Aug. 21, 2023.

Feder et al., U.S. Appl. No. 16/395,792, filed Apr. 26, 2019.

Non-Final Office Action from U.S. Appl. No. 16/395,792, dated Aug. 26, 2020.

Final Office Action from U.S. Appl. No. 16/395,792, dated Feb. 24, 2021.

Non-Final Office Action from U.S. Appl. No. 16/395,792, dated Nov. 9, 2021.

Final Office Action from U.S. Appl. No. 16/395,792, dated Apr. 26, 2022.

Feder et al., U.S. Appl. No. 17/953,238, filed Sep. 26, 2022.

Non-Final Office Action from U.S. Appl. No. 17/953,238, dated Jul. 5, 2023.

Rivard et al., U.S. Appl. No. 17/063,487, filed Oct. 5, 2020.

Non-Final Office Action from U.S. Appl. No. 17/063,487, dated Oct. 14, 2022.

Final Office Action from U.S. Appl. No. 17/063,487, dated May 19, 2023.

Rivard et al., U.S. Appl. No. 16/584,486, filed Sep. 26, 2019.

Notice of Allowance from U.S. Appl. No. 16/584,486, dated Oct. 21, 2020.

Corrected Notice of Allowance from U.S. Appl. No. 16/584,486, dated Nov. 18, 2020.

Corrected Notice of Allowance from U.S. Appl. No. 16/584,486, dated Dec. 24, 2020.

Rivard et al., U.S. Appl. No. 17/163,086, filed Jan. 29, 2021.

Non-Final Office Action for U.S. Appl. No. 17/163,086, dated Oct. 13, 2021.

Notice of Allowance from U.S. Appl. No. 17/163,086, dated Mar. 21, 2022.

Rivard et al., U.S. Appl. No. 17/857,906, filed Jul. 5, 2022.

Non-Final Office Action for U.S. Appl. No. 17/857,906, dated Feb. 9, 2023.

Final Office Action for U.S. Appl. No. 17/857,906, dated Sep. 11, 2023.

Rivard et al., U.S. Appl. No. 16/931,286, filed Jul. 16, 2020.

Non-Final Office Action for U.S. Appl. No. 16/931,286, dated May 11, 2021.

Final Office Action for U.S. Appl. No. 16/931,286, dated Dec. 29, 2021.

Notice of Allowance from U.S. Appl. No. 16/931,286, dated Jun. 7, 2022.

Rivard et al., U.S. Appl. No. 17/945,939, filed Sep. 15, 2022.

Non-Final Office Action for U.S. Appl. No. 17/945,939, dated Mar. 20, 2023.

Final Office Action for U.S. Appl. No. 17/945,939, dated Oct. 27, 2023.

Feder et al., U.S. Appl. No. 17/694,472, filed Mar. 14, 2022.

Non-Final Office Action from U.S. Appl. No. 17/694,472, dated May 29, 2024.

International Search Report and Written Opinion from International Application No. PCT/US15/59348, dated Feb. 2, 2016.

International Search Report and Written Opinion from International Application No. PCT/US15/59097, dated Jan. 4, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2015/060476, dated Feb. 10, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2015/058895, dated Apr. 11, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2015/059103, dated Dec. 21, 2015.

International Search Report and Written Opinion from International Application No. PCT/US2015/059105, dated Jul. 26, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2015/058896, dated Aug. 26, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2015/058891, dated Aug. 26, 2016.

International Search Report and Written Opinion from International Application No. PCT/US2016/050011, dated Nov. 10, 2016.

Wan et al., "CMOS Image Sensors With Multi-Bucket Pixels for Computational Photography," *IEEE Journal of Solid-State Circuits*, vol. 47, No. 4, Apr. 2012, pp. 1031-1042.

Chatterjee et al., "Clustering-Based Denoising With Locally Learned Dictionaries," *IEEE Transactions on Image Processing*, vol. 18, No. 7, Jul. 2009, pp. 1-14.

Burger et al., "Image denoising: Can plain Neural Networks compete with BM3D?," *Computer Vision and Pattern Recognition (CVPR)*, IEEE, 2012, pp. 4321-4328.

Kervann et al., "Optimal Spatial Adaptation for Patch-Based Image Denoising," *IEEE Transactions on Image Processing*, vol. 15, No. 10, Oct. 2006, pp. 2866-2878.

Foi et al., "Practical Poissonian-Gaussian noise modeling and fitting for single-image raw-data," *IEEE Transactions*, 2007, pp. 1-18.

International Search Report and Written Opinion from PCT Application No. PCT/US17/39946, dated Sep. 25, 2017.

European Office Communication and Exam Report from European Application No. 15856814.7, dated Dec. 14, 2017.

European Office Communication and Exam Report from European Application No. 15856267.8, dated Dec. 12, 2017.

European Office Communication and Exam Report from European Application No. 15856710.7, dated Dec. 21, 2017.

European Office Communication and Exam Report from European Application No. 15857675.1, dated Dec. 21, 2017.

European Office Communication and Exam Report from European Application No. 15856212.4, dated Dec. 15, 2017.

Notice of Allowance from U.S. Appl. No. 14/547,074, dated Aug. 19, 2016.

Supplemental Notice of Allowance from U.S. Appl. No. 14/547,074, dated Oct. 31, 2016.

Rivard et al., U.S. Appl. No. 14/547,077, filed Nov. 18, 2014.

Non-Final Office Action from U.S. Appl. No. 14/547,077, dated Dec. 9, 2016.

Final Office Action from U.S. Appl. No. 14/547,077, dated May 12, 2017.

Non-Final Office Action from U.S. Appl. No. 14/547,077, dated Dec. 28, 2017.

Notice of Allowance from U.S. Appl. No. 14/547,077, dated Jun. 1, 2018.

Rivard et al., U.S. Appl. No. 16/116,715, filed Aug. 29, 2018.

Non-Final Office Action from U.S. Appl. No. 16/116,715, dated Feb. 18, 2020.

Final Office Action from U.S. Appl. No. 16/116,715, dated Sep. 28, 2020.

Non-Final Office Action from U.S. Appl. No. 16/116,715, dated Jul. 12, 2021.

Final Office Action from U.S. Appl. No. 16/116,715, dated Mar. 7, 2022.

Rivard et al., U.S. Application No. 17/874, 102, filed Jul. 26, 2022.

Non-Final Office Action from U.S. Appl. No. 17/874,102, dated Jan. 23, 2023.

Final Office Action from U.S. Appl. No. 17/874,102, dated Jul. 20, 2023.

Rivard et al., U.S. Appl. No. 14/547,079, filed Nov. 18, 2014.

Non-Final Office Action from U.S. Appl. No. 14/547,079, dated Nov. 7, 2016.

Non-Final Office Action from U.S. Appl. No. 14/547,079, dated Mar. 15, 2017.

Final Office Action from U.S. Appl. No. 14/547,079, dated Jul. 17, 2017.

Non-Final Office Action from U.S. Appl. No. 14/547,079, dated Nov. 9, 2017.

(56)

References Cited**OTHER PUBLICATIONS**

Notice of Allowance from U.S. Appl. No. 14/547,079, dated May 3, 2018.

Rivard et al., U.S. Appl. No. 15/975,646, filed May 9, 2018.

Non-Final Office Action from U.S. Appl. No. 15/975,646, dated Sep. 27, 2018.

Final Office Action from U.S. Appl. No. 15/975,646, dated Mar. 25, 2019.

Advisory Action from U.S. Appl. No. 15/975,646, dated Apr. 23, 2019.

Notice of Allowance from U.S. Appl. No. 15/975,646, dated Aug. 9, 2019.

Rivard et al., U.S. Appl. No. 16/666,215, filed Oct. 28, 2019.

Non-Final Office Action from U.S. Appl. No. 16/666,215, dated Apr. 2, 2021.

Notice of Allowance from U.S. Appl. No. 16/666,215, dated Oct. 8, 2021.

Rivard et al., U.S. Appl. No. 17/569,400, filed Jan. 5, 2022.

Non-Final Office Action from U.S. Appl. No. 17/569,400, dated Jan. 11, 2023.

Final Office Action from U.S. Appl. No. 17/569,400, dated Jun. 26, 2023.

Non-Final Office Action from U.S. Appl. No. 17/569,400, dated Nov. 24, 2023.

Non-Final Office Action from U.S. Appl. No. 17/569,400, dated Mar. 13, 2024.

Notice of Allowance from U.S. Appl. No. 17/569,400, dated Jul. 31, 2024.

Rivard et al., U.S. Appl. No. 17/701,484, filed Mar. 22, 2022.

Non-Final Office Action from U.S. Appl. No. 17/701,484, dated Feb. 13, 2023.

Final Office Action from U.S. Appl. No. 17/701,484, dated Jul. 24, 2023.

Final Office Action from U.S. Appl. No. 17/701,484, dated Jul. 18, 2024.

Rivard et al., U.S. Appl. No. 14/702,549, filed May 1, 2015.

Non-Final Office Action from U.S. Appl. No. 14/702,549, dated Jan. 25, 2016.

Notice of Allowance from U.S. Appl. No. 14/702,549, dated Aug. 15, 2016.

Corrected Notice of Allowance from U.S. Appl. No. 14/702,549, dated Nov. 10, 2016.

Corrected Notice of Allowance from U.S. Appl. No. 14/702,549, dated Dec. 1, 2016.

Rivard et al., U.S. Appl. No. 15/352,510, filed Nov. 15, 2016.

Notice of Allowance from U.S. Appl. No. 15/352,510, dated Oct. 17, 2017.

Corrected Notice of Allowance from U.S. Appl. No. 15/352,510, dated Jan. 8, 2018.

Corrected Notice of Allowance from U.S. Appl. No. 15/352,510, dated Jan. 29, 2018.

Rivard et al., U.S. Appl. No. 15/687,278, filed Aug. 25, 2017.

Non-Final Office Action for U.S. Appl. No. 15/687,278, dated Apr. 13, 2018.

Kadir et al., "Non-parametric Estimation of Probability Distributions from Sampled Signals," Robotics Research Laboratory, Department of Engineering Science, University of Oxford, Jul. 4, 2005, pp. 1-22.

International Preliminary Examination Report from PCT Application No. PCT/US2018/066293, dated Jul. 2, 2020.

Examination Report from Indian Application No. 202027030380, dated May 30, 2022.

Sherrier et al., "Regionally Adaptive Histogram Equalization of the Chest," IEEE Transactions on Medical Imaging, vol. MI-6, No. 1, Mar. 1987, pp. 1-7.

International Search Report and Written Opinion from PCT Application No. PCT/US 19/13847, dated Apr. 12, 2019.

International Preliminary Examination Report from PCT Application No. PCT/US2019/13847, dated Jul. 30, 2020.

Office Action from Indian Patent Application No. 202027034317, dated Apr. 21, 2022.

Non-Final Office Action from U.S. Appl. No. 16/507,862, dated May 20, 2020.

Final Office Action from U.S. Appl. No. 16/507,862, dated Oct. 13, 2020.

Advisory Action from U.S. Appl. No. 16/507,862, dated Dec. 21, 2020.

Non-Final Office Action from U.S. Appl. No. 16/507,862, dated Feb. 5, 2021.

Notice of Allowance from U.S. Appl. No. 16/507,862, dated May 19, 2021.

Extended European Search Report from European Application No. 20835135.3, dated Jun. 14, 2023.

Canon, "Canon EOS 7D Mark II," Canon User Manual, 2014, 548 pages.

Canon, "Canon EOS 1200D," Canon User Manual, 2014, 342 pages.

Canon, "Canon PowerShot G1 X Mark II," Canon User Manual, 2014, 240 pages.

Canon, "Canon Powershot G7 X," Canon User Manual, 2014, 202 pages.

Canon, "Canon PowerShot S200," Canon User Manual, 2013, 183 pages.

Canon, "Canon PowerShot SX60 HS," Canon User Manual, 2014, 203 pages.

Canon, "Canon PowerShot SX520 HS," Canon User Manual, 2014, 139 pages.

Micron, "1/3.2-Inch 2-Megapixel SOC Digital Image Sensor Features," Micron Technology, MT9D111, 2012, 186 pages.

Google, "Google Search timeline Page," Google, 2018, 1 page.

Examination Report from Indian Application No. 202227003071, dated Jul. 1, 2022.

Examination Report from Canadian Application No. 3,144,478, dated Jan. 28, 2023.

Notice of Reasons for Refusal from Japanese Application No. 2021-576343, dated Jan. 31, 2023.

Ingle et al., "Samsung Proposes SEFET Sensor, Pixart Flips Color Filter and Microlens," Image Sensors World, Dec. 30, 2010, 7 pages.

Rivard et al., U.S. Appl. No. 17/696,717, filed Mar. 16, 2022.

Notice of Allowance from U.S. Appl. No. 17/696,717, dated Dec. 5, 2022.

Rivard et al., U.S. Appl. No. 18/121,579, filed Mar. 14, 2023.

Rivard et al., U.S. Appl. No. 18/646,581, filed Apr. 25, 2024.

Rivard et al., U.S. Appl. No. 18/930,881, filed Oct. 29, 2024.

Non-Final Office Action for U.S. Appl. No. 18/930,881, dated Dec. 23, 2024.

Notice of Allowance from U.S. Appl. No. 18/930,881, dated Feb. 4, 2025.

Rivard et al., U.S. Appl. No. 19/025,744, filed Jan. 16, 2025.

Rivard et al., U.S. Appl. No. 18/930,887, filed Oct. 29, 2024.

Notice of Allowance from U.S. Appl. No. 18/930,887, dated Dec. 20, 2024.

Rivard et al., U.S. Appl. No. 19/025,772, filed Jan. 16, 2025.

Rivard et al., U.S. Appl. No. 19/025,969, filed Jan. 16, 2025.

Rivard et al., U.S. Appl. No. 18/932,436, filed Oct. 30, 2024.

Non-Final Office Action for U.S. Appl. No. 18/932,436, dated Dec. 27, 2024.

Rivard et al., U.S. Appl. No. 19/025,856, filed Jan. 16, 2025.

Rivard et al., U.S. Appl. No. 18/930,891, filed Oct. 29, 2024.

Non-Final Office Action from U.S. Appl. No. 18/930,891, filed Jan. 30, 2025.

Rivard et al., U.S. Appl. No. 19/025,870, filed Jan. 16, 2025.

Feder et al., U.S. Appl. No. 18/957,506, filed Nov. 22, 2024.

Non-Final Office Action from U.S. Appl. No. 18/957,506, dated Jan. 1, 2025.

Feder et al., U.S. Appl. No. 19/025,885, filed Jan. 16, 2025.

Rivard et al., U.S. Appl. No. 16/244,982, filed Jan. 10, 2019.

Non-Final Office Action from U.S. Appl. No. 16/244,982, dated Apr. 1, 2020.

Final Office Action from U.S. Appl. No. 16/244,982, dated Nov. 27, 2020.

(56)

References Cited

OTHER PUBLICATIONS

Non-Final Office Action for U.S. Appl. No. 15/976,756, dated Jun. 27, 2019.

Notice of Allowance from U.S. Appl. No. 15/976,756, dated Oct. 4, 2019.

Rivard et al., U.S. Appl. No. 16/215,351, filed Dec. 10, 2018.

Non-Final Office Action from U.S. Appl. No. 16/215,351, dated Jan. 24, 2019.

Notice of Allowance from U.S. Appl. No. 16/215,351, dated Apr. 1, 2019.

Rivard et al., U.S. Appl. No. 16/290,763, filed Mar. 1, 2019.

Non-Final Office Action for U.S. Appl. No. 16/290,763, dated Jun. 26, 2019.

Notice of Allowance from U.S. Appl. No. 16/290,763, dated Oct. 10, 2019.

Rivard et al., U.S. Appl. No. 16/796,497, filed Feb. 20, 2020.

Non-Final Office Action for U.S. Appl. No. 16/796,497, dated Dec. 8, 2021.

Notice of Allowance from U.S. Appl. No. 16/796,497, dated May 26, 2022.

Rivard et al., U.S. Appl. No. 17/694,458, filed Mar. 14, 2022.

Notice of Allowance from U.S. Appl. No. 17/694,458, dated Mar. 1, 2023.

Rivard et al., U.S. Appl. No. 17/745,668, filed May 16, 2022.

Non-Final Office Action for U.S. Appl. No. 17/745,668, dated Jun. 8, 2023.

Rivard et al., U.S. Appl. No. 16/221,289, filed Dec. 14, 2018.

Non-Final Office Action from U.S. Appl. No. 16/221,289, dated May 15, 2020.

Final Office Action from U.S. Appl. No. 16/221,289, dated Nov. 24, 2020.

Notice of Allowance from U.S. Appl. No. 16/221,289, dated May 12, 2021.

Supplemental Notice of Allowance from U.S. Appl. No. 16/221,289, dated Jul. 16, 2021.

Rivard et al., U.S. Appl. No. 16/552,649, filed Aug. 27, 2019.

Non-Final Office Action from U.S. Appl. No. 16/552,649, dated Jun. 18, 2021.

Notice of Allowance from U.S. Appl. No. 16/552,649, dated Dec. 17, 2021.

Rivard et al., U.S. Appl. No. 16/206,241, filed Nov. 30, 2018.

Non-Final Office Action from U.S. Appl. No. 16/206,241, dated Jun. 20, 2019.

Non-Final Office Action from U.S. Appl. No. 16/206,241, dated Sep. 26, 2019.

Notice of Allowance from U.S. Appl. No. 16/206,241, dated Mar. 5, 2020.

Rivard et al., U.S. Appl. No. 16/547,358, filed Aug. 21, 2019.

Non-Final Office Action from U.S. Appl. No. 16/547,358, dated Mar. 23, 2020.

Notice of Allowance from U.S. Appl. No. 16/547,358, dated Aug. 31, 2020.

Rivard et al., U.S. Appl. No. 17/108,867, filed Dec. 1, 2020.

Non-Final Office Action from U.S. Appl. No. 17/108,867, dated Aug. 20, 2021.

Final Office Action from U.S. Appl. No. 17/108,867, dated Jan. 28, 2022.

Advisory Action from U.S. Appl. No. 17/108,867, dated Jul. 18, 2022.

Rivard et al., U.S. Appl. No. 17/543,519, filed Dec. 6, 2021.

Notice of Allowance from U.S. Appl. No. 17/543,519, dated Oct. 19, 2022.

Notice of Allowance from U.S. Appl. No. 17/543,519, dated Feb. 8, 2023.

Supplemental Notice of Allowance from U.S. Appl. No. 17/543,519, dated Apr. 26, 2023.

Rivard et al., U.S. Appl. No. 17/875,263, filed Jul. 27, 2022.

Non-Final Office Action from U.S. Appl. No. 17/875,263, dated Mar. 27, 2023.

Final Office Action from U.S. Appl. No. 17/875,263, dated Sep. 7, 2023.

Rivard et al., U.S. Appl. No. 17/969,593, filed Oct. 19, 2022.

Non-Final Office Action from U.S. Appl. No. 17/969,593, dated May 17, 2023.

Rivard et al., U.S. Appl. No. 16/460,807, filed Jul. 2, 2019.

Non-Final Office Action for U.S. Appl. No. 16/460,807, dated Aug. 20, 2020.

Final Office Action for U.S. Appl. No. 16/460,807, dated Mar. 1, 2021.

Non-Final Office Action from U.S. Appl. No. 16/460,807, dated Aug. 30, 2021.

Final Office Action for U.S. Appl. No. 16/460,807, dated Mar. 2, 2022.

Rivard et al., U.S. Appl. No. 17/874,086, filed Jul. 26, 2022.

Non-Final Office Action from U.S. Appl. No. 17/874,086, dated Feb. 2, 2023.

Rivard et al., U.S. Appl. No. 16/211,931, filed Dec. 6, 2018.

Non-Final Office Action from U.S. Appl. No. 16/211,931, dated Apr. 5, 2019.

Notice of Allowance from U.S. Appl. No. 16/211,931, dated Aug. 2, 2019.

Rivard et al., U.S. Appl. No. 16/677,385, filed Nov. 7, 2019.

Non-Final Office Action from U.S. Appl. No. 16/677,385, dated Jan. 8, 2021.

Final Office Action from U.S. Appl. No. 16/677,385, dated Sep. 17, 2021.

Notice of Allowance from U.S. Appl. No. 16/677,385, dated Mar. 22, 2022.

Corrected Notice of Allowance from U.S. Appl. No. 16/677,385, dated Apr. 6, 2022.

Rivard et al., U.S. Appl. No. 17/518,473, filed Nov. 3, 2021.

Non-Final Office Action from U.S. Appl. No. 17/518,473, dated Dec. 9, 2022.

Final Office Action from U.S. Appl. No. 17/518,473, dated Aug. 18, 2023.

Advisory Action from U.S. Appl. No. 17/518,473, dated Nov. 1, 2023.

Rivard et al., U.S. Appl. No. 17/694,441, filed Mar. 14, 2022.

Non-Final Office Action from U.S. Appl. No. 17/694,441, dated Jan. 23, 2023.

Final Office Action from U.S. Appl. No. 17/694,441, dated Oct. 2, 2023.

Rivard et al., U.S. Appl. No. 17/868,536, filed Jul. 19, 2022.

Notice of Allowance from U.S. Appl. No. 17/868,536, dated Sep. 29, 2023.

Rivard et al., U.S. Appl. No. 14/534,079, filed Nov. 5, 2014.

Non-Final Office Action from U.S. Appl. No. 14/534,079, dated Jan. 29, 2015.

Notice of Allowance from U.S. Appl. No. 14/534,079, dated May 11, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/534,079, dated Aug. 19, 2015.

Rivard et al., U.S. Appl. No. 14/534,089, filed Nov. 5, 2014.

Non-Final Office Action from U.S. Appl. No. 14/534,089, dated Feb. 25, 2015.

Notice of Allowance from U.S. Appl. No. 14/534,089, dated Jun. 23, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/534,089, dated Aug. 25, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/534,089, dated Sep. 23, 2015.

Rivard et al., U.S. Appl. No. 14/535,274, filed Nov. 6, 2014.

Non-Final Office Action from U.S. Appl. No. 14/535,274, dated Feb. 3, 2015.

Notice of Allowance from U.S. Appl. No. 14/535,274, dated May 26, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/535,274, dated Aug. 25, 2015.

Rivard et al., U.S. Appl. No. 14/535,279, filed Nov. 6, 2014.

Non-Final Office Action from U.S. Appl. No. 14/535,279, dated Feb. 5, 2015.

(56)

References Cited**OTHER PUBLICATIONS**

Notice of Allowance from U.S. Appl. No. 14/535,279, dated Aug. 31, 2015.

Rivard et al., U.S. Appl. No. 14/535,282, filed Nov. 6, 2014.

Non-Final Office Action from U.S. Appl. No. 14/535,282, dated Jan. 30, 2015.

Notice of Allowance from U.S. Appl. No. 14/535,282, dated Jun. 23, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/535,282, dated Sep. 18, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/535,282, dated Oct. 7, 2015.

Rivard et al., U.S. Appl. No. 14/536,524, filed Nov. 7, 2014.

Non-Final Office Action from U.S. Appl. No. 14/536,524, dated Mar. 3, 2015.

Notice of Allowance from U.S. Appl. No. 14/536,524, dated Jun. 29, 2015.

Supplemental Notice of Allowance from U.S. Appl. No. 14/536,524, dated Aug. 19, 2015.

Supplemental Notice of Allowance from U.S. Appl. No. 14/536,524, dated Sep. 18, 2015.

Feder et al., U.S. Appl. No. 14/535,285, filed Nov. 6, 2014.

Non-Final Office Action from U.S. Appl. No. 14/535,285, dated Feb. 3, 2015.

Notice of Allowance from U.S. Appl. No. 14/535,285, dated Jul. 31, 2015.

Notice of Allowance from U.S. Appl. No. 14/535,285, dated Oct. 15, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/535,285, dated Nov. 16, 2015.

Kindle et al., U.S. Appl. No. 14/547,074, filed Nov. 18, 2014.

Non-Final Office Action from U.S. Appl. No. 14/547,074, dated Feb. 26, 2016.

Supplemental Notice of Allowance from U.S. Appl. No. 15/354,935, dated Dec. 1, 2017.

Feder et al., U.S. Appl. No. 15/814,238, filed Nov. 15, 2017.

Non-Final Office Action from U.S. Appl. No. 15/814,238, dated Feb. 8, 2018.

Notice of Allowance for U.S. Appl. No. 15/814,238 dated Oct. 4, 2018.

Corrected Notice of Allowance for U.S. Appl. No. 15/814,238 dated Nov. 13, 2018.

Feder et al., U.S. Appl. No. 16/217,848, filed Dec. 12, 2018.

Notice of Allowance from U.S. Appl. No. 16/217,848, dated Jul. 31, 2019.

Corrected Notice of Allowance from U.S. Appl. No. 16/217,848, dated Sep. 24, 2019.

Corrected Notice of Allowance from U.S. Appl. No. 16/217,848, dated Oct. 31, 2019.

Feder et al., U.S. Appl. No. 16/684,389, filed Nov. 14, 2019.

Notice of Allowance from U.S. Appl. No. 16/684,389, dated Oct. 29, 2020.

Corrected Notice of Allowance from U.S. Appl. No. 16/684,389, dated Nov. 27, 2020.

Corrected Notice of Allowance from U.S. Appl. No. 16/684,389, dated Dec. 23, 2020.

Feder et al., U.S. Appl. No. 17/171,800, filed Feb. 9, 2021.

Non-Final Office Action for U.S. Appl. No. 17/171,800, dated Aug. 18, 2022.

Notice of Allowance from U.S. Appl. No. 17/171,800, dated May 26, 2023.

Non-Final Office Action for U.S. Appl. No. 17/171,800, dated Oct. 4, 2023.

Feder et al., U.S. Appl. No. 14/340,557, filed Jul. 24, 2014.

Non-Final Office Action from U.S. Appl. No. 14/340,557, dated Jan. 21, 2016.

Final Office Action from U.S. Appl. No. 14/340,557, dated Sep. 16, 2016.

Notice of Allowance from U.S. Appl. No. 14/340,557, dated Mar. 3, 2017.

Corrected Notice of Allowance from U.S. Appl. No. 14/340,557, dated Jul. 27, 2017.

Feder et al., U.S. Appl. No. 15/622,520, filed Jun. 14, 2017.

Non-Final Office Action from U.S. Appl. No. 15/622,520, dated Jan. 10, 2018.

Notice of Allowance from U.S. Appl. No. 15/622,520, dated Jul. 18, 2018.

Feder et al., U.S. Appl. No. 16/147,206, filed Sep. 28, 2018.

Non-Final Office Action from U.S. Appl. No. 16/147,206, dated Oct. 18, 2019.

Notice of Allowance from U.S. Appl. No. 16/147,206, dated Jun. 19, 2020.

Corrected Notice of Allowance from U.S. Appl. No. 16/147,206, dated Aug. 27, 2020.

Feder et al., U.S. Appl. No. 17/023,159, filed Sep. 16, 2020.

Non-Final Office Action from U.S. Appl. No. 17/023,159, dated Sep. 2, 2021.

Final Office Action from U.S. Appl. No. 17/023,159, dated Apr. 15, 2022.

Feder et al., U.S. Appl. No. 17/945,922, filed Sep. 15, 2022.

Non-Final Office Action from U.S. Appl. No. 17/945,922, dated Mar. 31, 2023.

Final Office Action from U.S. Appl. No. 17/945,922, dated Sep. 19, 2023.

Rivard et al., U.S. Appl. No. 15/201,283, filed Jul. 1, 2016.

Non-Final Office Action from U.S. Appl. No. 15/201,283, dated Dec. 7, 2016.

Notice of Allowance from U.S. Appl. No. 15/201,283, dated Mar. 23, 2017.

Notice of Allowance from U.S. Appl. No. 15/201,283, dated Jul. 19, 2017.

Rivard et al., U.S. Appl. No. 15/636,324, filed Jun. 28, 2017.

Non-Final Office Action for U.S. Appl. No. 15/636,324, dated Oct. 18, 2018.

Final Office Action for U.S. Appl. No. 15/636,324, dated Mar. 22, 2019.

Non-Final Office Action for U.S. Appl. No. 15/636,324, dated Apr. 18, 2019.

Notice of Allowance from U.S. Appl. No. 15/636,324, dated Jul. 2, 2019.

Corrected Notice of Allowance from U.S. Appl. No. 15/636,324, dated Aug. 20, 2019.

Corrected Notice of Allowance from U.S. Appl. No. 15/636,324, dated Sep. 5, 2019.

Rivard et al., U.S. Appl. No. 16/271,604, filed Feb. 8, 2019.

Non-Final Office Action from U.S. Appl. No. 16/271,604, dated Apr. 5, 2019.

Notice of Allowance from U.S. Appl. No. 16/271,604, dated Jul. 2, 2019.

Corrected Notice of Allowance from U.S. Appl. No. 16/271,604, dated Aug. 8, 2019.

Notice of Allowance from U.S. Appl. No. 17/321,166, dated Sep. 20, 2023.

Feder et al., U.S. Appl. No. 14/843,896, filed Sep. 2, 2015.

Non-Final Office Action from U.S. Appl. No. 14/843,896, dated Jan. 14, 2016.

Notice of Allowance from U.S. Appl. No. 14/843,896, dated Jun. 9, 2016.

Feder et al., U.S. Appl. No. 15/253,721, filed Aug. 31, 2016.

Non-Final Office Action from U.S. Appl. No. 15/253,721, dated Jun. 9, 2017.

Final Office Action from U.S. Appl. No. 15/253,721, dated Nov. 3, 2017.

Notice of Allowance from U.S. Appl. No. 15/253,721, dated Jan. 4, 2018.

Feder et al., U.S. Appl. No. 15/913,742, filed Mar. 6, 2018.

Non-Final Office Action from U.S. Appl. No. 15/913,742, dated Jul. 31, 2019.

Final Office Action from U.S. Appl. No. 15/913,742, dated Nov. 21, 2019.

Non-Final Office Action from U.S. Appl. No. 15/913,742, dated Jun. 18, 2020.

(56)

References Cited**OTHER PUBLICATIONS**

Final Office Action from U.S. Appl. No. 15/913,742, dated Dec. 16, 2020.

Non-Final Office Action from U.S. Appl. No. 15/913,742, dated Jun. 9, 2021.

Final Office Action from U.S. Appl. No. 15/913,742, dated Feb. 14, 2022.

Feder et al., U.S. Appl. No. 17/865,299, filed Jul. 14, 2022.

Non-Final Office Action from U.S. Appl. No. 17/865,299, dated Feb. 22, 2024.

Final Office Action from U.S. Appl. No. 17/865,299, dated Sep. 12, 2024.

Rivard et al., U.S. Appl. No. 13/999,343, filed Feb. 11, 2014.

Notice of Allowance for U.S. Appl. No. 13/999,343, dated Jul. 17, 2015.

Rivard et al., U.S. Appl. No. 14/887,211, filed Oct. 19, 2015.

Non-Final Office Action from U.S. Appl. No. 14/887,211, dated Feb. 18, 2016.

Final Office Action from U.S. Appl. No. 14/887,211, dated Sep. 23, 2016.

Non-Final Office Action from U.S. Appl. No. 14/887,211, dated Jan. 17, 2017.

Notice of Allowance from U.S. Appl. No. 14/887,211, dated May 1, 2017.

Notice of Allowance from U.S. Appl. No. 14/887,211, dated Sep. 13, 2017.

Rivard et al., U.S. Appl. No. 15/808,753, filed Nov. 9, 2017.

Non-Final Office Action from U.S. Appl. No. 15/808,753, dated Feb. 23, 2018.

Notice of Allowance from U.S. Appl. No. 15/808,753, dated Jul. 10, 2018.

Rivard et al., U.S. Appl. No. 16/147,149, filed Sep. 28, 2018.

Notice of Allowance from U.S. Appl. No. 16/147,149, dated Jan. 14, 2019.

Rivard et al., U.S. Appl. No. 16/296,038, filed Mar. 7, 2019.

Notice of Allowance from U.S. Appl. No. 16/296,038, dated Apr. 12, 2019.

Rivard et al., U.S. Appl. No. 16/518,811, filed Jul. 22, 2019.

Notice of Allowance from U.S. Appl. No. 16/518,811, dated Sep. 25, 2019.

Rivard et al., U.S. Appl. No. 16/744,735, filed Jan. 16, 2020.

Notice of Allowance from U.S. Appl. No. 16/744,735, dated Aug. 18, 2021.

Rivard et al., U.S. Appl. No. 17/518,436, filed Nov. 3, 2021.

Non-Final Office Action from U.S. Appl. No. 17/518,436, dated Jul. 11, 2022.

Notice of Allowance from U.S. Appl. No. 17/518,436, dated Mar. 28, 2023.

Rivard et al., U.S. Appl. No. 18/213,198, filed Jun. 22, 2023.

Non-Final Office Action from U.S. Appl. No. 18/213,198, dated Feb. 1, 2024.

Feder et al., U.S. Appl. No. 13/999,678, filed Mar. 14, 2014.

Non-Final Office Action from U.S. Appl. No. 13/999,678, dated Aug. 12, 2015.

Final Office Action from U.S. Appl. No. 13/999,678, dated Mar. 28, 2016.

Non-Final Office Action from U.S. Appl. No. 13/999,678, dated Dec. 20, 2016.

Notice of Allowance from U.S. Appl. No. 13/999,678, dated Jun. 29, 2017.

Feder et al., U.S. Appl. No. 15/354,935, filed Nov. 17, 2016.

Non-Final Office Action from U.S. Appl. No. 15/354,935, dated Feb. 8, 2017.

Notice of Allowance from U.S. Appl. No. 15/354,935, dated Aug. 23, 2017.

Examination Report from Indian Application No. 201927010939, dated Jun. 9, 2021.

Summons to Attend Oral Proceedings from European Application No. 15856267.8, dated Sep. 3, 2021.

Photoshop, "Photoshop Help/Levels adjustment," Photoshop Help, retrieved on Aug. 20, 2021 from <https://web.archive.org/web/20141018232619/http://helpx.adobe.com:80/photoshop/using/levels-adjustment.html>, 3 pages.

Photoshop, "Photoshop Help/Levels adjustment," Photoshop Help, 2014, retrieved on Aug. 20, 2021 from <https://web.archive.org/web/20141018232619/http://helpx.adobe.com:80/photoshop/using/levels-adjustment.html>, 3 pages.

Office Action from Japanese Patent Application No. 2020-121537, dated Oct. 19, 2021.

Extended European Search Report from European Application No. 21196442.4, dated Dec. 13, 2021.

Huo et al., "Robust Automatic White Balance algorithm using Gray Color Points in Images," IEEE Transactions on Consumer Electronics, vol. 52, No. 2, May 2006, pp. 541-546.

Examination Report from European Application No. 15857386.5, dated Dec. 12, 2021.

Final Office Action from Japanese Patent Application No. 2017-544282, dated Mar. 1, 2022.

Notice of Final Rejection from Japanese Patent Application No. 2017-544282, dated Mar. 1, 2022.

Examination Report from Indian Application No. 201927010939, dated Mar. 25, 2022.

Examination Report from Indian Application No. 202027018945, dated Mar. 17, 2022.

Office Action from Japanese Patent Application No. 2021-076679, dated Aug. 2, 2022.

Office Action from Japanese Patent Application No. 2021-079285, dated Aug. 2, 2022.

Office Action from Japanese Patent Application No. 2021-096499, dated Sep. 6, 2022.

Office Action from Japanese Patent Application No. 2021-154653, dated Sep. 13, 2022.

Office Action from Chinese Patent Application No. 202110773625.1, dated Nov. 2, 2022.

Summons to Attend Oral Proceedings from European Application No. 16 915 389.7, dated Oct. 21, 2022.

Office Action from Australian Patent Application No. 2020299585, dated Nov. 24, 2022.

Srivastava et al., U.S. Appl. No. 15/870,689, filed Jan. 12, 2018.

Non-Final Office Action from U.S. Appl. No. 15/870,689, dated Mar. 17, 2019.

Final Office Action from U.S. Appl. No. 15/870,689, dated Sep. 5, 2019.

Advisory Action from U.S. Appl. No. 15/870,689, dated Nov. 14, 2019.

Notice of Allowance from U.S. Appl. No. 15/870,689, dated Dec. 17, 2019.

Examination Report from European Application No. 15 857 386.5, dated Jan. 3, 2023.

Decision for Rejection from Japanese Patent Application No. 2021-154653, dated May 23, 2023.

Second Office Action from Chinese Patent Application No. 202110773625.1, dated May 7, 2023.

Examination Report from European Application No. 18864431.4, dated Apr. 21, 2023.

Decision for Rejection from Japanese Patent Application No. 2021-076679, dated Jun. 6, 2023.

Wang et al., "A High Dynamic Range CMOS APS Image Sensor," Semantic Scholar, 2001, pp. 1-4, retrieved from <https://www.semanticscholar.org/paper/A-High-Dynamic-Range-CMOS-APS-Image-Sensor-Wang/a824e97836438887089a3c62f4d9be77d47c9067>.

Bock et al., "A Wide-VGA CMOS Image Sensor with Global Shutter and Extended Dynamic Range," International Image Sensor Society, 2005, pp. 222-225, retrieved from <https://www.imagesensors.org/Past%20Workshops/2005%20Workshop/2005%20Papers/56%20Bock%20et%20al.pdf>.

Micron, "1/3-Inch, Wide-VGA CMOS Digital Image Sensor," Micron Technology, 2006, 15 pages, retrieved from <https://media.digikey.com/pdf/Data%20Sheets/Micron%20Technology%20Inc%20PDFs/MT9V022.pdf>.

(56)

References Cited**OTHER PUBLICATIONS**

Neo Film School, "A Final Nail in the Coffin of Film Cinematography!" Neo Film School Blog, Mar. 7, 2011, 4 pages, retrieved from <https://neofilmschool.wordpress.com/2011/03/07/a-final-nail-in-the-coffin-of-film-cinematography/>.

Cmosis, "Cmosis Outlining Low Noise/High Dynamic Image Sensor Concept," Photonics Online, Nov. 29, 2010, 1 page, retrieved from <https://www.photonicsonline.com/doc/cmosis-outlining-low-noisehigh-dynamic-image-0001?VNETCOOKIE=NO>.

Koifman et al., "Image Sensors World," Blog, Dec. 30, 2010, 62 pages, retrieved from <http://image-sensors-world.blogspot.com/2010/>.

Office Action from Chinese Patent Application No. 202111599811.4, dated Feb. 23, 2024, 7 pages.

Aksoy et al., "A Dataset of Flash and Ambient Illumination Pairs from the Crowd," ECCV, 2018, pp. 1-16.

Cornell, B., U.S. Appl. No. 15/162,326, filed May 23, 2016.

Non-Final Office Action from U.S. Appl. No. 15/162,326, dated Sep. 27, 2017.

Notice of Allowance from U.S. Appl. No. 15/162,326, dated Jan. 29, 2018.

Notice of Allowance from U.S. Appl. No. 15/162,326, dated Apr. 11, 2018.

Adnan et al., "Efficient Kernal Fusion Techniques for Massive Video Data Analysis on GPGPUs," arXiv, Sep. 15, 2015, 10 pages, retrieved from <https://arxiv.org/abs/1509.04394>.

Liu et al., "An Efficient SAR Processor Based on GPU via CUDA," 2nd International Congress on Image and Signal Processing, Oct. 17, 2009, pp. 1-5.

Sung, Hsiang-Wei et al., "OpenCV Optimization on Heterogeneous Multi-Core Systems for Gesture Recognition Applications," IEEE 45th International Conference on Parallel Processing Workshops {ICPPW}, Aug. 16, 2016, pp. 59-65.

U.S. Appl. No. 14/178,305, filed Feb. 12, 2014.

Final Office Action from U.S. Appl. No. 14/178,305, dated May 18, 2015.

Non-Final Office Action from U.S. Appl. No. 14/178,305, dated Aug. 11, 2014.

Le et al., U.S. Appl. No. 16/278,581, filed Feb. 18, 2019.

Zhen et al., U.S. Appl. No. 16/277,630, filed Feb. 15, 2019.

Le et al., U.S. Appl. No. 16/278,543, filed Feb. 18, 2019.

European Office Communication and Exam Report from European Application No. 15857386.5, dated Jan. 11, 2018.

Kim et al., "A CMOS Image Sensor Based on Unified Pixel Architecture With Time-Division Multiplexing Scheme for Color and Depth Image Acquisition," IEEE Journal of Solid-State Circuits, vol. 47, No. 11, Nov. 2012, pp. 2834-2845.

European Office Communication and Exam Report from European Application No. 15857748.6, dated Jan. 10, 2018.

Extended European Search Report from European Application No. 15891394.7 dated Jun. 19, 2018.

International Preliminary Examination Report from PCT Application No. PCT/US2017/39946, dated Jan. 10, 2019.

International Search Report and Written Opinion from International Application No. PCT/US 18/54014, dated Dec. 26, 2018.

Office Action from Chinese Patent Application No. 201580079444.1, dated Aug. 1, 2019.

Examination Report from European Application No. 15 856 814.7, dated Aug. 20, 2019.

Examination Report from European Application No. 15 857 675.1, dated Aug. 23, 2019.

Examination Report from European Application No. 15 856 710.7, dated Sep. 9, 2019.

Examination Report from European Application No. 15 857 386.5, dated Sep. 17, 2019.

Examination Report from European Application No. 15 857 748.6, dated Sep. 26, 2019.

Office Action from Japanese Patent Application No. 2017-544279, dated Oct. 23, 2019.

Office Action from Japanese Patent Application No. 2017-544280, dated Oct. 29, 2019.

Office Action from Japanese Patent Application No. 2017-544283, dated Oct. 29, 2019.

Office Action from Japanese Patent Application No. 2017-544547, dated Nov. 5, 2019.

Office Action from Japanese Patent Application No. 2017-544281, dated Nov. 26, 2019.

Extended European Search Report from European Application No. 16915389.7, dated Dec. 2, 2019.

Office Action from Japanese Patent Application No. 2017-544284, dated Dec. 10, 2019.

Office Action from Japanese Patent Application No. 2017-544282, dated Jan. 7, 2020.

Office Action from Chinese Patent Application No. 201780053926.9, dated Jan. 16, 2020.

Extended European Search Report from European Application No. 17821236.1, dated Jan. 24, 2020.

Petschnigg et al., "Digital Photography with Flash and No-Flash Image Pairs," ACM Transactions of Graphics, vol. 23, Aug. 2004, pp. 664-672.

International Preliminary Examination Report from PCT Application No. PCT/US2018/054014, dated Apr. 16, 2020.

Office Action from Chinese Patent Application No. 201680088945.0, dated May 21, 2020.

Office Action from Japanese Patent Application No. 2017-544284, dated Aug. 18, 2020.

International Search Report and Written Opinion from PCT Application No. PCT/US2020/040478, dated Sep. 25, 2020.

Summons to Attend Oral Proceedings from European Application No. 15 856 710.7, dated Sep. 18, 2020.

Office Action from Japanese Patent Application No. 2017-544281, dated Oct. 27, 2020.

Office Action from Chinese Patent Application No. 201780053926.9, dated Oct. 13, 2020.

Office Action from Japanese Patent Application No. 2017-544280, dated Jun. 30, 2020.

Second Office Action from Chinese Patent Application No. 201680088945.0, dated Dec. 17, 2020.

Office Action from Japanese Patent Application No. 2017-544282, dated Jan. 5, 2021.

Office Action from Japanese Patent Application No. 2017-544283, dated Jan. 12, 2021.

Decision to Refuse from European Application No. 15856710.7, dated Mar. 15, 2021.

Examination Report from Indian Application No. 201827049041, dated Mar. 19, 2021.

Decision to Refuse from European Application No. 15856212.4, dated Mar. 22, 2021.

Examination Report from European Application No. 16915389.7, dated Feb. 25, 2021.

Examination Report from European Application No. 15857386.5, dated Feb. 8, 2021.

Decision for Rejection and Decision of Dismissal of Amendment for Japanese Application No. 2017-544280, dated May 25, 2021.

Extended European Search Report from European Application No. 18864431.4, dated Jun. 1, 2021.

Kaufman et al., "Content-Aware Automatic Photo Enhancement," Computer Graphics Forum, vol. 31, No. 08, 2012, pp. 2528-2540.

Battiatto et al., "Automatic Image Enhancement by Content Dependent Exposure Correction," EURASIP Journal on Applied Signal Processing, 2004, pp. 1849-1860.

Mangiat et al., "Automatic scene relighting for video conferencing," IEEE 16th Annual International Conference on Image Processing (ICIP), Nov. 2009, pp. 2781-2784.

Weyrich et al., "Analysis of human faces using a measurement-based skin reflectance model," ACM Transactions on Graphics, vol. 25, No. 3, 2006, pp. 1013-1024.

Extended European Search Report from European Application No. 21169039.1, dated Jun. 16, 2021.

Office Action from Japanese Patent Application No. 2017-544284, dated Jul. 13, 2021.

(56)

References Cited**OTHER PUBLICATIONS**

Office Action from Japanese Patent Application No. 2017-544284, dated Jul. 3, 2021.

Office Action from Chinese Patent Application No. 202010904659.5, dated Jul. 28, 2021.

Extended European Search Report from European Application No. 21175832.1, dated Aug. 27, 2021.

International Search Report and Written Opinion from PCT Application No. PCT/US18/61547, dated Jan. 25, 2019.

Rivard et al., U.S. Appl. No. 16/519,244, filed Jul. 23, 2019.

Rivard et al., U.S. Appl. No. 15/815,463, filed Nov. 16, 2017.

Non-Final Office Action from U.S. Appl. No. 15/815,463, dated Mar. 21, 2019.

Notice of Allowance from U.S. Appl. No. 15/815,463, dated Jul. 10, 2019.

Rivard et al., U.S. Appl. No. 16/663,015, filed Oct. 24, 2019.

Rivard et al., U.S. Appl. No. 16/662,965, filed Oct. 24, 2019.

International Preliminary Examination Report from PCT Application No. PCT/US18/61547, dated May 28, 2020.

Non-Final Office Action from U.S. Appl. No. 16/663,015, dated Jul. 9, 2020.

Extended European Search Report from European Application No. 18878357.5, dated May 31, 2021.

Final Office Action from U.S. Appl. No. 16/663,015, dated Feb. 4, 2021.

Non-Final Office Action from U.S. Appl. No. 16/663,015, dated Jul. 22, 2021.

Notice of Allowance from U.S. Appl. No. 16/663,015, dated Feb. 15, 2022.

Corrected Notice of Allowance from U.S. Appl. No. 16/663,015, dated May 11, 2022.

Rivard et al., U.S. Appl. No. 17/824,773, filed May 25, 2022.

Feder et al., U.S. Appl. No. 14/503,210, filed Sep. 30, 2014.

Restriction Requirement from U.S. Appl. No. 14/503,210, dated Oct. 16, 2015.

Non-Final Office Action from U.S. Appl. No. 14/503,210, dated Jan. 12, 2016.

Notice of Allowance from U.S. Appl. No. 14/503,210, dated May 31, 2016.

Feder et al., U.S. Appl. No. 14/503,224, filed Sep. 30, 2014.

Non-Final Office Action from U.S. Appl. No. 14/503,224, dated Sep. 2, 2015.

Notice of Allowance from U.S. Appl. No. 14/503,224, dated Feb. 3, 2016.

Rivard et al., U.S. Appl. No. 13/573,252, filed Sep. 4, 2012.

Restriction Requirement from U.S. Appl. No. 13/573,252, dated Apr. 21, 2014.

Non-Final Office Action from U.S. Appl. No. 13/573,252, dated Jul. 10, 2014.

Notice of Allowance from U.S. Appl. No. 13/573,252, dated Oct. 22, 2014.

Rivard et al., U.S. Appl. No. 14/568,045, filed Dec. 11, 2014.

Restriction Requirement from U.S. Appl. No. 14/568,045, dated Jan. 15, 2015.

Non-Final Office Action from U.S. Appl. No. 14/568,045, dated Mar. 24, 2015.

Final Office Action from U.S. Appl. No. 14/568,045, dated Sep. 18, 2015.

Notice of Allowance from U.S. Appl. No. 14/568,045, dated Apr. 26, 2016.

Notice of Allowance from U.S. Appl. No. 14/568,045, dated Jan. 12, 2016.

Corrected Notice of Allowance from U.S. Appl. No. 14/568,045, dated Feb. 19, 2016.

Corrected Notice of Allowance from U.S. Appl. No. 14/568,045, dated Jun. 30, 2016.

Feder et al., U.S. Appl. No. 14/517,731, filed Oct. 17, 2014.

Non-Final Office Action from U.S. Appl. No. 14/517,731, dated Oct. 6, 2015.

Notice of Allowance from U.S. Appl. No. 14/517,731, dated May 19, 2016.

Rivard et al., U.S. Appl. No. 14/534,068, filed Nov. 5, 2014.

Non-Final Office Action from U.S. Appl. No. 14/534,068, dated Feb. 17, 2015.

Notice of Allowance from U.S. Appl. No. 14/534,068, dated Jul. 29, 2015.

Corrected Notice of Allowance from U.S. Appl. No. 14/534,068, dated Sep. 22, 2015.

Rivard et al., U.S. Appl. No. 14/543,782, filed Nov. 17, 2014.

Non-Final Office Action from U.S. Appl. No. 14/543,782, dated Dec. 3, 2015.

Notice of Allowance from U.S. Appl. No. 14/543,782, dated Jul. 7, 2016.

Rivard et al., U.S. Appl. No. 15/289,039, filed Oct. 7, 2016.

Non-Final Office Action from U.S. Appl. No. 15/289,039, dated May 11, 2017.

Notice of Allowance from U.S. Appl. No. 15/289,039, dated Oct. 12, 2017.

Rivard et al., U.S. Appl. No. 15/863,785, filed Jan. 5, 2018.

Non-Final Office Action from U.S. Appl. No. 15/863,785, dated Apr. 11, 2018.

Notice of Allowance from U.S. Appl. No. 15/863,785, dated Sep. 11, 2018.

Non-Final Office Action from U.S. Appl. No. 16/244,982, dated May 13, 2021.

Final Office Action from U.S. Appl. No. 16/244,982, dated Dec. 20, 2021.

Non-Final Office Action from U.S. Appl. No. 16/244,982, dated Jun. 24, 2022.

Final Office Action from U.S. Appl. No. 16/244,982, dated Feb. 14, 2023.

Non-Final Office Action from U.S. Appl. No. 16/244,982, dated Aug. 3, 2023.

Rivard et al., U.S. Appl. No. 18/388,158, filed Nov. 8, 2023.

Rivard et al., U.S. Appl. No. 19/025,941, filed Jan. 16, 2025.

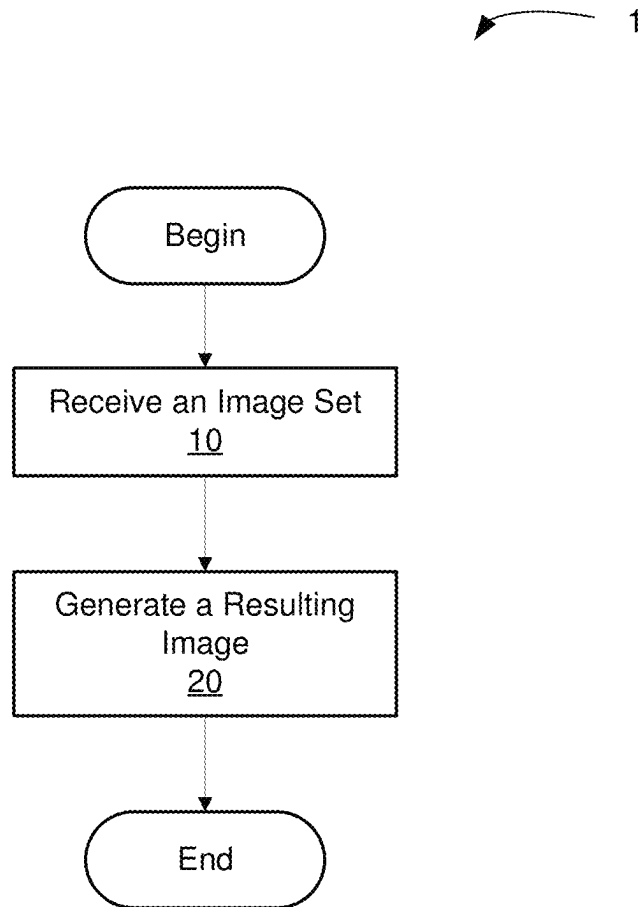
Rivard et al., U.S. Appl. No. 18/935,319, filed Nov. 1, 2024.

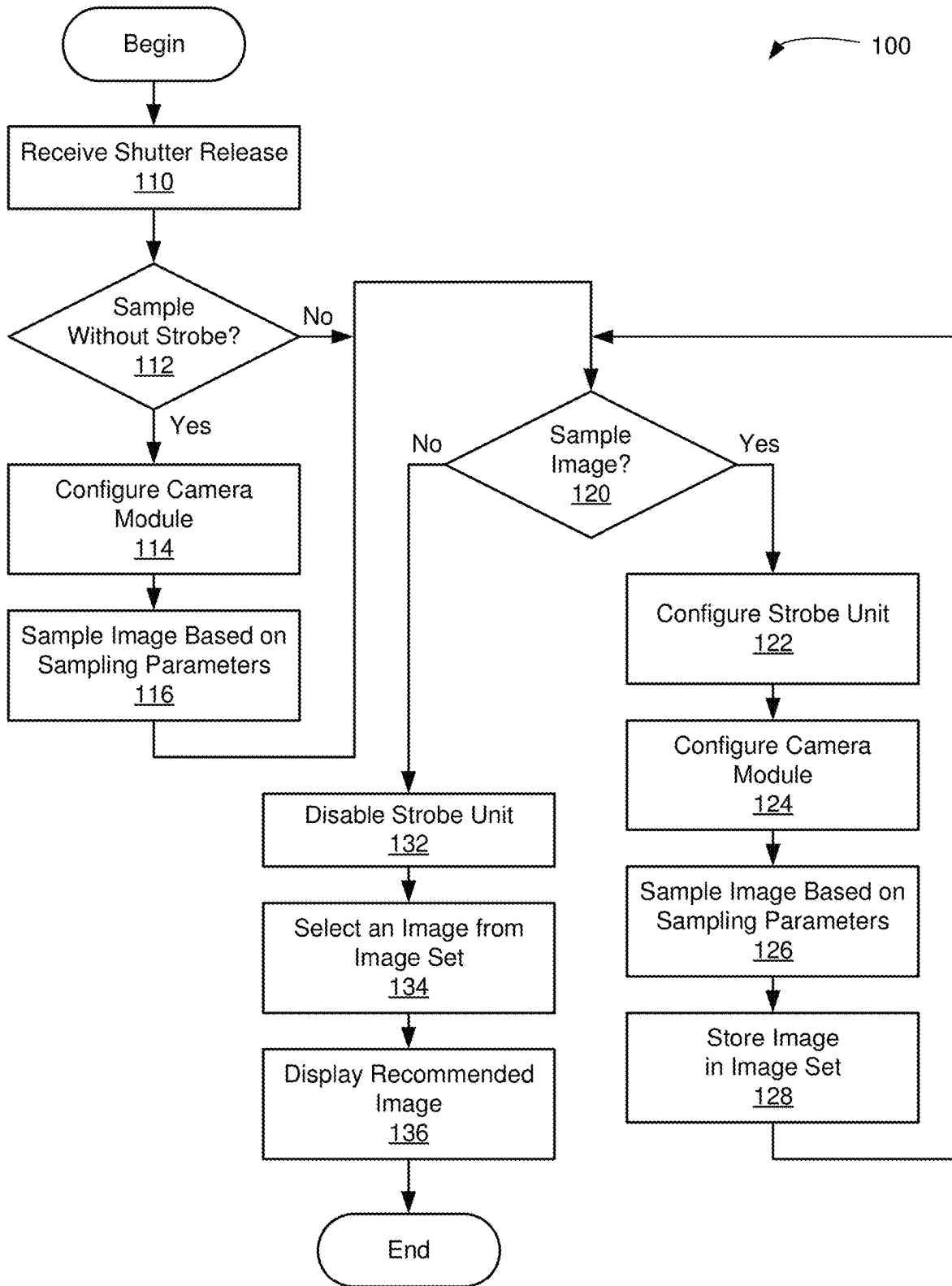
Non-Final Office Action for U.S. Appl. No. 18/935,319, dated Dec. 12, 2024.

Rivard et al., U.S. Appl. No. 19/025,910, filed Jan. 16, 2025.

Non-Final Office Action from U.S. Appl. No. 18/388,158 dated May 19, 2025.

* cited by examiner

**Figure 1A**

**Figure 1B**

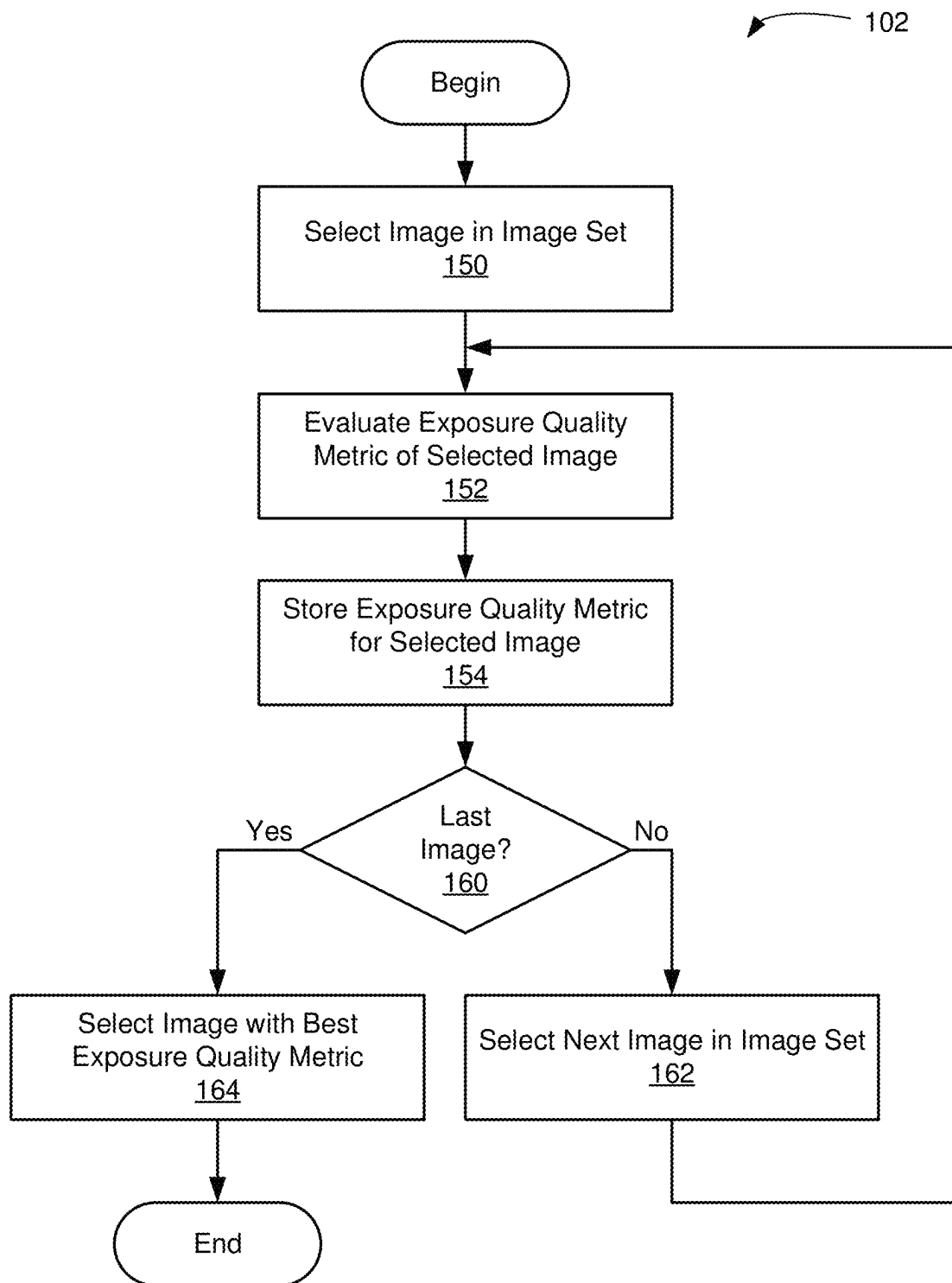


Figure 1C

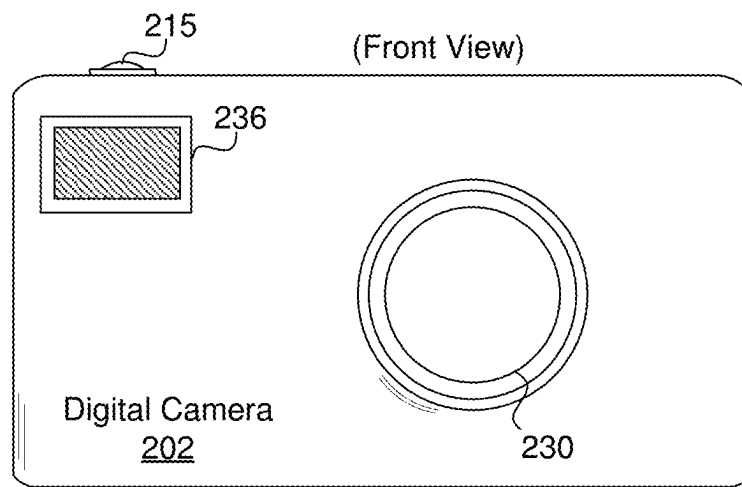


Figure 2A

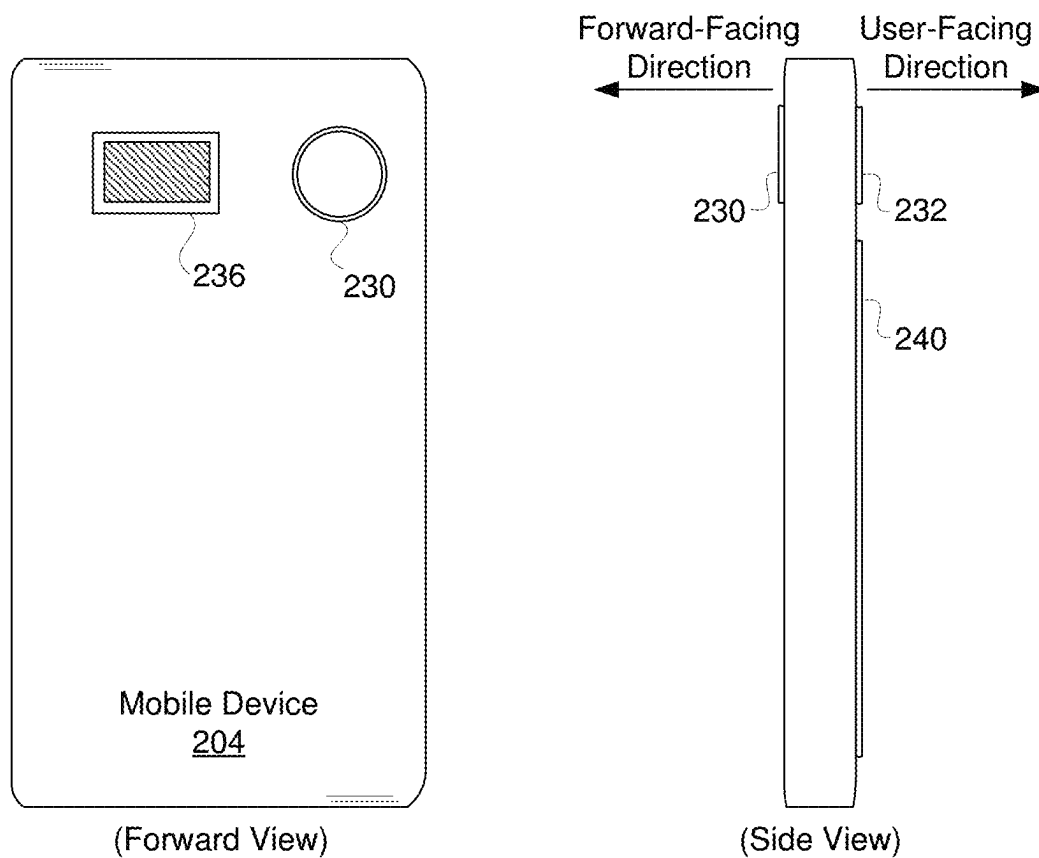


Figure 2B

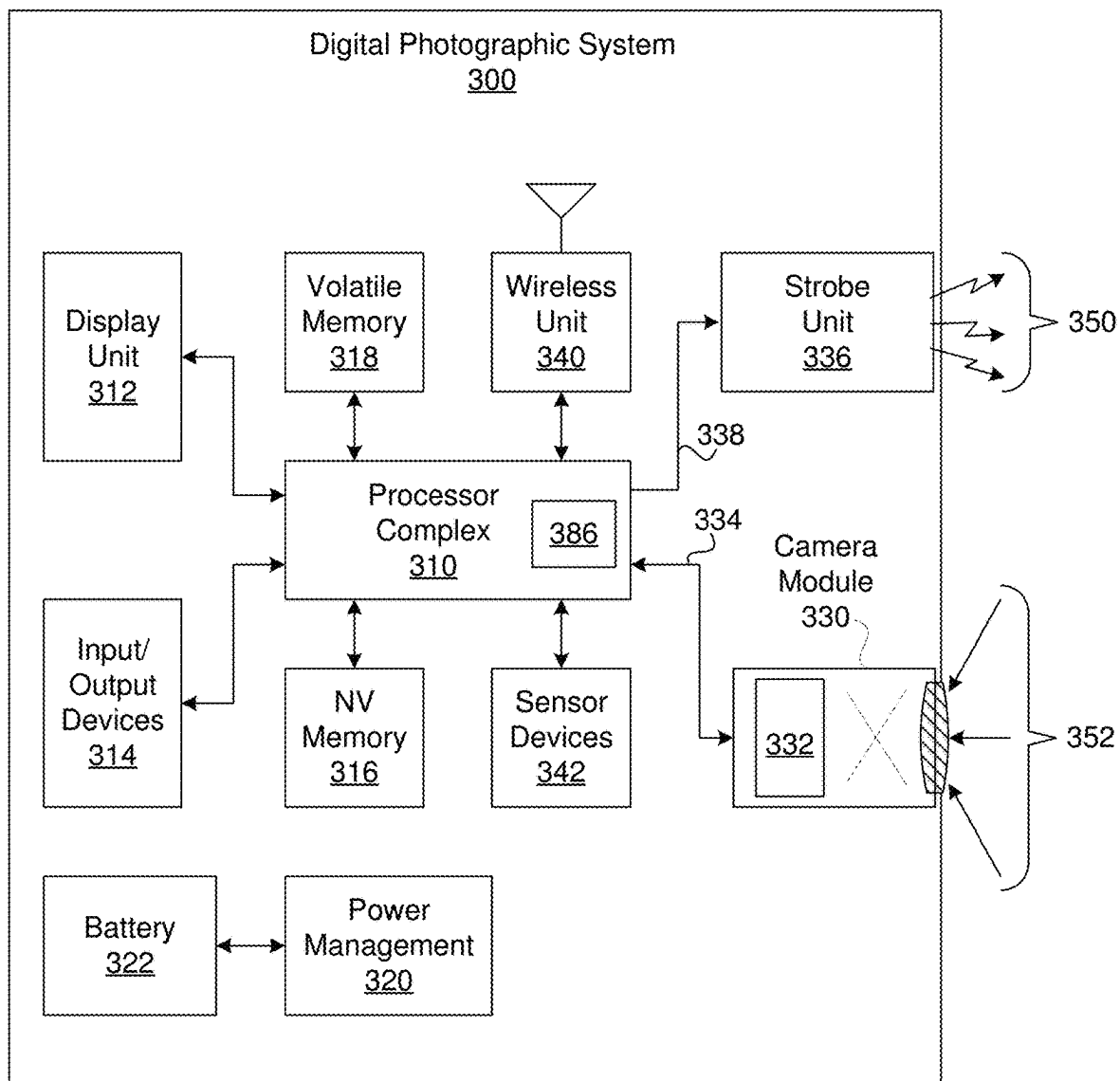
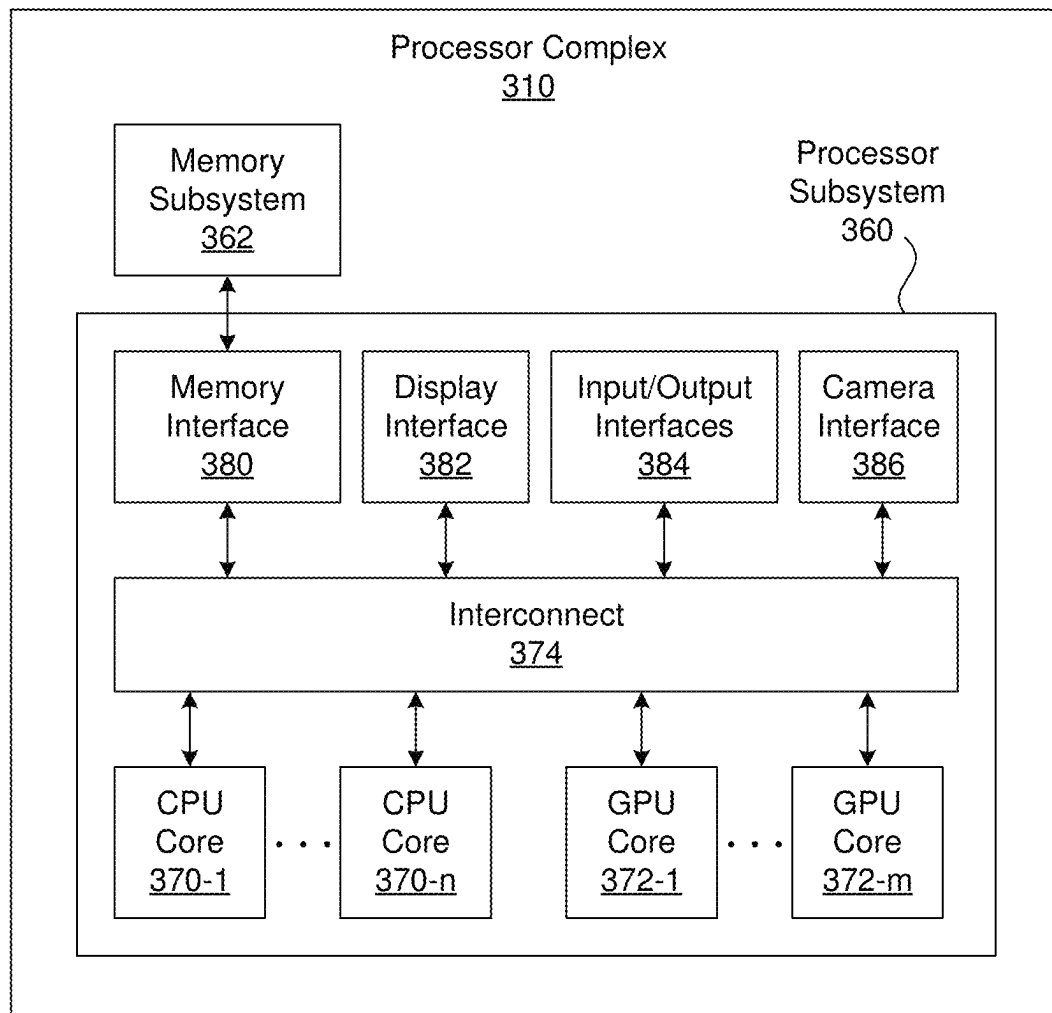


Figure 3A

**Figure 3B**

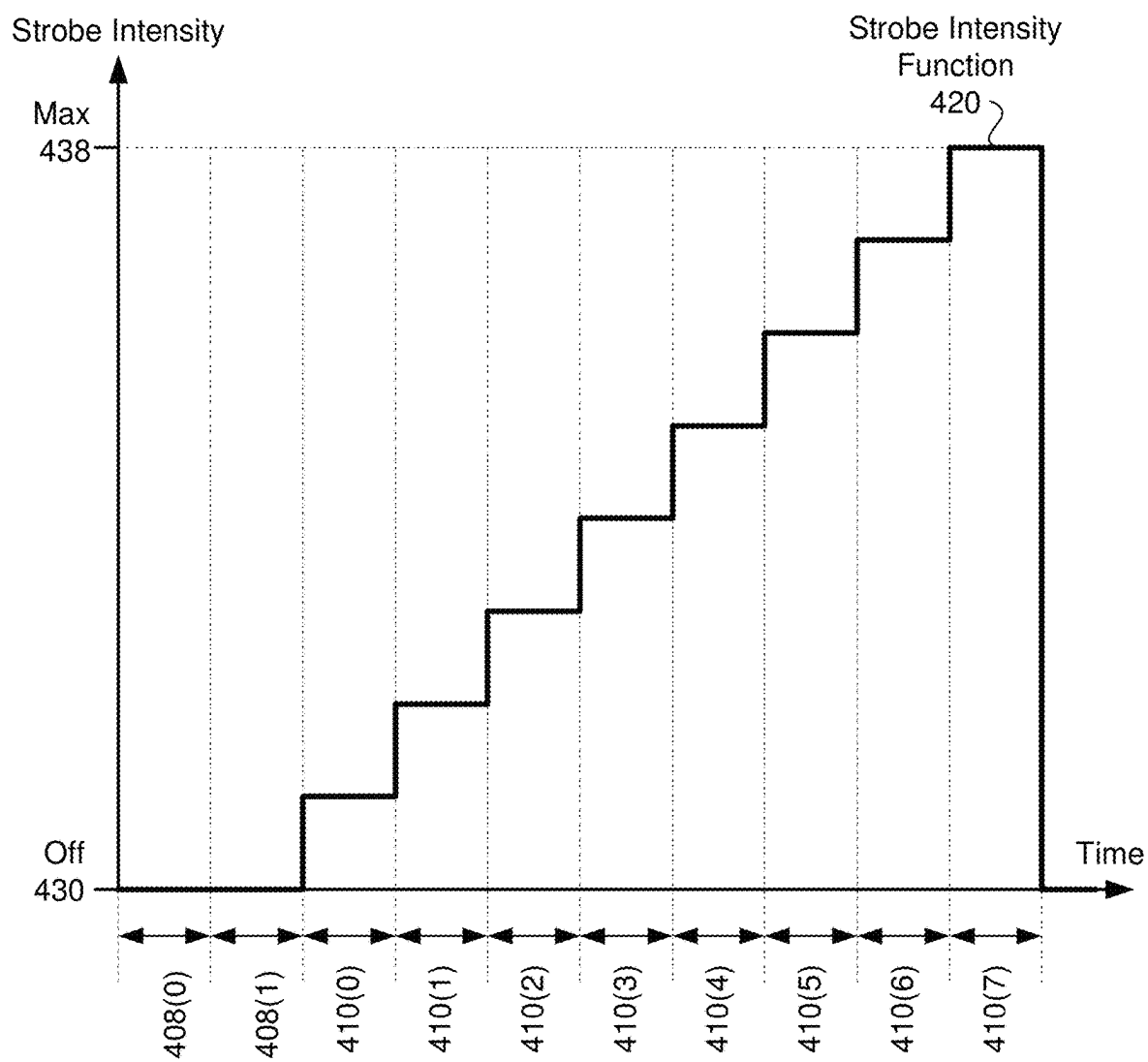


Figure 4A

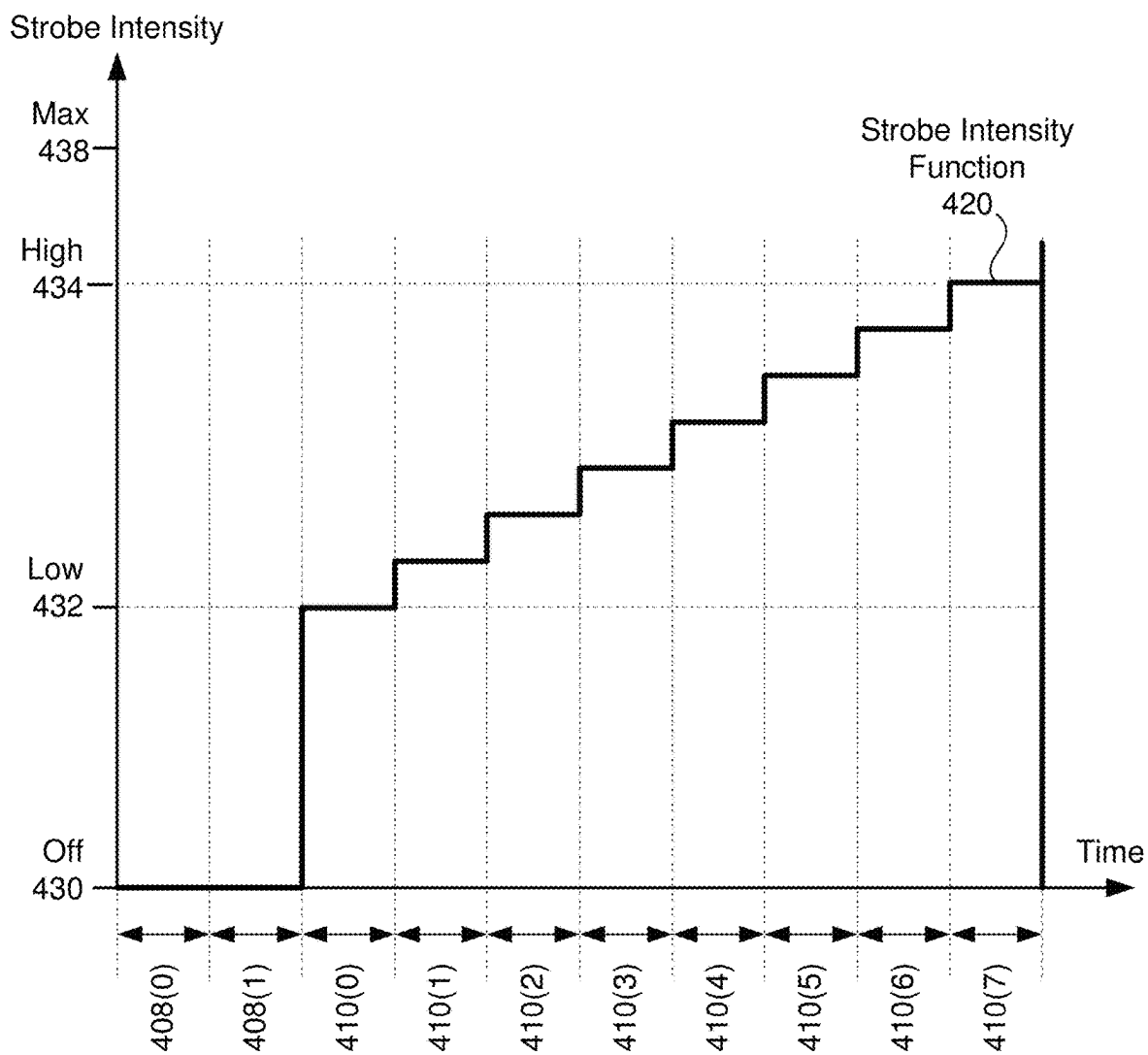


Figure 4B

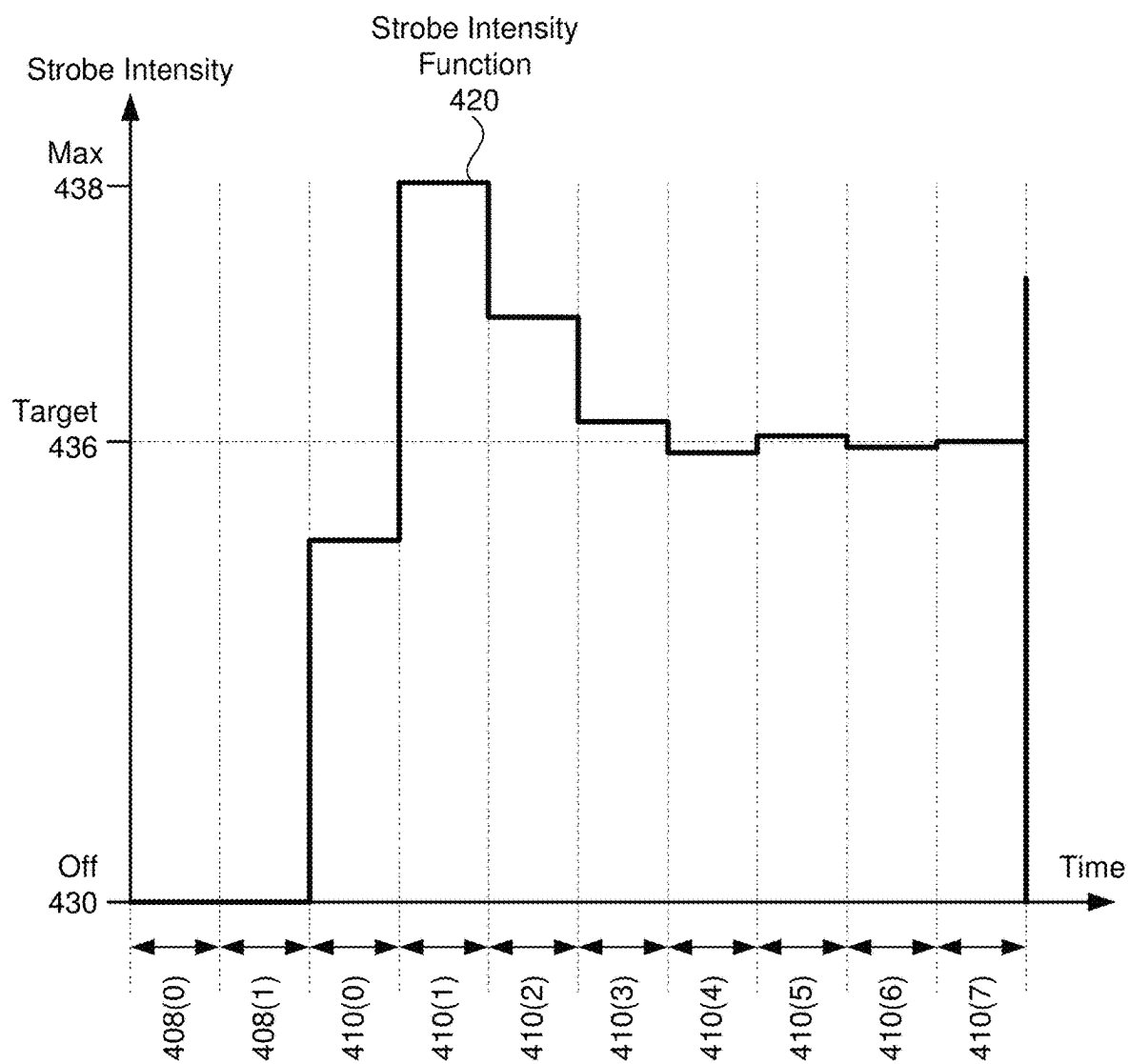


Figure 4C

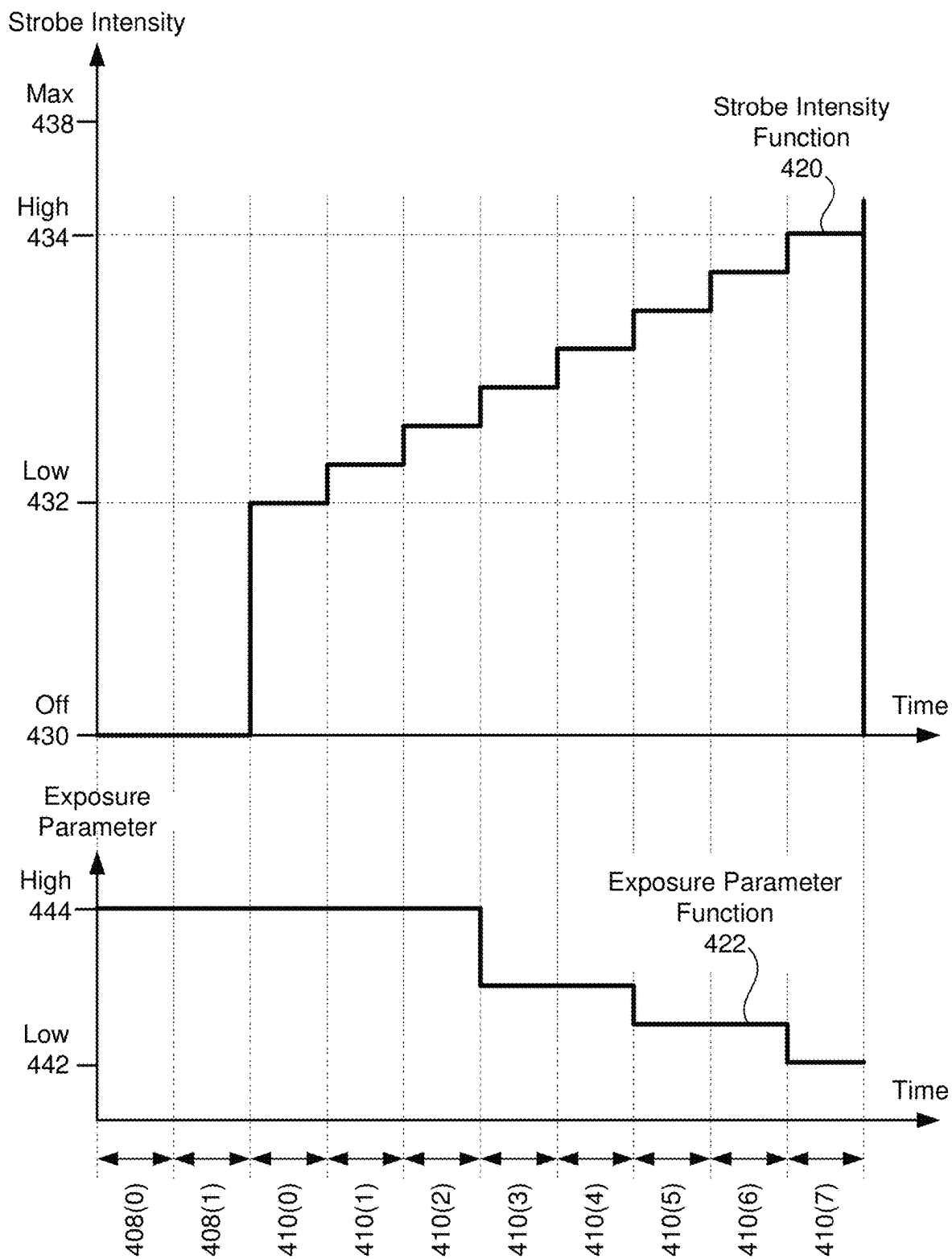


Figure 4D

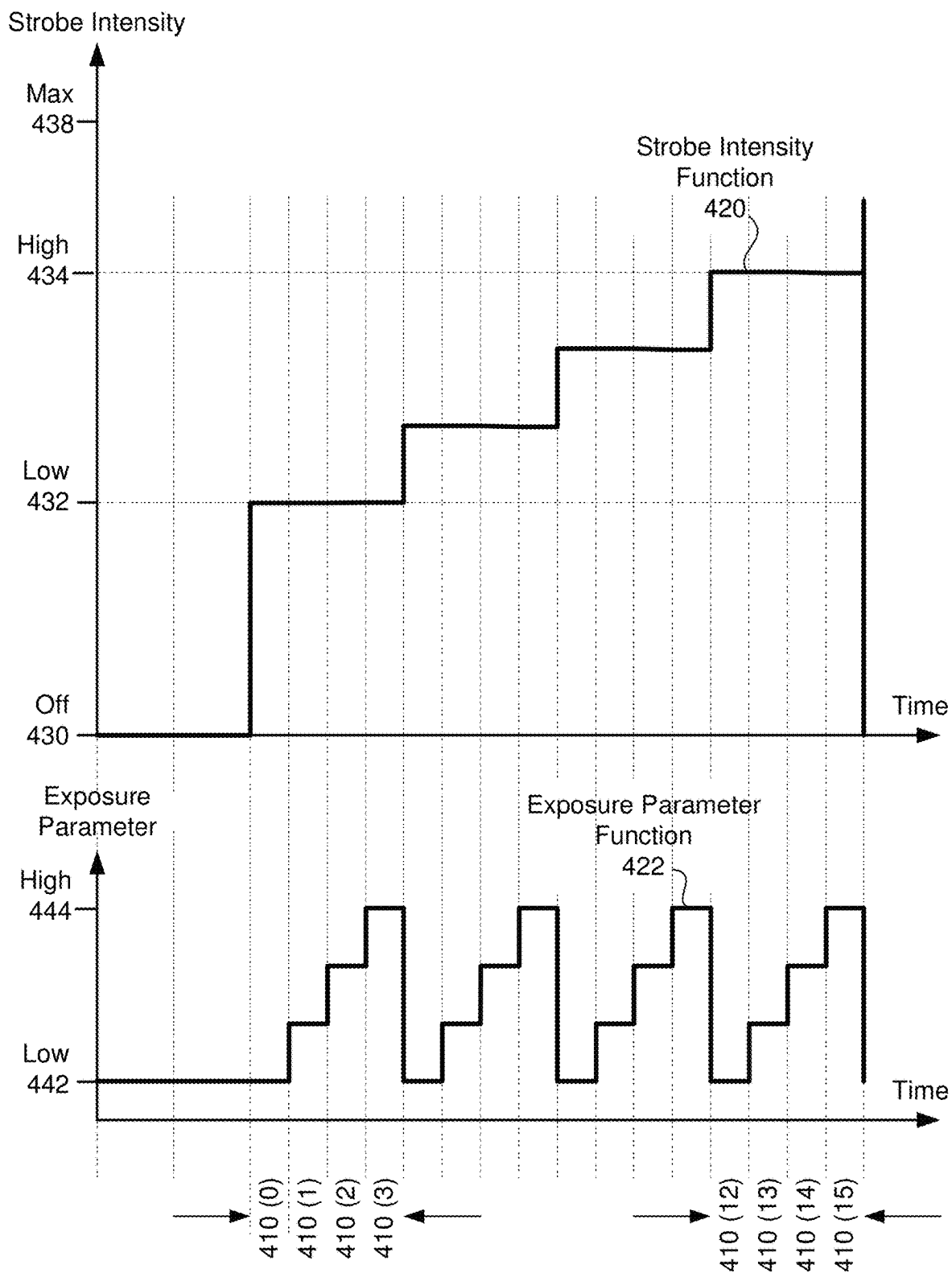


Figure 4E

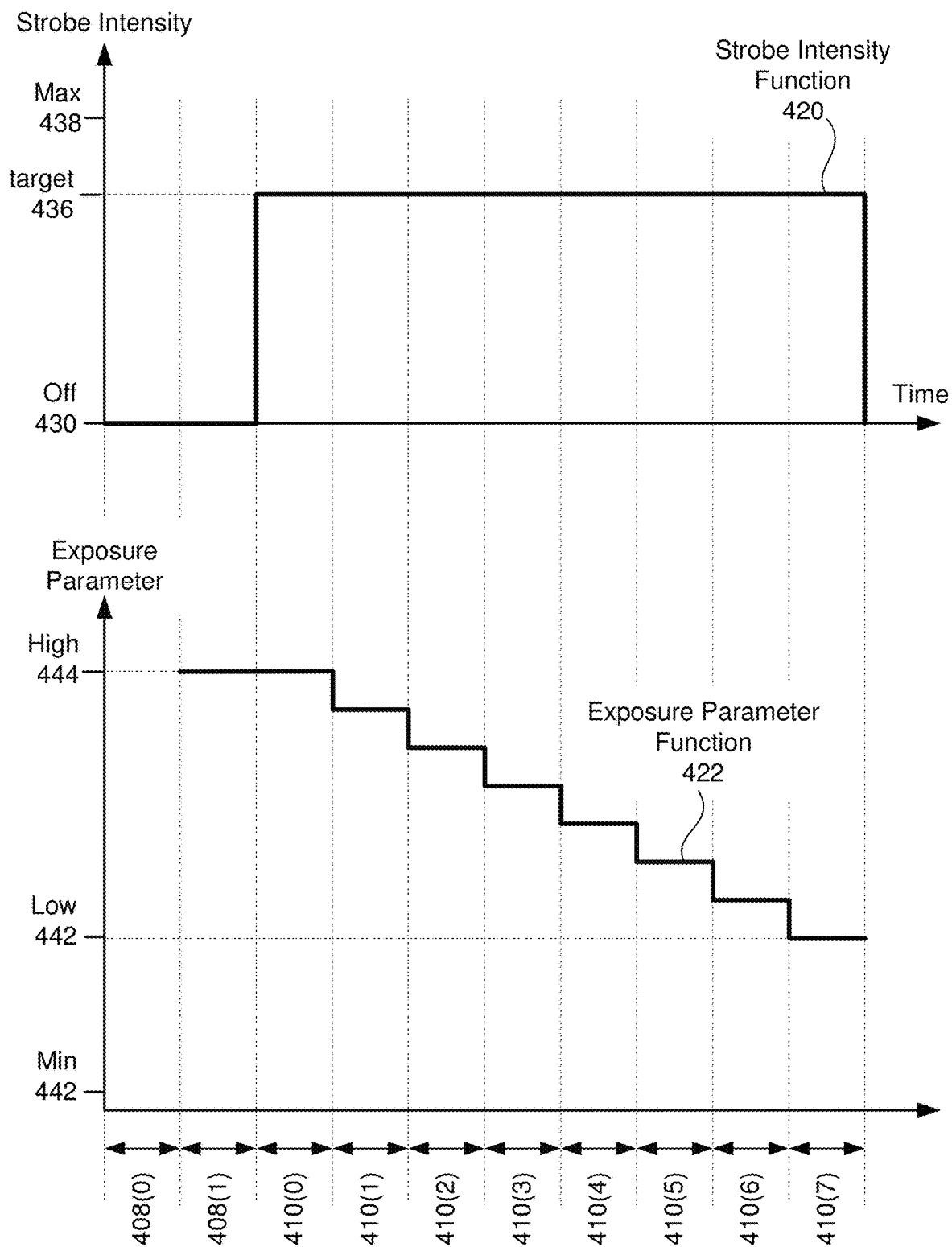


Figure 4F

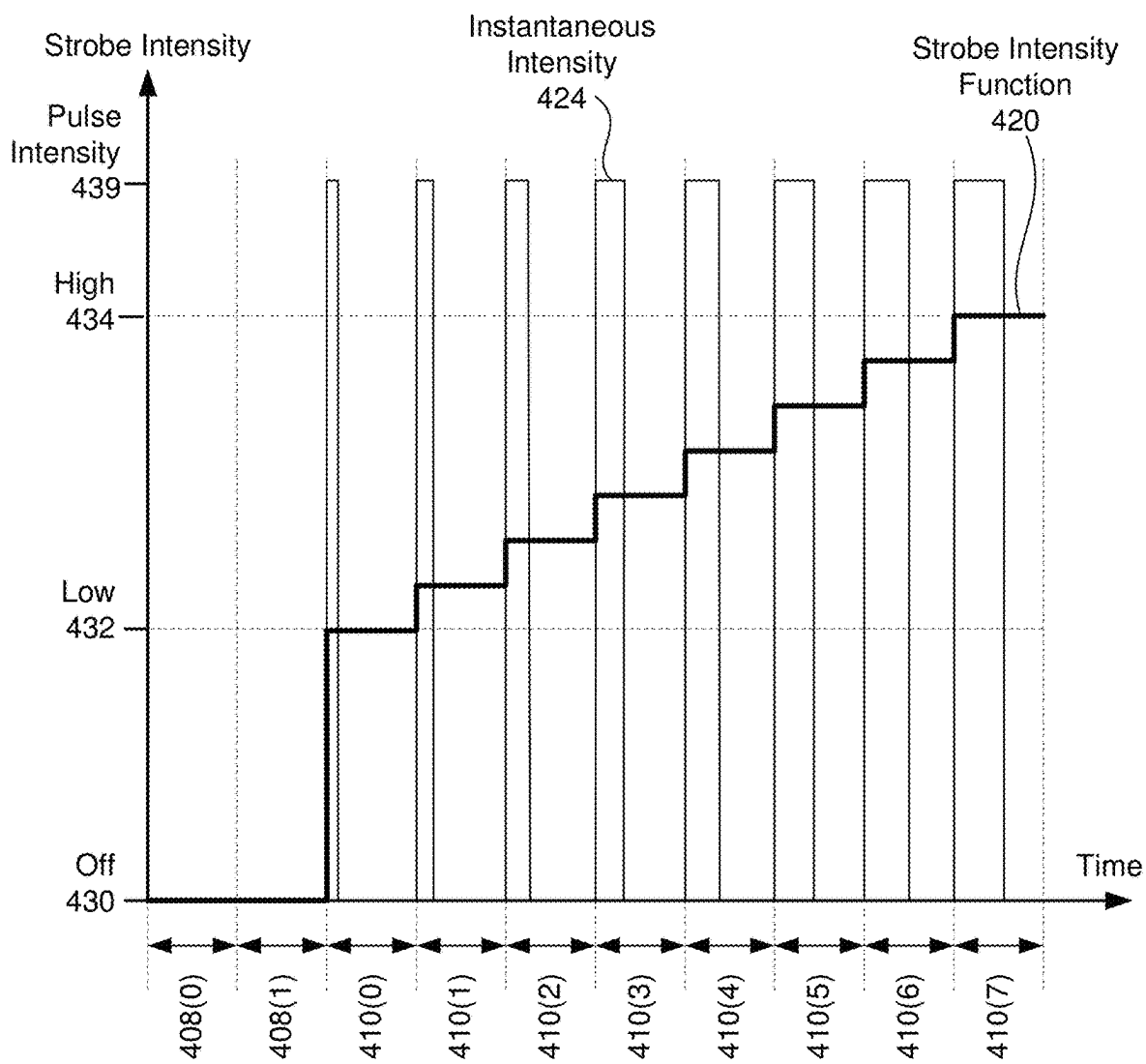


Figure 4G

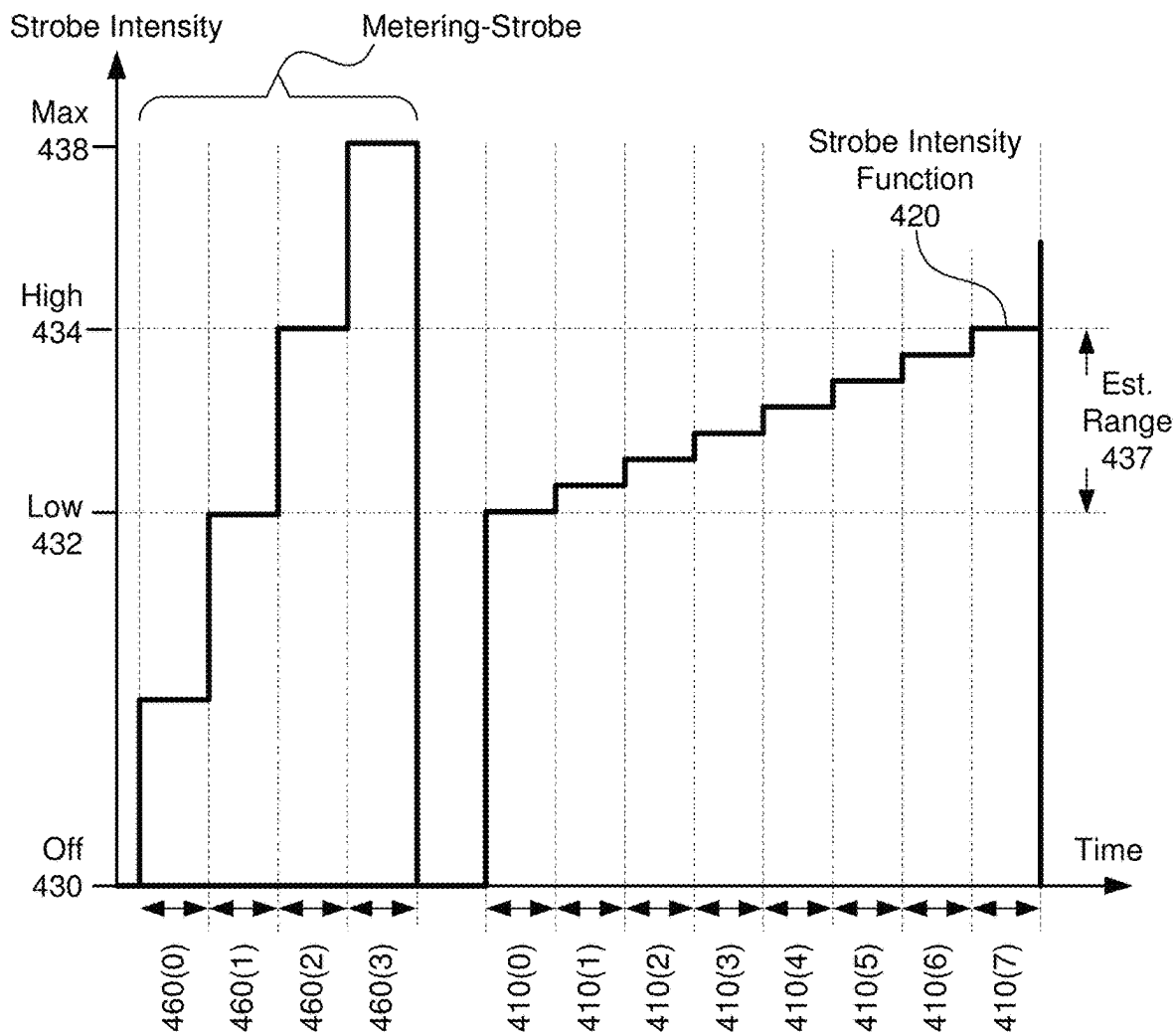


Figure 4H

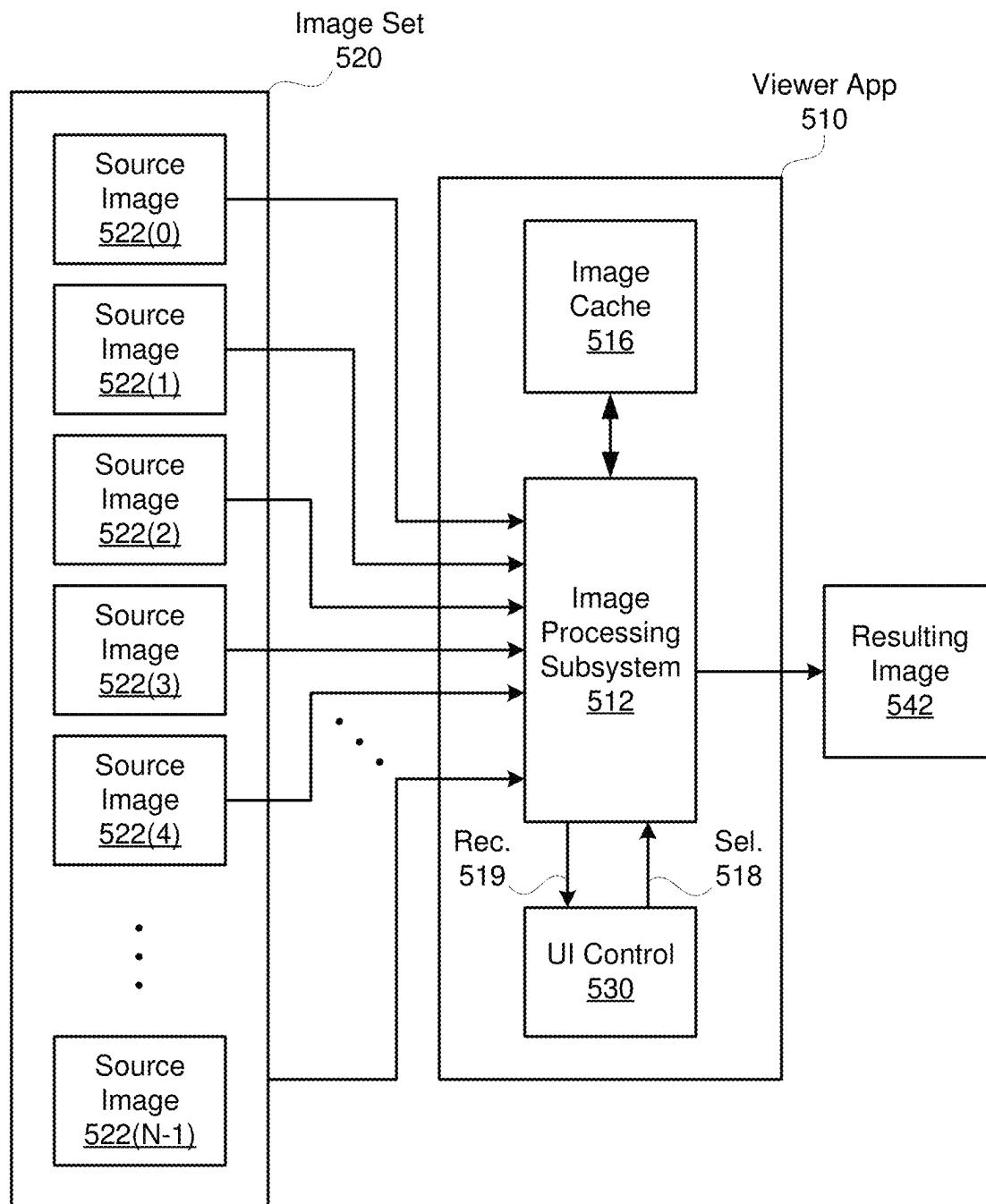


Figure 5A

Application Window
540

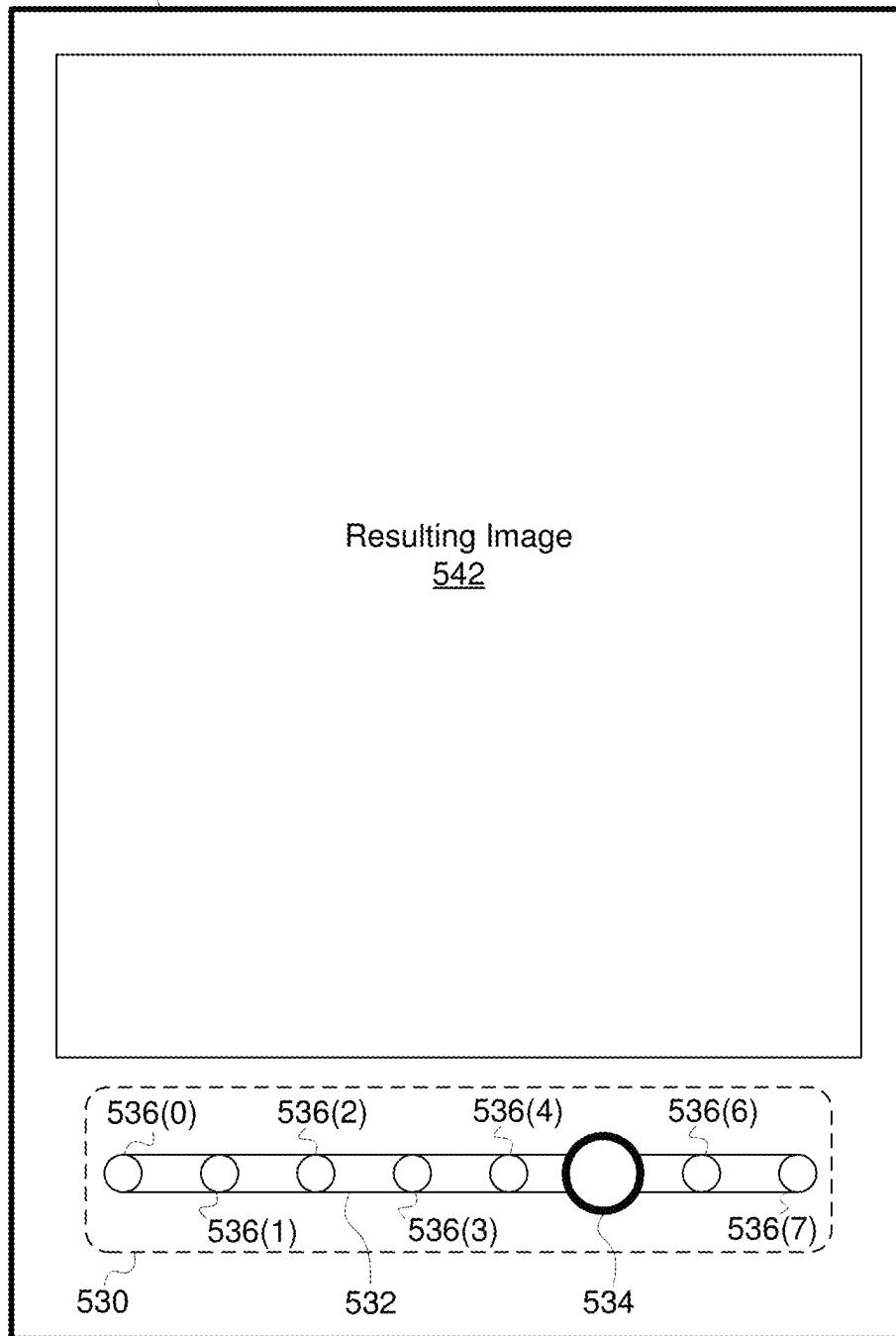


Figure 5B

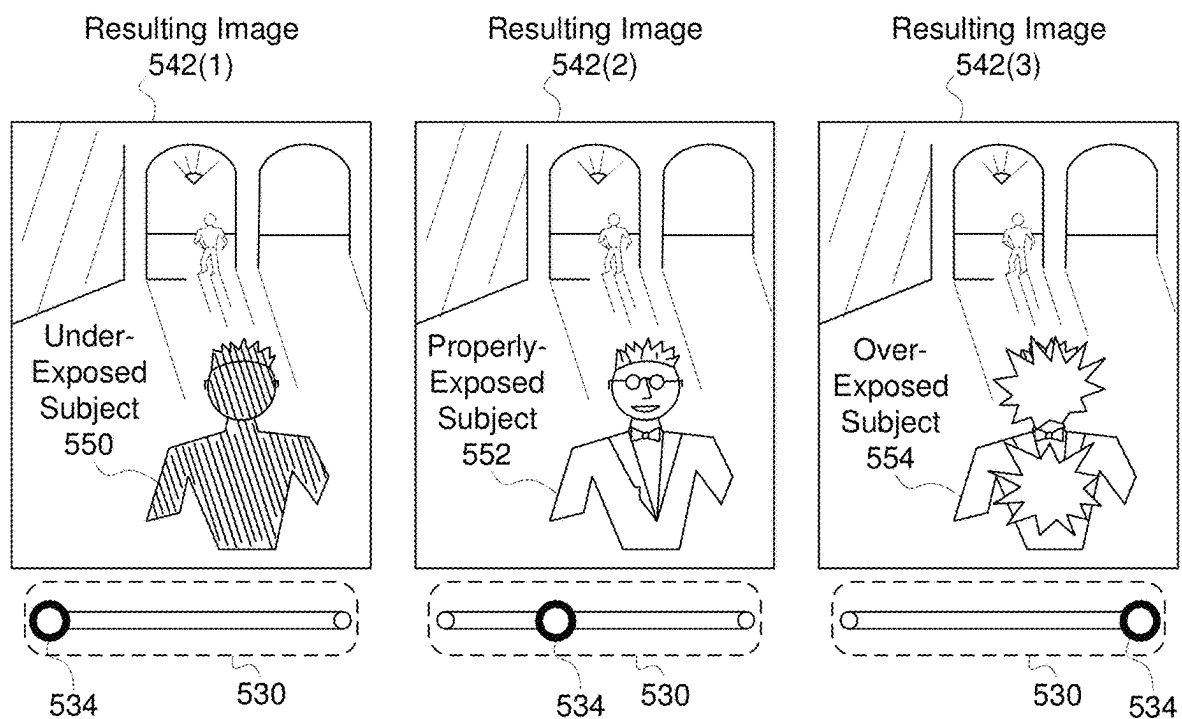


Figure 5C

Application Window

540

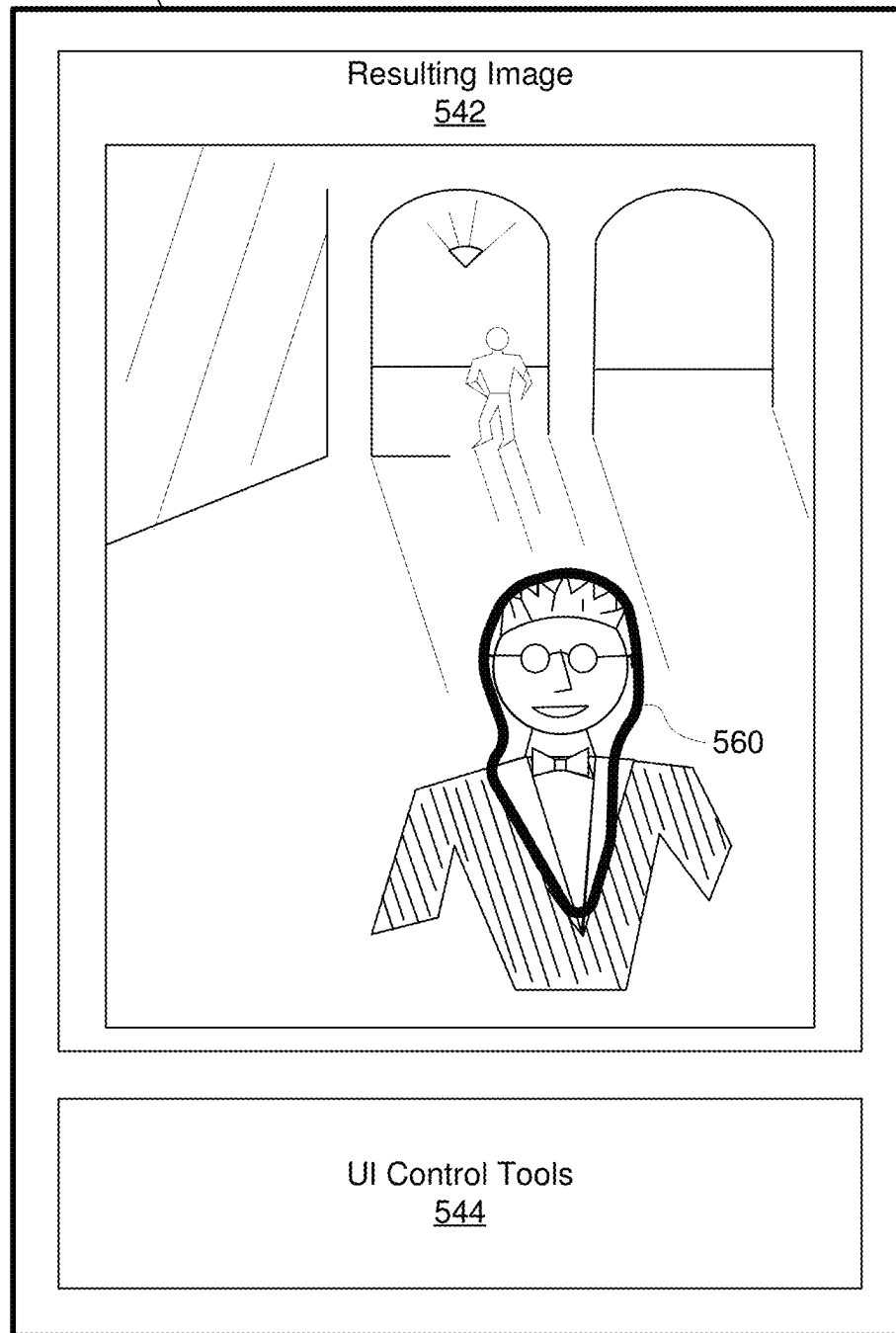


Figure 5D

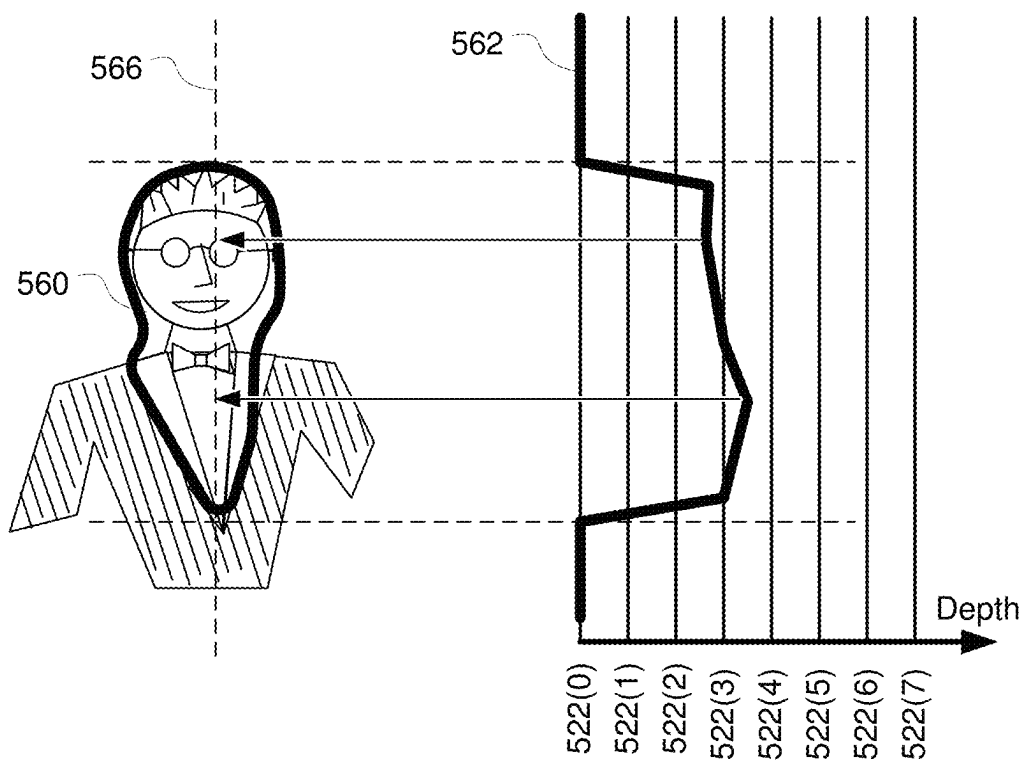


Figure 5E

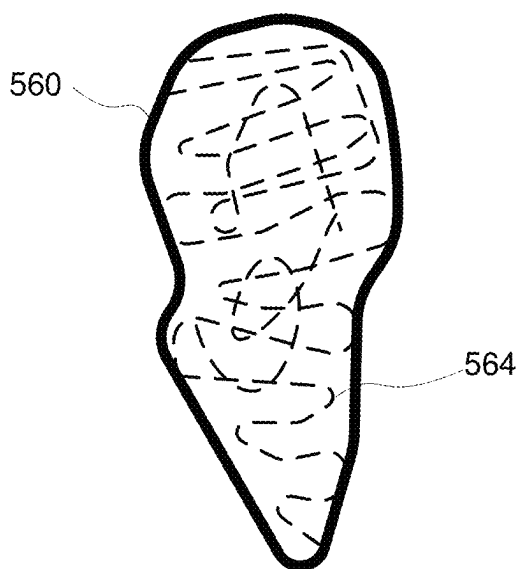


Figure 5F

SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE

RELATED APPLICATIONS

The present application is a continuation in part, by virtue of the removal of subject matter (that was either expressly disclosed or incorporated by reference in one or more priority applications), with the purpose of claiming priority to and including herewith the full express and incorporated disclosure of U.S. patent application Ser. No. 14/543,782, now U.S. Pat. No. 9,509,919, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," filed Nov. 17, 2014, which, at the time of the aforementioned Nov. 17, 2014 filing, incorporated by reference U.S. application Ser. No. 14/178,305, filed Feb. 12, 2014 (abandoned in favor of the above application), entitled "SYSTEMS AND METHOD FOR GENERATING A DIGITAL IMAGE" (which is incorporated by reference in its entirety).

To accomplish the above, the present application is a continuation in part of, and claims priority to, U.S. patent application Ser. No. 18/388,158, entitled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," filed Nov. 8, 2023, which is a continuation-in-part of, and claims priority to U.S. patent application Ser. No. 17/868,536 (abandoned in favor of the above application), entitled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," filed Jul. 19, 2022, which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 17/518,473 (abandoned in favor of the above application), filed Nov. 3, 2021, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 16/677,385, now U.S. Pat. No. 11,394,895, filed Nov. 7, 2019, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 16/211,931, now U.S. Pat. No. 10,491,834, filed Dec. 6, 2018, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," now U.S. Pat. No. 10,491,834, which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 15/863,785, filed Jan. 5, 2018, now U.S. Pat. No. 10,178,323 titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 15/289,039, filed Oct. 7, 2016, now U.S. Pat. No. 9,894,289, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE," which in turn is a continuation of and claims priority to U.S. patent application Ser. No. 14/543,782, filed Nov. 17, 2014, now U.S. Pat. No. 9,509,919, titled "SYSTEM AND METHOD FOR GENERATING A DIGITAL IMAGE."

FIELD OF THE INVENTION

The present invention relates generally to digital photographic systems, and more specifically to a system, method, and computer program product for generating a digital image.

BACKGROUND

Traditional digital photography systems are inherently limited by the dynamic range of a capturing image sensor. One solution to such limitation is the use of high dynamic-range (HDR) photography. HDR photography involves cap-

turing multiple exposures of a same scene, where each of the exposures is exposed differently, and then merging the multiple captures to create an image with a larger dynamic range.

SUMMARY

A system, method, and computer program product for generating a resulting image from a set of images is disclosed. The method comprises receiving an image set that includes a first image of a photographic scene based on a first set of sampling parameters and a second image of the photographic scene based on a second set of sampling parameters, and generating a resulting image based on the first image and the second image according to depth values in a selection depth map. Each distinct depth value in the selection depth map corresponds to a different image in the image set.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates a flow chart of a method for generating a resulting image from an image set comprising two or more images sampled under ambient illumination and/or strobe illumination, in accordance with one embodiment;

FIG. 1B illustrates a flow chart of a method for generating an image set comprising two or more images sampled under strobe illumination, in accordance with one embodiment;

FIG. 1C illustrates a flow chart of a method for selecting one image from an image set based on exposure quality, in accordance with one embodiment;

FIG. 2A illustrates a digital camera, configured to implement one or more aspects of the present invention;

FIG. 2B illustrates a mobile device, configured to implement one or more aspects of the present invention;

FIG. 3A illustrates a digital photographic system, configured to implement one or more aspects of the present invention;

FIG. 3B illustrates a processor complex within the digital photographic system, in accordance with one embodiment;

FIG. 4A illustrates a linear profile for a strobe intensity function associated with sampling an image set, in accordance with one embodiment;

FIG. 4B illustrates a range-limited profile for a strobe intensity function associated with sampling an image set, in accordance with one embodiment;

FIG. 4C illustrates an adaptive profile for a strobe intensity function associated with sampling an image set, in accordance with one embodiment;

FIG. 4D illustrates a range-limited profile for a strobe intensity function and a monotonic profile for an exposure parameter function associated with sampling a set of digital photographs, in accordance with one embodiment;

FIG. 4E illustrates a range-limited profile for a strobe intensity function and a non-monotonic profile for an exposure parameter function associated with sampling a set of digital photographs, in accordance with one embodiment;

FIG. 4F illustrates a constant intensity profile for a strobe intensity function and a monotonic profile for exposure parameter function associated with sampling a set of digital photographs, in accordance with one embodiment;

FIG. 4G illustrates a range-limited profile for a strobe intensity function based on variable width pulses, in accordance with one embodiment;

FIG. 4H illustrates determining a range for a range-limited profile, in accordance with one embodiment;

3

FIG. 5A illustrates a viewer application configured to generate a resulting image based on an image set, in accordance with one embodiment;

FIG. 5B illustrates an exemplary user interface associated with a viewer application, in accordance with one embodiment;

FIG. 5C illustrates a resulting image with differing levels of strobe exposure, in accordance with one embodiment;

FIG. 5D illustrates a resulting image generated from source images and a selection depth map, in accordance with one embodiment;

FIG. 5E illustrates a selection depth map relative to a stack of source images, in accordance with one embodiment; and

FIG. 5F illustrates specifying a selection depth map, in accordance with one embodiment.

DETAILED DESCRIPTION

Embodiments of the present invention enable a digital photographic system to generate a digital image (or simply “image”) of a photographic scene subjected to strobe illumination. Exemplary digital photographic systems include, without limitation, digital cameras and mobile devices such as smart phones that are configured to include a digital camera module and a strobe unit. A given photographic scene is a portion of an overall scene sampled by the digital photographic system.

Two or more images are sequentially sampled by the digital photographic system to generate an image set. Each image within the image set is generated in conjunction with different strobe intensity, different exposure parameters, or a combination thereof. Exposure parameters may include sensor sensitivity (“ISO” parameter), exposure time (shutter speed), aperture size (f-stop), and focus distance. In certain embodiments, one or more exposure parameters, such as aperture size, may be constant and not subject to determination. For example, aperture size may be constant based on a given lens design associated with the digital photographic system. At least two of the images comprising the image set are sampled in conjunction with a strobe unit, such as a light-emitting diode (LED) strobe unit, configured to contribute illumination to the photographic scene.

In one embodiment, exposure parameters are initially determined and held constant for each image in the image set. The exposure parameters may be initially determined based on ambient lighting conditions. If insufficient ambient lighting is available, such as for extremely dark scenes, then exposure parameters may be determined based on a mid-range strobe intensity. For example, mid-range strobe intensity may be selected as fifty-percent of a maximum strobe intensity for the strobe unit. The strobe unit is configured to modulate strobe intensity to provide a range of illumination contribution among the images within the image set. For example, the image set may comprise ten images, each with monotonically increasing illumination from the strobe unit. The first of ten images within the image set may be relatively under-exposed with respect to strobe illumination, while the tenth image may be over-exposed with respect to strobe illumination. However, one or more images between the first image and the tenth image will likely be appropriately-exposed with respect to strobe illumination. An image with appropriate illumination may be selected automatically or manually selected by a user. The image with appropriate illumination is then a resulting image for the photographic scene.

4

In another embodiment, images within the image set are sampled according to varying exposure parameters while the strobe unit is enabled to illuminate the photographic scene. One or more of the exposure parameters may be varied among sequential images comprising the image set. For example, exposure time may be increased (or decreased) among sequential images within the image set. Alternatively, sensor sensitivity may be increased (or decreased) among sequential images within the image set. Furthermore, two or more of the exposure parameters may be varied together among sequential images comprising the image set. For example, exposure time may be decreased while sensor sensitivity is increased. The net effect may preserve overall exposure, while decreasing blur as strobe intensity increases and exposure time decreases. As strobe intensity increases, sensor sensitivity may be increased. Because subject illumination in the photographic scene increases as the strobe intensity increases, increasing sensor sensitivity in sequential images will not likely introduce substantial additional image noise.

Sequential images may be sampled rapidly relative to motion and changes within the photographic scene. For example, the sequential images may be sampled at a rate of at least five sampled per second. In high performance systems, the images may be sampled at greater than fifty samples per second. With an inter-image time of less than two-hundred milliseconds, each image within the image set will depict substantially identical subject matter in common photographic scenarios. In one embodiment, the strobe unit is configured to modulate strobe intensity in time-synchronization with respect to image sampling. For example, the strobe unit may be configured to maintain a specified first strobe intensity during an exposure time for a first image in the image set and maintain a second strobe intensity during an exposure time of a second image in the image set. A transition from the first strobe intensity to the second strobe intensity is synchronized in time with completion of sampling the first image and initiating sampling of the second image.

In one embodiment, an image is selected from the image set based on exposure metrics generated for each image within the image set. In another embodiment, an image is selected by a user. An image may be recommended to the user based on the exposure metrics, but the user may select a different image based on individual choice. The user may view each image in the image set and select the image through a viewer system, configured to present the image set as a sequence of images that may be browsed using a continuous position control, such as a slider control.

FIG. 1A illustrates a flow chart of a method 1 for generating a resulting image from an image set comprising two or more images sampled under ambient illumination and/or strobe illumination, in accordance with one embodiment. Although method 1 is described in conjunction with the systems of FIGS. 2A-3B, persons of ordinary skill in the art will understand that any system that performs method 1 is within the scope and spirit of embodiments of the present invention. In one embodiment, a digital photographic system, such as digital photographic system 300 of FIG. 3A, is configured to perform method 1. The digital photographic system 300 may be implemented within a digital camera, such as digital camera 202 of FIG. 2A, or a mobile device, such as mobile device 204 of FIG. 2B.

Method 1 begins at step 10, where the digital photographic system receives an image set. The image set may include a plurality of images including at least a first image and a second image. In one embodiment, the first image may

5

be captured by the digital photographic system based on a first set of sampling parameters, and the second image may be captured by the digital photographic system based on a second set of sampling parameters. The set of sampling parameters may include, but is not limited to, a shutter speed, an aperture setting, a strobe setting (i.e., a flash setting), a strobe intensity, an image sensor sensitivity (e.g., ISO setting), and the like. The images in the image set may include one or more images captured with ambient light (i.e., without a flash) and/or one or more images captured with strobe illumination (i.e., with a flash).

At step 20, a resulting image is generated based on the first image and the second image according to depth values in a selection depth map. Each distinct depth value in the selection depth map corresponds to a different image in the image set. For example, a first image captured with ambient light may correspond to a minimum value (e.g., distinct value of 0) in the selection depth map. A second image captured with strobe illumination may correspond to a higher value (e.g., distinct value of 1) in the selection depth map. Each different image may be assigned to a distinct depth. Additional images captured with strobe illumination may correspond to even higher distinct values in the selection depth map and so forth. The image set may consist entirely of images captured with ambient illumination (e.g., images captured without flash but with increasing exposure time/decreasing shutter speed, etc.), entirely of images captured with strobe illumination (e.g., images captured with increasing strobe intensity, etc.), or some combination of the two. Exemplary image sets may include tens or hundreds of images. Other image sets may include just two images, or more than two images.

As used herein, images captured with ambient illumination may comprise images captured while a strobe unit is disabled. In contrast, images captured with strobe illumination may comprise images captured while a strobe unit is enabled for at least a portion of the exposure time period such that the captured image data includes color information that includes light from the strobe unit reflected off at least one object in the photographic scene.

In one embodiment, the selection depth map includes a two-dimensional (2D) array of depth values and a location of a particular depth value in the 2D array corresponds to a particular pixel in the resulting image. In other words, the selection depth map may include an array having the same number of elements as the resolution of the images in the image set. The array may have a number of rows equal to the vertical resolution of the images in the image set and a number of columns equal to the horizontal resolution of the images in the image set. Alternatively, the array may have fewer elements than the resolution of each image in the image set and a particular depth value for a pixel in the resulting image is interpolated from elements of the array. For example a particular depth value may be calculated using a bilinear interpolation of array elements.

In another embodiment, the resulting image may be displayed in a user interface associated with a viewer application. For example, a digital photographic system may include an LCD display and a viewer application configured to display a user interface on the LCD display. The viewer application may display a resulting image and enable a user to modify the selection depth map to blend a plurality of images in the image set to generate the resulting image. In one embodiment, the viewer application may detect user input associated with the user interface that indicates a selection path within the resulting image. For example, the user may use an input device such as a mouse or stylus to

6

draw a selection path over the resulting image. Alternatively, if the display device includes a touch-sensitive interface, the user may use touch input or multi-touch input to draw the selection path over the resulting image. The selection path may comprise a set of points relative to the pixel locations of the resulting image. The set of points may be connected to determine a plurality of pixels that intersect with the selection path. The selection path may intersect the same pixel more than once and may cross itself based on the user input.

In one embodiment, a selection region is selected based on the selection path. For example, a surface that includes all points in the selection path may be selected as the selection region. The points may include all points covered by stroking a geometric shape, such as a circle, along the selection path. Further, in another embodiment, a source image in the set of images may be analyzed to find edges in the source image. Any technically feasible edge detection algorithm may be used to find edges in the source image. Once edges are detected in the source image, a surface that includes all points in the selection path may be selected as the selection region, wherein the surface edges correspond to the predetermined edges in the source image. In one embodiment, a nearest edge is selected for each point in the selection path to generate a set of nearest edges in the source image. The set of nearest edges are then analyzed to determine a closed surface that includes all of the edges in the set of nearest edges, where the closed surface is then selected as the selection region. Alternatively, the closed surface may define a selection affinity boundary with a tapering region of influence.

Depth values in the selection depth map corresponding to the selection region are then adjusted based on the selection path. For example, in one embodiment, a depth value for a particular pixel is incremented each time the selection path overlaps with that pixel. In other words, as a user draws the selection path over the resulting image, the depth values in the selection depth map will be accumulated based on the intersection of the selection path with those pixels in the resulting image. In another embodiment, depth values in the selection depth map corresponding to the selection region based on a tapering region of influence around the selection path. In other words, a window function may increase (or decrease) depth values for pixels close to the selection path, but not intersected by the selection path, as well as pixels intersected by the selection path. The amount that a depth value is incremented (or decremented) may be based on the distance of that pixel from the selection path, with pixels closer to or intersected by the selection path being incremented by a greater amount than pixels further away from the selection path. Such adjusting of the selection depth map may be visualized similar to an airbrush tool in a Paint program. In such embodiments, the depth values may be incremented by fractional depth values that indicate a blend (e.g., alpha blend, etc.) between two source images in the image set. In one input mode, depth values are increased, while in another input mode, depth values are decreased.

FIG. 1B illustrates a flow chart of a method 100 for generating an image set comprising two or more images sampled under strobe illumination, in accordance with one embodiment. Although method 100 is described in conjunction with the systems of FIGS. 2A-3B, persons of ordinary skill in the art will understand that any system that performs method 100 is within the scope and spirit of embodiments of the present invention. In one embodiment, a digital photographic system, such as digital photographic system 300 of FIG. 3A, is configured to perform method 100. The digital

photographic system 300 may be implemented within a digital camera, such as digital camera 202 of FIG. 2A, or a mobile device, such as mobile device 204 of FIG. 2B.

Method 100 begins in step 110, where the digital photographic system receives a shutter release command. The shutter release command may be generated explicitly by a user pressing a physical button or virtual button, or the shutter release command may be generated by a timer, motion sensor, voice control, remote control, or any other technically feasible mechanism.

If, in step 112, one or more images should be sampled without strobe illumination, then method 100 proceeds to step 114. In certain configurations, one or more images may be sampled without strobe illumination and added to the image set. An image sampled without strobe illumination is referred to herein as an ambient image. For certain photographic scenes, an ambient image may represent a better choice for a resulting image than other images within the image set sampled with strobe illumination. In certain implementations, whether to sample an ambient image is a design decision that applies to all image sets. In other implementations, a decision to sample an ambient image is made by a user and indicated through a user interface (UI) control. In still other implementations, a decision to sample an ambient image is made automatically based on photographic scene illumination conditions; for example, if the photographic scene is sufficiently dark, then no ambient image is sampled.

In step 114, the digital photographic system configures a camera module, such as camera module 330, to sample an ambient image according to ambient illumination conditions. Configuring the camera module may include determining exposure parameters for a current photographic scene and transmitting the exposure parameters to the camera module. In certain embodiments, the camera module determines exposure parameters for the photographic scene, such as in response to a command to meter a scene. Sampling parameters include the exposure parameters and strobe parameters, which include strobe intensity and may include strobe color. The strobe unit is disabled in this step and corresponding strobe parameters may be set to reflect that the strobe unit is disabled.

In step 116, the digital photographic system samples one or more ambient images based on the sampling parameters. In one embodiment, the one or more ambient images are sampled in conjunction with an exposure parameter sweep, such as an exposure time sweep. The one or more ambient images may be stored within the image set or combined to generate one or more high-dynamic range (HDR) images, which may be stored within the image set. In certain embodiments, the one or more HDR images are stored within the image set, but the one or more ambient images are not.

Returning to step 112, if one or more images should not be sampled without strobe illumination, then the method 100 proceeds to step 120.

If, in step 120, an image should be sampled, then the method 100 proceeds to step 122. A given image set should comprise at least two images sampled with strobe illumination. An image should be sampled if another image is needed to complete a given image set. In one embodiment, the image set comprises a fixed number of images, and another image should be sampled until the fixed number of images has been sampled. In other embodiments, the image set comprises a variable number of images, the number of images being determined adaptively based on exposure of sampled images comprising the image set. For example, the

number of images may continue to increase, up to a maximum number of images, until an image having proper exposure is sampled in conjunction with varying the sampling parameters. Once the image having proper exposure is sampled, another image need not be sampled.

In step 122, the digital photographic system configures the strobe unit based on strobe parameters. Configuring the strobe unit may include enabling the strobe unit to generate illumination according to the strobe parameters. The strobe parameters specify a strobe intensity function that defines strobe intensity as a function corresponding to the images within the image set. In one embodiment, the strobe intensity function defines a predetermined strobe intensity for sequential images within the image set. In another embodiment, the strobe intensity function adaptively generates strobe intensity corresponding to a given image based on at least one previously sampled image or a previously determined exposure.

In certain embodiments, the strobe parameters specify a strobe illumination color, which may be specified as a ratio of red, green, and blue intensity, a color temperature, a color hue, or any other technically feasible color specification. When enabled, the strobe unit is configured to generate strobe illumination according to the specified strobe illumination color. In one embodiment, an ambient illumination color measurement is performed to determine which color to specify as a strobe illumination color. Any technically feasible technique may be implemented to perform the ambient illumination color measurement, including, without limitation, scene color averaging techniques, illuminator color detection techniques, and the like. Measurement data comprising image frames may be collected by a digital camera module. By causing the strobe unit to generate strobe illumination that is consistent in color with ambient illumination, subjects within the photographic scene that are substantially illuminated by the strobe illumination may appear to have proper and consistent color relative to other objects in the scene that are instead substantially illuminated by ambient illumination sources.

In step 124, the digital photographic system configures the camera module to be ready to sample an image according to exposure parameters, which may be determined once and applied to each image within the image set sampled with strobe illumination, or determined per image within the image set based on an exposure parameter function for the image set. Configuring the camera module may include writing registers within the camera module according to the exposure parameters.

The digital camera module includes an image sensor with a sensor sensitivity ranging from low-sensitivity (an ISO value of 100 or less) for generally bright photographic scenes to high-sensitivity (an ISO value of 1600 or more) for generally dark photographic scenes. Exposure time may range from one millisecond or less to more than a second. Determining exposure parameters typically comprises finding a combination of at least sensor sensitivity and exposure time that is estimated to be appropriate for a given intensity distribution, absolute overall scene brightness, or any combination thereof. The camera module may adaptively sample multiple image frames in finding the combination. As sensitivity is increased and/or exposure time is increased, image noise also increases. Image quality degradation due to noise typically establishes a combination of maximum exposure time and maximum sensitivity that should not be exceeded. If sufficient ambient illumination is not available, the digital camera module is unable to sample a usable image, and in some cases may simply generate a dark noise field. A certain

implementation or design of a digital camera module may have better low-light performance than another implementation or design, and therefore a determination of whether sufficient ambient illumination is available is implementation-dependent.

In one embodiment, the exposure parameters are determined based on ambient illumination. Any technically feasible technique may be implemented to determine the exposure parameters. Persons skilled in the art will understand that such techniques are well-known in the art and, in certain scenarios, depend on a specific implementation of a digital photographic system.

In another embodiment, the exposure parameters are determined based on ambient illumination if sufficient ambient illumination is available or determined based on metering strobe illumination if sufficient ambient illumination is not available. The metering strobe illumination is provided by the strobe unit (or a similar unit), configured to provide mid-range strobe intensity while the exposure parameters are being determined. Alternatively, the strobe unit may be configured to provide a range of strobe intensity to provide a larger search space for determining exposure parameters. Any technically feasible technique may be implemented to determine whether sufficient ambient illumination is available, including, but not limited to, techniques that are associated with a particular implementation of a given digital camera module.

In step 126, the digital photographic system causes the digital camera module to sample an image based on current sampling parameters. In certain embodiments, a new set of sampling parameters may be determined based on the sampled image to be applied to sampling a subsequent image. For example, if the sampled image is under exposed, the new set of sampling parameters may provide for increased exposure time or an increased ISO value.

In step 128, the digital photographic system stores the image in the image set. The image set may be stored within NV memory 316, volatile memory 318, or the image set may be stored to a remote storage system, such as through wireless unit 340. Each image within the image set may be referred to herein as a source image.

Returning to step 120, if the last image for the image set has been sampled, then the method 100 proceeds to step 132, where the digital photographic system disables the strobe unit. In step 134, the digital photographic system evaluates images comprising the image set to select a recommended image from the image set having appropriate exposure. In one embodiment, step 134 comprises method 102 of FIG. 1C. In step 136, the digital photographic system displays the recommended image. As discussed below in FIGS. 5A through 5C, a user may select a different image within the image set than the recommended image.

In an alternative embodiment, an exposure quality metric, described below in FIG. 1C, is evaluated and stored in step 128 for a corresponding image sampled in step 126. In step 134, a recommended image is selected from the image set based on at least one exposure quality metric associated with each image in the image set. For example, if the exposure quality metric is defined to be a cost function that assigns an increasing cost penalty to increasingly undesirable exposures, then the recommended image is selected to have the lowest corresponding cost penalty (exposure quality metric) among the images in the image set.

FIG. 1C illustrates a flow chart of a method 102 for selecting one image from an image set based on exposure quality, in accordance with one embodiment. Although method 102 is described in conjunction with the systems of

FIGS. 2A-3B, persons of ordinary skill in the art will understand that any system that performs method 102 is within the scope and spirit of embodiments of the present invention. In one embodiment, a digital photographic system, such as digital photographic system 300 of FIG. 3A, is configured to perform method 102. The digital photographic system may be implemented within a digital camera, such as digital camera 202 of FIG. 2A, or a mobile device, such as mobile device 204 of FIG. 2B.

Method 102 begins in step 150, where the digital photographic system selects an image in the image set of FIG. 1C. In one embodiment, the image set is organized as an ordered sequence of images, and the selected image comprises a first image of the ordered sequence of images. In step 152, the digital photographic system evaluates exposure quality of the selected image. In one embodiment, a cost function that assigns a high cost penalty to over-exposed pixels within the selected image may be implemented to evaluate an exposure quality metric. The cost function may assign a constant cost penalty to each over-exposed pixel or a progressively increasing cost function based on a count for over-exposed pixels within the selected image. Any other technically feasible technique for evaluating an exposure quality metric may also be implemented without departing the scope and spirit of the present invention.

In step 154, the digital photographic system stores the exposure quality metric associated with the selected image. If, in step 160, the selected image is not the last image within the image set to be selected and evaluated, then the method proceeds to step 162, where the digital photographic system selects a next image in the image set before proceeding back to step 152.

Returning to step 160, if the selected image is the last image within the image set to be selected and evaluated, then the method proceeds to step 164, where the digital photographic system selects an image within the image set having the best exposure quality metric. For example, if the exposure quality metric is defined to be a cost function configured to assign an increasingly higher cost to increasingly undesirable exposures, then an image is selected to have the lowest corresponding cost (exposure quality metric) among the images in the image set. The selected image in this step may comprise a recommended image.

FIG. 2A illustrates a digital camera 202, configured to implement one or more aspects of the present invention. Digital camera 202 includes a digital photographic system, such as digital photographic system 300 of FIG. 3A, configured to generate an image set by sampling a photographic scene as described in conjunction with method 100 of FIG. 1B. A digital camera unit within the digital photographic system is coupled to a lens 230, through which each image comprising the image set is sampled.

Digital camera 202 includes a strobe unit 236, and may include a shutter release button 215 for triggering a photographic sample event, whereby digital camera 202 samples two or more images comprising an image set. Any other technically feasible shutter release command may trigger the photographic sample event, such as a timer trigger or remote control receiver configured to generate a shutter release command. Embodiments of the present invention advantageously enable a user to photograph a scene using a single shutter release command, and subsequently select an image sampled according to a strobe intensity that best satisfies user aesthetic requirements for the photographic scene. In contrast, a conventional digital camera typically samples a single image illuminated by strobe illumination per shutter release trigger, commonly forcing the user to either manu-

11

ally photograph the photographic scene multiple times, or to suffer poor image quality resulting over-exposure or under-exposure.

FIG. 2B illustrates a mobile device **204**, configured to implement one or more aspects of the present invention. Mobile device **204** includes a digital photographic system, such as digital photographic system **300** of FIG. 3A, configured to generate an image set by sampling a scene as described in conjunction with method **100** of FIG. 1B. A shutter release command may be generated through a mechanical button or a virtual button, which may be activated by a touch gesture on a touch entry display system **240** within mobile device **204**.

In one embodiment, the touch entry display system **240** is disposed on the opposite side of mobile device **204** relative to the lens **230**. In certain embodiments, the mobile device **204** includes a user-facing digital camera coupled to lens **232** and a user-facing strobe unit. The user-facing digital camera and user-facing strobe unit are configured to sample an image set in accordance with method **100** of FIG. 1B.

FIG. 3A illustrates a digital photographic system **300**, configured to implement one or more aspects of the present invention. Digital photographic system **300** includes a processor complex **310** coupled to a camera module **330** and a strobe unit **336**. Digital photographic system **300** may also include, without limitation, a display unit **312**, a set of input/output devices **314**, non-volatile memory **316**, volatile memory **318**, a wireless unit **340**, and sensor devices **342**, each coupled to processor complex **310**. In one embodiment, a power management subsystem **320** is configured to generate appropriate power supply voltages for each electrical load element within digital photographic system **300**. A battery **322** may be configured to supply electrical energy to power management subsystem **320**. Battery **322** may implement any technically feasible energy storage system, including primary or rechargeable battery technologies. In one embodiment, lens **230** of FIGS. 2A and 2B is coupled to camera module **330**, and strobe unit **236** comprises strobe unit **336**.

In one embodiment, strobe unit **336** is integrated into digital photographic system **300** and configured to provide strobe illumination **350** during an image sample event performed by digital photographic system **300**. In an alternative embodiment, strobe unit **336** is implemented as an independent device from digital photographic system **300** and configured to provide strobe illumination **350** during an image sample event performed by digital photographic system **300**. Strobe unit **336** may comprise one or more LED devices. In certain embodiments, two or more strobe units are configured to synchronously generate strobe illumination in conjunction with sampling an image.

In one embodiment, strobe unit **336** is directed through a strobe control signal **338** to either emit strobe illumination **350** or not emit strobe illumination **350**. The strobe control signal **338** may implement any technically feasible signal transmission protocol. Strobe control signal **338** may indicate a strobe parameter, such as strobe intensity or strobe color, for directing strobe unit **336** to generate a specified intensity and/or color of strobe illumination **350**. As shown, strobe control signal **338** may be generated by processor complex **310**. Alternatively, strobe control signal **338** may be generated by camera module **330** or by any other technically feasible system element.

In one usage scenario, strobe illumination **350** comprises at least a portion of overall illumination in a photographic scene being photographed by camera module **330**. Optical scene information **352**, which may include strobe illumina-

12

tion **350** reflected from objects in the photographic scene, is focused as an optical image onto an image sensor **332**, within camera module **330**. Image sensor **332** generates an electronic representation of the optical image. The electronic representation comprises spatial color intensity information, which may include different color intensity samples, such as for red, green, and blue light. The spatial color intensity information may also include samples for white light. Alternatively, the color intensity samples may include spatial color intensity information for cyan, magenta, and yellow light. Persons skilled in the art will recognize that other and further sets of spatial color intensity information may be implemented. The electronic representation is transmitted to processor complex **310** via interconnect **334**, which may implement any technically feasible signal transmission protocol.

Input/output devices **314** may include, without limitation, a capacitive touch input surface, a resistive tablet input surface, one or more buttons, one or more knobs, light-emitting devices, light detecting devices, sound emitting devices, sound detecting devices, or any other technically feasible device for receiving user input and converting the input to electrical signals, or converting electrical signals into a physical signal. In one embodiment, input/output devices **314** include a capacitive touch input surface coupled to display unit **312**.

Non-volatile (NV) memory **316** is configured to store data when power is interrupted. In one embodiment, NV memory **316** comprises one or more flash memory devices. NV memory **316** may be configured to include programming instructions for execution by one or more processing units within processor complex **310**. The programming instructions may implement, without limitation, an operating system (OS), UI modules, image processing and storage modules, one or more modules for sampling an image set through camera module **330**, one or more modules for presenting the image set through display unit **312**. The programming instructions may also implement one or more modules for merging images or portions of images within the image set, aligning at least portions of each image within the image set, or a combination thereof. One or more memory devices comprising NV memory **316** may be packaged as a module configured to be installed or removed by a user. In one embodiment, volatile memory **318** comprises dynamic random access memory (DRAM) configured to temporarily store programming instructions, image data such as data associated with an image set, and the like, accessed during the course of normal operation of digital photographic system **300**.

Sensor devices **342** may include, without limitation, an accelerometer to detect motion and/or orientation, an electronic gyroscope to detect motion and/or orientation, a magnetic flux detector to detect orientation, a global positioning system (GPS) module to detect geographic position, or any combination thereof.

Wireless unit **340** may include one or more digital radios configured to send and receive digital data. In particular, wireless unit **340** may implement wireless standards known in the art as "WiFi" based on Institute for Electrical and Electronics Engineers (IEEE) standard 802.11, and may implement digital cellular telephony standards for data communication such as the well-known "3G" and "4G" suites of standards. Wireless unit **340** may further implement standards and protocols known in the art as LTE (long term evolution). In one embodiment, digital photographic system **300** is configured to transmit one or more digital photographs, sampled according to techniques taught herein, to an

13

online or “cloud-based” photographic media service via wireless unit **340**. The one or more digital photographs may reside within either NV memory **316** or volatile memory **318**. In such a scenario, a user may possess credentials to access the online photographic media service and to transmit the one or more digital photographs for storage and presentation by the online photographic media service. The credentials may be stored or generated within digital photographic system **300** prior to transmission of the digital photographs. The online photographic media service may comprise a social networking service, photograph sharing service, or any other network-based service that provides storage and transmission of digital photographs. In certain embodiments, one or more digital photographs are generated by the online photographic media service based on an image set sampled according to techniques taught herein. In such embodiments, a user may upload source images comprising an image set for processing by the online photographic media service.

In one embodiment, digital photographic system **300** comprises a plurality of camera modules **330**. Such an embodiment also includes at least one strobe unit **336** configured to illuminate a photographic scene, sampled as multiple views by the plurality of camera modules **330**. The plurality of camera modules **330** may be configured to sample a wide angle view (greater than forty-five degrees of sweep among cameras) to generate a panoramic photograph. The plurality of camera modules **330** may also be configured to sample two or more narrow angle views (less than forty-five degrees of sweep among cameras) to generate a stereoscopic photograph.

Display unit **312** is configured to display a two-dimensional array of pixels to form an image for display. Display unit **312** may comprise a liquid-crystal display, an organic LED display, or any other technically feasible type of display. In certain embodiments, display unit **312** is able to display a narrower dynamic range of image intensity values than a complete range of intensity values sampled over a set of two or more images comprising the image set. Here, images comprising the image set may be merged according to any technically feasible HDR blending technique to generate a synthetic image for display within dynamic range constraints of display unit **312**. In one embodiment, the limited dynamic range specifies an eight-bit per color channel binary representation of corresponding color intensities. In other embodiments, the limited dynamic range specifies a twelve-bit per color channel binary representation.

FIG. 3B illustrates a processor complex **310** within digital photographic system **300** of FIG. 3A, according to one embodiment of the present invention. Processor complex **310** includes a processor subsystem **360** and may include a memory subsystem **362**. In one embodiment, processor complex **310** comprises a system on a chip (SoC) device that implements processor subsystem **360**, and memory subsystem **362** comprising one or more DRAM devices coupled to processor subsystem **360**. In one implementation of the embodiment, processor complex **310** comprises a multi-chip module (MCM) encapsulating the SoC device and the one or more DRAM devices.

Processor subsystem **360** may include, without limitation, one or more central processing unit (CPU) cores **370**, a memory interface **380**, input/output interfaces unit **384**, and a display interface unit **382**, each coupled to an interconnect **374**. The one or more CPU cores **370** may be configured to execute instructions residing within memory subsystem **362**, volatile memory **318**, NV memory **316**, or any combination thereof. Each of the one or more CPU cores **370** may be

14

configured to retrieve and store data via interconnect **374** and memory interface **380**. Each of the one or more CPU cores **370** may include a data cache, and an instruction cache. Two or more CPU cores **370** may share a data cache, an instruction cache, or any combination thereof. In one embodiment, a cache hierarchy is implemented to provide each CPU core **370** with a private cache layer, and a shared cache layer.

Processor subsystem **360** may further include one or more graphics processing unit (GPU) cores **372**. Each GPU core **372** comprises a plurality of multi-threaded execution units that may be programmed to implement graphics acceleration functions. GPU cores **372** may be configured to execute multiple thread programs according to well-known standards such as OpenGL™, OpenCL™, CUDA™, and the like. In certain embodiments, at least one GPU core **372** implements at least a portion of a motion estimation function, such as a well-known Harris detector or a well-known Hessian-Laplace detector. Such a motion estimation function may be used for aligning images or portions of images within the image set.

Interconnect **374** is configured to transmit data between and among memory interface **380**, display interface unit **382**, input/output interfaces unit **384**, CPU cores **370**, and GPU cores **372**. Interconnect **374** may implement one or more buses, one or more rings, a cross-bar, a mesh, or any other technically feasible data transmission structure or technique. Memory interface **380** is configured to couple memory subsystem **362** to interconnect **374**. Memory interface **380** may also couple NV memory **316**, volatile memory **318**, or any combination thereof to interconnect **374**. Display interface unit **382** is configured to couple display unit **312** to interconnect **374**. Display interface unit **382** may implement certain frame buffer functions such as frame refresh. Alternatively, display unit **312** may implement frame refresh. Input/output interfaces unit **384** is configured to couple various input/output devices to interconnect **374**.

In certain embodiments, camera module **330** is configured to store exposure parameters for sampling each image in an image set. When directed to sample an image set, the camera module **330** samples the image set according to the stored exposure parameters. A software module executing within processor complex **310** may generate and store the exposure parameters prior to directing the camera module **330** to sample the image set.

In other embodiments, camera module **330** is configured to store exposure parameters for sampling an image in an image set, and the camera interface unit **386** within the processor complex **310** is configured to cause the camera module **330** to first store exposure parameters for a given image comprising the image set, and to subsequently sample the image. In one embodiment, exposure parameters associated with images comprising the image set are stored within a parameter data structure. The camera interface unit **386** is configured to read exposure parameters from the parameter data structure for a given image to be sampled, and to transmit the exposure parameters to the camera module **330** in preparation of sampling an image. After the camera module **330** is configured according to the exposure parameters, the camera interface unit **386** directs the camera module **330** to sample an image. Each image within an image set may be sampled in this way. The data structure may be stored within the camera interface unit **386**, within a memory circuit within processor complex **310**, within volatile memory **318**, within NV memory **316**, or within any

15

other technically feasible memory circuit. A software module executing within processor complex 310 may generate and store the data structure.

In one embodiment, the camera interface unit 386 transmits exposure parameters and commands to camera module 330 through interconnect 334. In certain embodiments, the camera interface unit 386 is configured to directly control the strobe unit 336 by transmitting control commands to the strobe unit 336 through strobe control signal 338. By directly controlling both the camera module 330 and the strobe unit 336, the camera interface unit 386 may cause the camera module 330 and the strobe unit 336 to perform their respective operations in precise time synchronization. In this context, synchronization applies to coordinating at least method steps 122, 124, and 126 of method 100. That is, synchronizing the steps of configuring the camera module 330 prior to sampling an image, configuring the strobe unit 336 to generate appropriate strobe illumination, and directing the camera module 330 to sample a photographic scene subjected to strobe illumination.

Additional set-up time or execution time associated with each step may reduce overall sampling performance. Therefore, a dedicated control circuit, such as the camera interface unit 386, may be implemented to substantially minimize set-up and execution time associated with each step and any intervening time between steps.

In other embodiments, a software module executing within processor complex 310 directs the operation and synchronization of camera module 330 and the strobe unit 336, with potentially reduced performance.

In one embodiment, camera interface unit 386 is configured to accumulate statistics while receiving image data from the camera module 330. In particular, the camera interface unit 386 may accumulate exposure statistics for a given image while receiving image data for the image through interconnect 334. Exposure statistics may include an intensity histogram, a count of over-exposed pixels, a counter of under-exposed pixels, an intensity-weighted sum of pixel intensity, or any combination thereof. The camera interface unit 386 may present the exposure statistics as memory-mapped storage locations within a physical or virtual address space defined by a processor, such as a CPU core 370, within processor complex 310.

In certain embodiments, camera interface unit 386 accumulates color statistics for estimating scene white-balance. Any technically feasible color statistics may be accumulated for estimating white balance, such as a sum of intensities for different color channels comprising red, green, and blue color channels. The sum of color channel intensities may then be used to perform a white-balance color correction on an associated image, according to a white-balance model such as a gray-world white-balance model. In other embodiments, curve-fitting statistics are accumulated for a linear or a quadratic curve fit used for implementing white-balance correction on an image. In one embodiment, camera interface unit 386 accumulates spatial color statistics for performing color-matching between or among images, such as between or among an ambient image and one or more images sampled with strobe illumination. As with the exposure statistics, the color statistics may be presented as memory-mapped storage locations within processor complex 310.

In one embodiment, camera module 330 transmits strobe control signal 338 to strobe unit 336, enabling strobe unit 336 to generate illumination while the camera module 330 is sampling an image. In another embodiment, camera module 330 samples an image illuminated by strobe unit 336

16

upon receiving an indication from camera interface unit 386 that strobe unit 336 is enabled. In yet another embodiment, camera module 330 samples an image illuminated by strobe unit 336 upon detecting strobe illumination within a photographic scene via a rapid rise in scene illumination.

FIG. 4A illustrates a linear profile for a strobe intensity function 420 associated with sampling an image set, in accordance with one embodiment. The strobe intensity function 420 represents average strobe intensity during an exposure time for an image sampled within a corresponding time interval 410. Strobe intensity may vary between off 430 and a maximum intensity 438. The maximum intensity 438 is a characteristic of a given implementation of a strobe unit, such as strobe unit 336 of FIG. 3A. Strobe intensity function 420 is depicted herein as scaled according to the maximum intensity 438.

As shown, the strobe intensity function 420 includes a set of increasing strobe intensity values up to and including maximum intensity 438. An image comprising the image set may be sampled during each time interval 410 under strobe illumination. An image, such as an ambient image, may also be sampled during a time interval 408 prior to time interval 410(0). In one embodiment, sequential images comprising the image set are sampled in sequential time intervals 410. While each time interval 408, 410 is shown to be of substantially equal duration, each time interval 408, 410 may vary.

In one embodiment, steps 122, 124, 126, and 128 of method 100 are performed for each time interval 410. Steps 132, 134, and 136 may be performed subsequent to time interval 410(7), or more generally, subsequent to a final time interval for a given image set.

FIG. 4B illustrates a range-limited profile for a strobe intensity function 420 associated with sampling an image set, in accordance with one embodiment. As shown, the range-limited profile modifies the strobe intensity function 420 relative to FIG. 4A by limiting the total range of strobe intensity between a low intensity value 432 and a high intensity value 434. By limiting the strobe intensity range, finer strobe intensity resolution within a range estimated to be appropriate for a given photographic scene may be available within images comprising the image set. The low intensity value 432 and the high intensity value 434 may be determined using any technically feasible technique, such as through a metering-strobe technique illustrated below in FIG. 4H. In certain operating modes, such as a portrait photographic mode, the high intensity value 434 is constrained based on ambient illumination within a photographic scene. Constraining the high intensity value 434 based on ambient illumination may avoid excessive strobe intensity, which may be unpleasant to persons being subjected to the strobe illumination. The low intensity value 432 may then be determined to provide an appropriate range of strobe intensity for images within the image set.

FIG. 4C illustrates an adaptive profile for a strobe intensity function 420 associated with sampling an image set, in accordance with one embodiment. As shown, the adaptive profile modifies the strobe intensity function 420 relative to FIG. 4A by adaptively determining a subsequent strobe intensity value based on at least one previously sampled image. By adaptively determining a subsequent strobe intensity value for sequentially sampled images, a final image may be sampled having a target intensity value 436 that provides an estimated proper strobe exposure for the final image. In one embodiment, adaptively determining a subsequent strobe intensity value is performed during step 122 of method 100.

Estimating proper strobe exposure may be performed using any technically feasible techniques, including techniques that attempt to minimize over-exposed pixels while maximizing the strobe intensity. As shown, the strobe intensity function 420 represent an exemplary binary-style search performed to minimize over-exposed pixels in sequential time intervals 410 to converge on a target intensity value 436 that provides a minimum of over-exposed pixels while maximizing strobe intensity.

In certain embodiments, camera interface unit 386 is configured to generate exposure statistics, which are analyzed to adaptively determine a subsequent strobe intensity. For example, strobe intensity function 420 may follow a binary search profile, with strobe intensity modulated based on a count of over-exposed pixels and/or under-exposed pixels. Alternatively, the strobe intensity may be modulated according to the binary search profile based on an intensity histogram, which indicates general over-exposure, general under-exposure, and median intensity.

In one embodiment, images comprising the image set are ordered sequentially in sorted order according to strobe intensity. The images within the image set may then be viewed according to a monotonic strobe intensity, which may provide a more intuitive depiction of the strobe intensity range. Any technically feasible technique may be implemented to order the images according to monotonic strobe intensity. For example, image order within the image set may be represented as a sorted list of strobe intensity, whereby each element of the sorted list includes a reference to a corresponding image. Sequentially referencing images within the image set may therefore be performed by sequentially accessing images referred to by list elements. In one embodiment, certain images that are excessively over-exposed or under-exposed may be discarded from the image set. In general, images within a given image set may be ordered according to a monotonic parameter sequence, such as an increasing strobe intensity sequence, regardless of a particular profile for strobe intensity function 420.

FIG. 4D illustrates a range-limited profile for a strobe intensity function 420 and a monotonic profile for an exposure parameter function 422 associated with sampling a set of digital photographs, in accordance with one embodiment. As shown, the range-limited profile for strobe intensity function 420 of FIG. 4B is executed in conjunction with exposure parameter function 422. The exposure parameter function 422 may represent any exposure parameter over time intervals 408 and 410. In one embodiment, the exposure parameter function 422 specifies exposure time varying between a high value 444 (longer exposure) to a low value 442 (shorter exposure).

The high value 444 and the low value 442 may each be selected using any technically feasible technique. In one embodiment, the high value 444 is selected to over-expose an ambient photographic scene, metered during time interval 408(0), by a predetermined amount. For example, the high value 444 may be selected over-expose the ambient photographic scene by one conventional photographic exposure stop. Similarly, the low value 442 may be selected to under-expose the ambient photographic scene by a predetermined amount, such as one conventional photographic exposure stop. Regions within the photographic scene predominantly illuminated by ambient illumination will be over-exposed in time interval 410(0), under-exposed in time interval 410(7), and properly-exposed mid-way between time intervals 410(0) and 410(7). Sequential images within the image set corresponding to time intervals 410(0) through

410(7) will advantageously provide a range of exposure variation for both ambient-illuminated regions and strobe-illuminated regions.

In certain embodiments, two exposure parameters may be varied together in sequential time intervals. For example, exposure time may be reduced, while increasing sensor sensitivity to provide a trade-off between sensor image noise and blur due to motion in sequential images comprising the image set.

FIG. 4E illustrates a range-limited profile for a strobe intensity function 420 and a non-monotonic profile for an exposure parameter function 422 associated with sampling a set of digital photographs, in accordance with one embodiment. As shown, the range-limited profile for strobe intensity function 420 and exposure parameter function 422 of FIG. 4D are both modified to provide a two-dimensional sampling of both strobe intensity and an exposure parameter. For each strobe intensity value ranging between low intensity value 432 and high intensity value 434, an image is sampled over each of four different exposure parameter values, as indicated by exposure parameter function 422. Images within the image set may be displayed based on a combination of strobe intensity function 420 and exposure parameter function 422, each separately selected. In this way, an image may be selected from the image set based on substantially separately varying strobe exposure and ambient exposure.

FIG. 4F illustrates a constant intensity profile for a strobe intensity function 420 and a monotonic profile for exposure parameter function 422 associated with sampling a set of digital photographs, in accordance with one embodiment. As shown, the strobe intensity profile indicates a substantially constant strobe intensity of target 436, which may be determined using any technically feasible technique to sample images having proper exposure for regions within a photographic scene predominately illuminated by strobe illumination. The value of target 436 may be determined for an exposure parameter between low value 442 and high value 444 to ensure at least one of the images sampled during time intervals 410 may have proper exposure with respect to strobe illumination.

FIG. 4G illustrates a range-limited profile for a strobe intensity function 420 based on variable width pulses, in accordance with one embodiment. An instantaneous intensity 424 illustrates instantaneous strobe intensity as a function of time, while the strobe intensity function 420 represents average strobe intensity during an exposure time for an image sampled within a corresponding time interval 410. Average strobe intensity corresponds to total accumulated instantaneous intensity 424 during a corresponding time interval 410. As shown, the value of strobe intensity function 420 for a given time interval 410 is proportional to a pulse width of a pulse having pulse intensity 439. In one embodiment, each pulse comprising the instantaneous intensity 424 is positioned in time to be within an exposure time for a corresponding image being sampled.

In one embodiment, the strobe unit 336 of FIG. 3A comprises a gas-discharge strobe, such as a xenon tube strobe, and pulse intensity 439 represents the instantaneous intensity of the gas-discharge strobe while actively firing. In one embodiment, time interval 410 represents less than a fifty-millisecond interval.

While a single pulse is shown here in conjunction with achieving a particular average strobe intensity, alternative embodiments may implement strobe intensity averaging over multiple pulses per time interval 410. For example, an LED illuminator device may implement pulse-width modu-

19

lation (PWM), or other average energy modulation techniques, to achieve a specified average strobe intensity. In such embodiments, multiple pulses should be generated during an associated exposure time. In general, an arbitrary function for instantaneous intensity **424** may be implemented to achieve an average strobe intensity during a given associated exposure time during a corresponding time interval **410**. For example, a smooth ramp may be implemented as a function for instantaneous intensity **424**, the smooth ramp providing increasing average strobe intensity for during sequential, equal time intervals **410**. In certain embodiments, strobe intensity is controlled by controlling a current source configured to drive an illuminator device. In other embodiments, strobe intensity is controlled using a feedback loop that adjusts current driving the illuminator device to match measured optical strobe intensity generated by the strobe unit.

FIG. **4H** illustrates determining an intensity range for a range-limited profile, in accordance with one embodiment. Determining the range comprises sampling strobe exposure for a photographic scene subjected strobe illumination during metering-strobe time intervals **460**. Strobe intensity is varied during time intervals **460** to estimate the low intensity value **432**, below which the photographic scene is underexposed with respect to strobe illumination, and the high value, above which the photographic scene is over-exposed with respect to strobe illumination.

In one embodiment, the maximum intensity **438** is constrained based on ambient illumination rather than strobe unit capacity. In darker settings, the maximum intensity **438** is constrained to an intensity that a person will not find unpleasant. In brighter settings, the maximum intensity **438** may instead be constrained by strobe unit capacity.

In certain embodiments, a digital photographic system is configured to continuously sample ambient images within a circular buffer, and to identify a subset of the ambient images within the circular buffer to be stored as identified ambient images within the image set based on time proximity to a shutter release command. Upon receiving the shutter release command, the identified ambient images are associated with time intervals **408** of FIGS. **4A** through **4H**, as though sampled according to steps **110** through **116** of method **100** of FIG. **1B**. Method **100** may proceed to step **120** upon identifying the ambient images.

FIG. **5A** illustrates a viewer application **510** configured to generate a resulting image **542** based on an image set **520**, in accordance with one embodiment. The image set **520** includes two or more source images **522**, which may be generated by performing method **100** of FIG. **1B**.

In one embodiment, the resulting image **542** represents a source image **522** that is selected from the image set **520**. The source image **522** may be selected according to any technically feasible technique. For example, a given source image **522** may be selected automatically based on exposure quality, as described previously in method **100** and in method **102** of FIG. **1C**. Alternatively, a given source image **522** may be selected manually through a UI control **530**, discussed in greater detail below in FIG. **5B**. The UI control **530** generates a selection parameter **518** that indicates the manual selection. An image processing subsystem **512** is configured to generate the resulting image **542** by either instantiating the selected source image **522** or marking the selected source image **522** within the image set **520** as the resulting image **542**. In certain embodiments, the image processing subsystem **512** automatically selects a source image and transmits a corresponding recommendation **519** to the UI control **530**. The recommendation **519** indicates,

20

through the UI control **530**, which source image was automatically selected. A user may keep the recommendation or select a different source image to be the resulting image **542** using the UI control **530**.

In an alternative embodiment, viewer application **510** is configured to combine two or more source images **522** to generate a resulting image **542**. The two or more source images **522** may be mutually aligned by the image processing subsystem **512** prior to being combined. Selection parameter **518** may include a weight assigned to each of two source images **522**. The weight may be used to perform a transparency/opacity blend (known as an alpha blend) between two source images **522**.

In certain embodiments, source image **522(0)** is sampled under exclusively ambient illumination, with the strobe unit off. Source image **522(0)** is generated to be white-balanced, according to any technically feasible white balancing technique. Source images **522(1)** through **522(N-1)** are sampled under strobe illumination, which may be of a color that is discordant with respect to ambient illumination. Source images **522(1)** through **522(N-1)** may be white-balanced according to the strobe illumination color. Discordance in strobe illumination color may cause certain regions to appear incorrectly colored with respect to other regions in common photographic settings. For example, in a photographic scene with foreground subjects predominantly illuminated by white strobe illumination and white-balanced accordingly, background subjects that are predominantly illuminated by incandescent lights may appear excessively orange or even red. In one embodiment, spatial color correction is implemented within image processing subsystem **512** to match the color of regions within a selected source image **522** to that of source image **522(0)**. Spatial color correction implements regional color-matching to ambient-illuminated source image **522(0)**. The regions may range in overall scene coverage from individual pixels, to blocks of pixels, to whole frames. In one embodiment, each pixel in a color-corrected image includes a weighted color correction contribution from at least a corresponding pixel and an associated block of pixels.

In certain implementations, viewer application **510** includes an image cache **516**, configured to include a set of cached images corresponding to the source images **522**, but rendered to a lower resolution than source images **522**. The image cache **516** provides images that may be used to readily and efficiently generate or display resulting image **542** in response to real-time changes to selection parameter **518**. In one embodiment, the cached images are rendered to a screen resolution of display unit **312**. When a user manipulates the UI control **530** to select a source image **522**, a corresponding cached image may be displayed on the display unit **312**. Caching source images **522** may advantageously reduce power consumption associated with rendering a given source image **522** for display. Caching source images **522** may also improve performance by eliminating a rendering process needed to resize a given source image **522** for display each time UI control **530** detects that a user has selected a different source image **522**.

FIG. **5B** illustrates an exemplary user interface associated with the viewer application **510** of FIG. **5A**, in accordance with one embodiment. The user interface comprises an application window **540** configured to display the resulting image **542** based on a position of the UI control **530**. The viewer application **510** may invoke the UI control **530**, configured to generate the selection parameter **518** based on a position of a control knob **534**. The recommendation **519** may determine an initial position of the control knob **534**,

21

corresponding to a recommended source image **522** within the image set **520**. In one embodiment, the UI control **530** comprises a linear slider control with a control knob **534** configured to slide along a slide path **532**. A user may position the control knob **534** by performing a slide gesture. For example, the slide gesture may include touching the control knob **534** in a current position, and sliding the control knob **534** to a new position. Alternatively, the user may touch along the slide path **532** to move the control knob **534** to a new position defined by a location of the touch.

In one embodiment, positioning the control knob **534** into a discrete position **536** along the slide path **532** causes the selection parameter **518** to indicate selection of a corresponding source image **522**. For example, a user may move control knob **534** into discrete position **536(3)**, to indicate that source image **522(3)** is selected. The UI control **530** then generates selection parameter **518** to indicate that source image **522(3)** is selected. The image processing subsystem **512** responds to the selection parameter **518** by generating the resulting image **542** based on source image **522(3)**. The control knob **534** may be configured to snap to a closest discrete position **536** when released by a user withdrawing their finger.

In an alternative embodiment, the control knob **534** may be positioned between two discrete positions **536** to indicate that resulting image **542** should be generated based on two source images **522** corresponding to the two discrete positions **536**. For example, if the control knob **534** is positioned between discrete position **536(3)** and discrete position **536(4)**, then the image processing subsystem **512** generates resulting image **542** from source images **522(3)** and **522(4)**. In one embodiment, the image processing subsystem **512** generates resulting image **542** by aligning source images **522(3)** and **522(4)**, and performing an alpha-blend between the aligned images according to the position of the control knob **534**. For example, if the control knob **534** is positioned to be one quarter of the distance from discrete position **536(3)** to discrete position **536(4)** along slide path **532**, then an aligned image corresponding to source image **522(4)** may be blended with twenty-five percent opacity over a fully opaque aligned image corresponding to source image **522(3)**.

In one embodiment, UI control **530** is configured to include a discrete position **536** for each source image **522** within a given image set **520** being viewed. Each image set **520** stored within the digital photographic system **300** of FIG. 3A may include a different number of source images **522**, and UI control **530** may be configured to establish discrete positions **536** to correspond to the source images **522** for a given image set **520**.

In certain embodiments, the viewer application **510** may implement two or more independent UI controls. For example, the viewer application **510** may implement a first UI control to select a source image **522** according to strobe intensity, and a second UI control to select a source image **522** according to ambient exposure. The first UI control may select among different values of strobe intensity function **420** illustrated in FIG. 4E, while the second UI control may select among different values of exposure parameter function **422**. In this way, the viewer application **510** may enable a user to independently adjust exposure with respect to strobe intensity and exposure after an image set has been sampled. In one embodiment, another independent UI control is implemented to select among source images **522** sampled under different strobe illumination colors.

FIG. 5C illustrates a resulting image **542** with differing levels of strobe exposure, in accordance with one embodi-

22

ment. In this example, control knob **534** is configured to select source images **522** of FIG. 5A sampled under increasing strobe intensity from left to right. When the control knob **534** is in the left-most position, the selected source image **522** may correspond to an image captured within time interval **410(0)** of FIG. 4A, 4B, 4C, 4D, 4G, or 4H. With respect to FIG. 4E, the left-most control knob position may select a source image captured within time intervals **410(0)** through **410(3)**, with a specific source image **522** selected through a second UI control configured to select according to an exposure parameter. When the control knob **534** is in the right-most position, the selected source image **522** may correspond to an image captured within time interval **410(7)** of FIG. 4A, 4B, 4D, 4G, or 4H. In the right-most position, the selected source image **522** may correspond to time interval **410(1)** of FIG. 4C, or time intervals **410(12)** through **410(15)** of FIG. 4E.

As shown, resulting image **542(1)** includes an under-exposed subject **550** sampled under insufficient strobe intensity, resulting image **542(2)** includes a properly-exposed subject **552** sampled under appropriate strobe intensity, and resulting image **542(3)** includes an over-exposed subject **554** sampled under excessive strobe intensity. A determination of appropriate strobe intensity is sometimes subjective, and embodiments of the present invention advantageously enable a user to subjectively select an image having a desirable or appropriate strobe intensity after a picture has been taken, and without loss of image quality or dynamic range. In practice, a user is able to take what is apparently one photograph by asserting a single shutter-release. The single shutter-release causes the digital photographic system **300** of FIG. 3A to sample multiple images in rapid succession, where each of the multiple images is sampled under varying strobe intensity. In one embodiment, time intervals **410** of less than two-hundred milliseconds are defined herein to establish rapid succession. A resulting image set **520** enables the user to advantageously select a resulting image later, such as after a particular photographic scene of interest is no longer available. This is in contrast to prior art solutions that conventionally force a user to manually take different photographs and manually adjust strobe intensity over the different photographs. This manual prior art process typically introduces substantial inter-image delay, resulting in a loss of content consistency among sampled images.

FIG. 5D illustrates a resulting image **542** generated from source images **522** and a selection depth map, in accordance with one embodiment. The source images **522** comprising an image set **520** are conceptually stacked in increasing depth, where depth corresponds to strobe intensity for a given source image **522**. Increasing depth therefore corresponds to increasing (or decreasing) strobe intensity. The selection depth map includes depth samples organized as a two-dimensional array that corresponds to pixels or regions comprising resulting image **542**. A depth value in this context acts similarly to selection parameter **518**, but just for a corresponding pixel or region. In certain embodiments, the source images **522** are aligned within the stack.

As shown, selection region **560** encompasses a portion of a foreground subject. Outside the selection region **560**, pixel data may be taken from source image **522(0)** and represented in the resulting image **542**, where source image **522(0)** corresponds to a minimum depth value. Inside the selection region **560**, pixel data may be taken from source images **522(1)** through **522(N-1)**, based on depth values for a given pixel or region in a selection depth map corresponding to the resulting image **542**. Depth information comprising the selection depth map may be generated using any

23

technically feasible technique. One technique is illustrated below in FIGS. 5E and 5F, where swiping over a region multiple times, using an input device such as a mouse or stylus or touch input (such as with a touch-sensitive display), increases depth values associated with the region. In one embodiment, UI control tools 544 include a UI control that sets a minimum depth (minimum strobe intensity) for the selection depth map. UI control tools 544 may also include a UI control to clear the selection depth map.

FIG. 5E illustrates a selection depth map relative to a stack of source images 522, in accordance with one embodiment. Depth values for the selection region 560 along an intersecting plane 566 are shown as a depth profile 562. The depth profile 562 corresponds to sequentially organized source images 522(0) through 522(7) comprising a stack. A greater depth value along the depth profile 562 indicates a corresponding selection of a source image 522. A fractional depth value may indicate selection of two source images 522 and a blend between two source images 522.

FIG. 5F illustrates specifying a selection depth map, in accordance with one embodiment. As shown, a selection path 564 defines the selection region 560. In one embodiment, a touch-input device is configured to track selection path 564, which may be drawn by a user on the touch-input device. In one embodiment, a depth value for a given region is assigned a value that reflects an accumulation of selection path 564 intersections with the region. For example, a depth value for each pixel in the selection region 560 that intersects with the selection path 564 may be increased by one a specified increment each time the selection path 564 intersects that particular pixel. The specified increment may correspond to a fractional depth value (e.g., 0.25 of a distinct depth increment) separating discrete depths associated with distinct images. So, if the selection path 564 intersects a particular pixel in the selection region 560 three times, the depth value for that pixel may be increased by three specified increments. In one embodiment, the selection path 564 intersects a pixel if the selection path 564 crosses any edge of the pixel. In another embodiment, the selection path 564 intersects a pixel if the selection path crosses any edge of a pixel within n number of pixels from that pixel (i.e., the selection path may intersect with pixels within a certain distance from the selection path). Depth values may accumulate according to any technically feasible technique.

In another embodiment, the selection region 560 is defined by the selection path 564. For example, a closed surface that includes all points in the selection path 564 may be selected as the selection region 560. Further, a closed surface is determined based on the selection path 564 and a set of pre-determined edges in the resulting image 542 and selected as the selection region 560. Once the selection region 560 has been defined, depth values for pixels in the selection region 560 may be incremented uniformly based on the selection path 564. For example, a length of the selection path 564 may be determined by accumulating a distance between points in the selection path 564. Then, all depth values for pixels in the selection region 560 may be incremented by a uniform value calculated as the length of the selection path 564 divided by a normalization factor. In other words, the longer the selection path 564, the greater the increase in depth value for the selection region 560. The normalization factor may be selected such that each incremental step in depth values requires a significant increase in the length of the selection path 564. Such embodiments, enable a user to adjust the depth of a particular region uniformly such that the entire region comprises pixel data from a single image in the image stack.

24

In one embodiment, depth values around the selection path 564 may accumulate according to a tapering region of influence about the selection path 564 to generate a field of depth samples resembling the effect of an air brush. For example, a window function may be specified that determines a value to increase the depth value corresponding to a pixel based on the distance of that pixel from the selection path 564. For example, pixels intersected by the selection path 564 may be increased by three units of depth value, pixels within one pixel width from the selection path 564, but not intersected by the selection path 564, may be increased by two units of depth value, pixels within two pixel widths from the selection path 564 may be increased by one unit of depth value, and pixels greater than three units of depth value from the selection path 564 may not be increased at all. In another embodiment, the depth values may be fractional depth values. In these embodiments, the window function may return values to increase the depth value of between 0 and 1, where 1 is returned for pixels that are intersected by the selection path 564 and values between [0,1) being returned for pixels not intersected by the selection path 564. The effect of the tapering window function using fractional depth values would be to blend one image associated with a lower depth value with another image associated with a higher depth value around the selection path 564 in order to have a soft transition between the two source images.

One advantage of the present invention is that a user may photograph a scene using a single shutter release command, and subsequently select an image sampled according to a strobe intensity that best satisfies user aesthetic requirements for the photographic scene. The one shutter release command causes a digital photographic system to rapidly sample a sequence of images with a range of strobe intensity and/or color. For example, twenty or more full-resolution images may be sampled within one second, allowing a user to capture a potentially fleeting photographic moment with the advantage of strobe illumination. Embodiments of the present invention advantageously enable a given digital photographic system to capture a nearly optimal image, among images within an image set, given implementation-specific limitations and/or capabilities of the digital photographic system. Furthermore, a user may select an image from the image set according to their subjective aesthetic assessment. Another advantage of the present invention is that a user may easily specify spatially varying contributions of different images within the image set (e.g., the sequence of images) through a simple UI control.

While various embodiments have been described above with respect to a digital camera 202 and a mobile device 204, any device configured to perform the methods 1, 100, or 102 of FIG. 1A, 1B, or 1C is within the scope and spirit of the present invention. In certain embodiments, two or more digital photographic systems implemented in respective devices are configured to sample corresponding image sets in mutual time synchronization. A single shutter release command may trigger the two or more digital photographic systems.

While the foregoing is directed to embodiments of the present invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. An apparatus, comprising:
 - a camera;
 - a strobe;

25

a display;
 a touch interface;
 one or more non-transitory memories; and
 one or more processors in communication with the camera, the strobe, the display, the touch interface, and the one or more non-transitory memories, wherein the one or more processors execute instructions in the one or more non-transitory memories, to cause the apparatus to:

emit, utilizing the strobe, illumination during a first time interval based on a first set of strobe parameters including a first strobe intensity and a first strobe color;

sample, utilizing the camera, a first image of a photographic scene during the first time interval based on a first set of exposure parameters;

emit, utilizing the strobe, illumination during a second time interval based on a second set of strobe parameters including a second strobe intensity and a second strobe color;

sample, utilizing the camera, a second image of the photographic scene during the second time interval based on a second set of exposure parameters;

emit, utilizing the strobe, illumination during a third time interval based on a third set of strobe parameters including a third strobe intensity and a third strobe color;

sample, utilizing the camera, a third image of the photographic scene during the third time interval based on a third set of exposure parameters;

generate a resulting image by performing a white-balance correction; and

store the resulting image.

2. The apparatus of claim 1, wherein the apparatus is configured such that the first set of exposure parameters, the second set of exposure parameters, and the third set of exposure parameters, each include an exposure time and an exposure sensitivity.

3. The apparatus of claim 1, wherein the apparatus is configured such that the second strobe intensity is the same as the third strobe intensity.

4. An apparatus, comprising:

a camera;

a strobe;

a display;

a touch interface;

one or more non-transitory memories; and

one or more processors in communication with the camera, the strobe, the display, the touch interface, and the one or more non-transitory memories, wherein the one or more processors execute instructions in the one or more non-transitory memories, to cause the apparatus to:

emit, utilizing the strobe, illumination during a first time interval based on a first set of strobe parameters including a first strobe intensity and a first strobe color;

sample, utilizing the camera, a first image of a photographic scene during at least a portion of the first time interval based on a first set of exposure parameters;

emit, utilizing the strobe, illumination during a second time interval based on a second set of strobe parameters including a second strobe intensity and a second strobe color;

26

sample, utilizing the camera, a second image of the photographic scene during at least a portion of the second time interval based on a second set of exposure parameters;

emit, utilizing the strobe, illumination during a third time interval based on a third set of strobe parameters including a third strobe intensity and a third strobe color, such that the first time interval, the second time interval, and the third time interval are different;

sample, utilizing the camera, a third image of the photographic scene during at least a portion of the third time interval based on a third set of exposure parameters;

generate a resulting image by performing a white-balance correction, by combining at least portions of at least two of: the first image, the second image, and the third image;

and store the resulting image.

5. The apparatus of claim 4, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal.

6. The apparatus of claim 4, wherein the apparatus is configured such that the measured ambient illumination color corresponds to color statistics accumulated while the one or more processors receive image data from the camera.

7. The apparatus of claim 1, wherein the apparatus is configured such that the first set of exposure parameters is determined based on an ambient illumination.

8. The apparatus of claim 7, wherein the apparatus is configured such that the first set of exposure parameters, the second set of exposure parameters, and the third set of exposure parameters are equal.

9. The apparatus of claim 1, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, on the display, the resulting image.

10. The apparatus of claim 9, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, utilizing the display, a user interface slider control; detect, by the touch interface, a slide gesture directed to the slider control; and

in response to detecting the slide gesture, display, utilizing the display, the first image.

11. The apparatus of claim 1, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to

further cause the apparatus to: select an image from an image set by evaluating exposure quality metric values for images in the image set and selecting the image with a best exposure quality metric value.

12. The apparatus of claim 11, wherein the apparatus is configured such that each of the exposure quality metric values comprises a cost function value, and the best exposure quality metric value corresponds to a lowest cost function value.

13. The apparatus of claim 12, wherein the apparatus is configured such that the cost function value reflects a cost penalty to over-exposed pixels.

14. The apparatus of claim 1, wherein the apparatus is configured such that the camera includes an image sensor that samples the photographic scene based on an exposure time and an exposure sensitivity.

15. An apparatus, comprising:

a camera;

a strobe;

27

a display;
 a touch interface;
 one or more non-transitory memories; and
 one or more processors in communication with the camera, the strobe, the display, the touch interface, and the one or more non-transitory memories, wherein the one or more processors execute instructions in the one or more non-transitory memories, to cause the apparatus to:

emit, utilizing the strobe, illumination during a first time interval based on a first set of strobe parameters including a first strobe intensity and a first strobe color;

sample, utilizing the camera, a first image of a photographic scene during at least a portion of the first time interval based on a first set of exposure parameters;

emit, utilizing the strobe, illumination during a second time interval based on a second set of strobe parameters including a second strobe intensity and a second strobe color;

sample, utilizing the camera, a second image of the photographic scene during at least a portion of the second time interval based on a second set of exposure parameters;

emit, utilizing the strobe, illumination during a third time interval based on a third set of strobe parameters including a third strobe intensity and a third strobe color;

sample, utilizing the camera, a third image of the photographic scene during at least a portion of the third time interval based on a third set of exposure parameters;

sample, utilizing the camera, a fourth image of the photographic scene during at least a portion of a fourth time interval based on a fourth set of exposure parameters and without the strobe emitting illumination, such that the first time interval, the second time interval, the third time interval, and the fourth time interval are different;

identify a flash image based on at least one of: the first image, the second image, or the third image;

generate a white-balanced flash image by performing a white-balance correction on the flash image;

generate a white-balanced ambient image by performing a white-balance correction on the fourth image; generate a resulting image by combining at least portions of the white-balanced flash image and the white-balanced ambient image; and

store the resulting image.

16. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal to a measured ambient illumination color, and wherein the first set of exposure parameters and the fourth set of exposure parameters are determined based on an ambient illumination.

17. The apparatus of claim 15, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, utilizing the display, the resulting image; display, utilizing the display, a user interface slider control;

detect, utilizing the touch interface, a slide gesture directed to the slider control; and in response to detecting the slide gesture, display, utilizing the display, the first image.

28

18. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal to a measured ambient illumination color.

19. The apparatus of claim 18, wherein the apparatus is configured such that the first set of exposure parameters and the fourth set of exposure parameters are determined based on an ambient illumination.

20. The apparatus of claim 15, wherein the apparatus is configured such that the camera includes an image sensor that samples the photographic scene based on an exposure time and an exposure sensitivity.

21. The apparatus of claim 15, wherein the apparatus is configured such that the first image, the second image, and the third image are stored in an image set.

22. The apparatus of claim 15, wherein the apparatus is configured such that the first image, the second image, and the third image are stored in an image set from which the flash image is identified.

23. The apparatus of claim 15, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, utilizing the display, a user interface control; detecting, utilizing the touch interface, a gesture directed to the user interface control; and in response to detecting the gesture, alter a viewing parameter.

24. The apparatus of claim 15, wherein the apparatus is configured such that the second strobe intensity is different than the first strobe intensity.

25. The apparatus of claim 15, wherein the apparatus is configured such that the third strobe intensity is different than the first strobe intensity.

26. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe intensity, the second strobe intensity, and the third strobe intensity, are not equal.

27. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal.

28. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are not equal.

29. The apparatus of claim 15, wherein the apparatus is configured such that at least two of: the first strobe color, the second strobe color, and the third strobe color, are not equal.

30. The apparatus of claim 15, wherein the apparatus is configured such that the first set of exposure parameters, the second set of exposure parameters, and the third set of exposure parameters, each include an exposure time and an exposure sensitivity.

31. The apparatus of claim 15, wherein the apparatus is configured such that the second strobe intensity is the same as the third strobe intensity.

32. The apparatus of claim 15, wherein the apparatus is configured such that the first strobe color is equal to a measured ambient illumination color.

33. The apparatus of claim 32, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal.

34. The apparatus of claim 32, wherein the apparatus is configured such that the measured ambient illumination color corresponds to color statistics accumulated while the one or more processors receive image data from the camera.

35. The apparatus of claim 15, wherein the apparatus is configured such that the first set of exposure parameters is determined based on an ambient illumination.

36. The apparatus of claim 35, wherein the apparatus is configured such that the first set of exposure parameters, the

29

second set of exposure parameters, and the third set of exposure parameters are equal.

37. The apparatus of claim 15, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: select an image from an image set by evaluating exposure quality metric values for images in the image set and selecting the image based on an exposure quality metric value thereof.

38. The apparatus of claim 37, wherein the apparatus is configured such that the exposure quality metric values each includes a cost function value, and the exposure quality metric value of the selected image includes to a lowest cost function value.

39. The apparatus of claim 38, wherein the apparatus is configured such that the cost function value reflects a cost penalty to over-exposed pixels.

40. The apparatus of claim 15, wherein the apparatus is configured such that the flash image is identified based on at least two of: the first image, the second image, and the third image.

41. The apparatus of claim 1, wherein the apparatus is configured such that the at least two of: the first image, the second image, and the third image, includes the first image, the second image, and the third image.

42. The apparatus of claim 1, wherein the apparatus is configured such that white-balance correction is performed before the at least portions of the at least two of: the first image, the second image, and the third image, are combined.

43. The apparatus of claim 1, wherein the apparatus is configured such that white-balance correction is performed after the at least portions of the at least two of: the first image, the second image, and the third image, are combined.

44. An apparatus, comprising:

a camera;

a strobe;

a display;

a touch interface;

one or more non-transitory memories; and

one or more processors in communication with the camera, the strobe, the display, the touch interface, and the one or more non-transitory memories, wherein the one or more processors execute instructions in the one or more non-transitory memories, to cause the apparatus to:

emit, utilizing the strobe, illumination during a first time interval based on a first set of strobe parameters including a first strobe intensity and a first strobe color;

sample, utilizing the camera, a first image of a photographic scene during at least a portion of the first time interval based on a first set of exposure parameters;

emit, utilizing the strobe, illumination during a second time interval based on a second set of strobe parameters including a second strobe intensity and a second strobe color;

sample, utilizing the camera, a second image of the photographic scene during at least a portion of the second time interval based on a second set of exposure parameters;

emit, utilizing the strobe, illumination during a third time interval based on a third set of strobe parameters including a third strobe intensity and a third strobe color;

30

sample, utilizing the camera, a third image of the photographic scene during at least a portion of the third time interval based on a third set of exposure parameters;

sample, utilizing the camera, a fourth image of the photographic scene during at least a portion of a fourth time interval based on a fourth set of exposure parameters and without the strobe emitting illumination, such that the first time interval, the second time interval, the third time interval, and the fourth time interval are different;

identify a flash image based on at least one of: the first image, the second image, or the third image;

generate a combined image by combining at least portions of the flash image and the fourth image;

generate a resulting image by performing a white-balance correction on the combined image; and

store the resulting image.

45. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal to a measured ambient illumination color, and wherein the first set of exposure parameters and the fourth set of exposure parameters are determined based on an ambient illumination.

46. The apparatus of claim 44, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, utilizing the display, the resulting image;

display, utilizing the display, a user interface slider control;

detect, utilizing the touch interface, a slide gesture directed to the slider control; and

in response to detecting the slide gesture, display, utilizing the display, the first image.

47. The apparatus of claim 44, wherein the apparatus is configured such that the camera includes an image sensor that samples the photographic scene based on an exposure time and an exposure sensitivity.

48. The apparatus of claim 44, wherein the apparatus is configured such that the first image, the second image, and the third image are stored in an image set.

49. The apparatus of claim 44, wherein the apparatus is configured such that the first image, the second image, and the third image are stored in an image set from which the flash image is identified.

50. The apparatus of claim 44, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: display, utilizing the display, a user interface control;

detecting, utilizing the touch interface, a gesture directed to the user interface control; and

in response to detecting the gesture, alter an exposure parameter.

51. The apparatus of claim 44, wherein the apparatus is configured such that the second strobe intensity is different than the first strobe intensity.

52. The apparatus of claim 44, wherein the apparatus is configured such that the third strobe intensity is different than the first strobe intensity.

53. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe intensity, the second strobe intensity, and the third strobe intensity, are not equal.

54. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal.

31

55. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are not equal.

56. The apparatus of claim 44, wherein the apparatus is configured such that at least two of:

the first strobe color, the second strobe color, and the third strobe color, are not equal.

57. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal to a measured ambient illumination color.

58. The apparatus of claim 57, wherein the apparatus is configured such that the first set of exposure parameters and the fourth set of exposure parameters are determined based on an ambient illumination.

59. The apparatus of claim 44, wherein the apparatus is configured such that the first set of exposure parameters, the second set of exposure parameters, and the third set of exposure parameters, each include an exposure time and an exposure sensitivity.

60. The apparatus of claim 44, wherein the apparatus is configured such that the second strobe intensity is the same as the third strobe intensity.

61. The apparatus of claim 44, wherein the apparatus is configured such that the first strobe color is equal to a measured ambient illumination color.

62. The apparatus of claim 61, wherein the apparatus is configured such that the first strobe color, the second strobe color, and the third strobe color are equal.

63. The apparatus of claim 61, wherein the apparatus is configured such that the measured ambient illumination

32

color corresponds to color statistics accumulated while the one or more processors receive image data from the camera.

64. The apparatus of claim 44, wherein the apparatus is configured such that the first set of exposure parameters is determined based on an ambient illumination.

65. The apparatus of claim 64, wherein the apparatus is configured such that the first set of exposure parameters, the second set of exposure parameters, and the third set of exposure parameters are equal.

66. The apparatus of claim 44, wherein the one or more processors execute the instructions in the one or more non-transitory memories, to further cause the apparatus to: select an image from an image set by evaluating exposure quality metric values for images in the image set and selecting the image based on an exposure quality metric value thereof.

67. The apparatus of claim 66, wherein the apparatus is configured such that the exposure quality metric values each includes a cost function value, and the exposure quality metric value of the selected image includes to a lowest cost function value.

68. The apparatus of claim 67, wherein the apparatus is configured such that the cost function value reflects a cost penalty to over-exposed pixels.

69. The apparatus of claim 44, wherein the apparatus is configured such that the flash image is identified based on at least two of: the first image, the second image, and the third image.

* * * * *